Fourier Analysis and Applications

Lecture notes, Chapters 1–8 and Appendix

Based on E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*

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Chapter 1. The Genesis of Fourier Analysis

1.1 The vibrating string

Motivation. Fourier analysis was born from two physical problems: the vibrating string and the flow of heat. Both lead to linear partial differential equations whose separated solutions are sines and cosines. The general solution is then a superposition, i.e. a Fourier series, and the central question becomes: which functions can be written this way?

Simple harmonic motion. A mass m on a spring with constant k obeys $-ky(t) = my''(t)$. With $c = \sqrt{k/m}$ the solutions are $y(t) = a\cos ct + b\sin ct = A\cos(ct-\varphi)$, with **amplitude** $A = \sqrt{a^2+b^2}$, **period** $2\pi/c$ and **frequency** $c/2\pi$; here $a = y(0)$, $b = y'(0)/c$. In complex form $e^{ix} = \cos x + i\sin x$ (Euler), $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

Standing and travelling waves. A **travelling wave** has the form $u(x,t) = F(x-ct)$ (a profile moving with speed c). A **standing wave** has the form $u(x,t) = \varphi(x)\psi(t)$: the profile keeps its shape and only its amplitude varies in time.

1.1.1 Derivation of the wave equation

Model a string of length L, density ρ and tension τ fixed at its ends by N particles of mass ρh ($h = L/N$) at $x_n = nh$, connected by springs. Let $y_n(t)$ be the vertical displacement of the n-th particle. Hooke's law for the vertical components of the tensions gives

$$\rho h\, y_n''(t) = \frac{\tau}{h}\big(y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)\big).$$

As $h \to 0$ and $y_n(t) \to u(x,t)$, the right side divided by h tends to $\tau\,\partial_x^2 u$, giving the **wave equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\tau/\rho}. \tag{1.1}$$

(Changing $x \mapsto x/L$, $t \mapsto ct/L$ we may take $c = 1$, $L = \pi$.) **Invariance properties:** the equation is linear and invariant under translations in x and t.

1.1.2 Solution to the wave equation

d'Alembert's solution. Introduce $\xi = x+t$, $\eta = x-t$ and $v(\xi,\eta) = u(x,t)$. The wave equation becomes $\frac{\partial^2 v}{\partial\xi\partial\eta} = 0$, so $v = F(\xi) + G(\eta)$:

$$u(x,t) = F(x+t) + G(x-t),$$

the sum of two travelling waves. On the real line with $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$:

$$u(x,t) = \frac{f(x+t)+f(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy. \tag{1.2}$$

Check [Supplement]: $F+G = f$ and $F'-G' = g$ give $F = \frac{f}{2} + \frac{1}{2}\int_0^x g + C$, $G = \frac{f}{2} - \frac{1}{2}\int_0^x g - C$. On $[0,\pi]$ with $u(0,t) = u(\pi,t) = 0$, extend f, g to $[-\pi,\pi]$ as odd functions and then 2π-periodically; (1.2) then satisfies the boundary conditions.

Superposition of standing waves. Separating variables, $u(x,t) = \varphi(x)\psi(t)$ gives $\frac{\psi''}{\psi} = \frac{\varphi''}{\varphi} = \lambda$, a constant. Boundedness forces $\lambda = -m^2 \le 0$, so $\psi(t) = A\cos mt + B\sin mt$ and $\varphi(x) = \tilde{A}\cos mx + \tilde{B}\sin mx$; $\varphi(0) = \varphi(\pi) = 0$ forces $\varphi(x) = \sin mx$ with $m = 1, 2, \dots$. The standing waves are

$$u_m(x,t) = (A_m\cos mt + B_m\sin mt)\sin mx,$$

the m-th **harmonic** ($m=1$: **fundamental tone**; $m\geq 2$: **overtones**, of higher frequency $m/2\pi$). By linearity

$$u(x,t)=\sum_{m=1}^{\infty}(A_m\cos mt+B_m\sin mt)\sin mx \tag{1.3}$$

is a candidate solution, and the initial condition $u(x,0)=f(x)$ requires

$$f(x)=\sum_{m=1}^{\infty}A_m\sin mx. \tag{1.4}$$

Fourier's question: can every (reasonable) f on $[0,\pi]$ with $f(0)=f(\pi)=0$ be written as in (1.4)? If so, the orthogonality $\int_0^\pi \sin mx\sin nx\,dx=\frac{\pi}{2}\delta_{mn}$ forces

$$A_m=\frac{2}{\pi}\int_0^\pi f(x)\sin mx\,dx,$$

and similarly B_m is determined by $\partial_t u(x,0)=g(x)=\sum mB_m\sin mx$. Extending f oddly to $[-\pi,\pi]$, (1.4) is the special case of

$$F(x)=\sum_{m=-\infty}^{\infty}a_m e^{imx},\qquad a_m=\frac{1}{2\pi}\int_{-\pi}^{\pi}F(x)e^{-imx}\,dx. \tag{1.5}$$

1.1.3 Example: the plucked string

Pluck the string at $p\in(0,\pi)$ to height h with zero initial velocity: $f(x)=\frac{xh}{p}$ for $0\leq x\leq p$, $f(x)=\frac{h(\pi-x)}{\pi-p}$ for $p\leq x\leq\pi$, $g=0$. Then $B_m=0$ and, integrating by parts,

$$A_m=\frac{2h\sin mp}{m^2p(\pi-p)},\qquad u(x,t)=\sum_{m\geq 1}A_m\cos mt\sin mx.$$

Since $\sum|A_m|<\infty$, the series converges uniformly, and u is continuous. In general, though, (1.3) may fail to be twice differentiable, so it solves the wave equation only in a generalised sense (compare d'Alembert's formula).

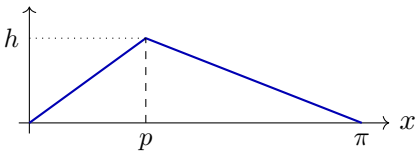


Figure. *The plucked string at $t=0$.*

1.2 The heat equation

Motivation. Heat flow leads to the heat and Laplace equations. Steady-state temperatures in a disc are again obtained by separating variables, this time in polar coordinates, and lead to Fourier series in the angle θ.

1.2.1 Derivation of the heat equation

For a plate $\mathbb{R}^2$ with temperature $u(x,y,t)$, specific heat σ and thermal conductivity κ: the heat in a small square S is $\sigma\iint_S u$, and the heat flux across ∂S is $\kappa\int_{\partial S}\partial u/\partial n$ (Fourier's law: heat flows from

hotter to colder, proportionally to the gradient). Equating the rate of change of heat in S with the flux, dividing by the area and shrinking S:

$$\frac{\partial u}{\partial t} = \frac{\kappa}{\sigma}\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\Big) = \frac{\kappa}{\sigma}\,\triangle u. \tag{1.6}$$

In a **steady state** $\partial u/\partial t = 0$, and u satisfies the **Laplace equation** $\triangle u = 0$; its solutions are **harmonic functions**.

1.2.2 Steady-state heat equation in the disc

Dirichlet problem. Find u with $\triangle u = 0$ in the unit disc $D = \{x^2 + y^2 < 1\}$ and $u = f$ on the circle C. In polar coordinates

$$\triangle u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

Separating $u = F(r)G(\theta)$, $\frac{r^2F'' + rF'}{F} = -\frac{G''}{G} = \lambda$. G is 2π-periodic, so $\lambda = m^2$ and $G(\theta) = e^{\pm im\theta}$, $m \geq 0$. Then $r^2F'' + rF' - m^2F = 0$ has solutions $F = r^{\pm m}$ $(m \geq 1)$, $F = 1, \log r$ $(m = 0)$; boundedness at 0 leaves $F = r^{|m|}$. So the candidate solution is

$$u(r,\theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \qquad f(\theta) = \sum a_m e^{im\theta}. \tag{1.7}$$

Substituting $a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)e^{-im\varphi}d\varphi$ and summing the geometric series formally,

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi) f(\varphi)\, d\varphi, \qquad P_r(\gamma) = \sum_{m\in\mathbb{Z}} r^{|m|}e^{im\gamma} = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2},$$

the **Poisson kernel**; this is made rigorous in Chapter 2.

1.3 Selected exercises

Motivation. Five exercises from the book that justify the steps taken for granted above: the form of the separated solutions, the discrete-to-continuous limit in the wave equation, the polar Laplacian, and the plucked-string coefficients.

Exercise 1.6 (Solutions of $f'' + c^2 f = 0$)

Let $c > 0$ and let f be a C^2 function on $\mathbb{R}$ with $f''(t) + c^2 f(t) = 0$. Prove that there are constants a, b with $f(t) = a\cos ct + b\sin ct$ for all t.

Solution.

Step 1. Let $g(t) = f(t)\cos ct - c^{-1}f'(t)\sin ct$ and $h(t) = f(t)\sin ct + c^{-1}f'(t)\cos ct$, both C^1. Using $f'' = -c^2 f$,

$$g'(t) = f'\cos ct - cf\sin ct - c^{-1}f''\sin ct - f'\cos ct = -cf\sin ct + cf\sin ct = 0,$$
$$h'(t) = f'\sin ct + cf\cos ct + c^{-1}f''\cos ct - f'\sin ct = cf\cos ct - cf\cos ct = 0.$$

Step 2. By the mean value theorem $g \equiv a$ and $h \equiv b$ are constant on $\mathbb{R}$ (with $a = f(0)$, $b = f'(0)/c$).

Step 3. Then, for every t,

$$a\cos ct + b\sin ct = f(\cos^2 ct + \sin^2 ct) + c^{-1}f'(-\sin ct\cos ct + \sin ct\cos ct) = f(t).$$

This justifies the form of ψ and φ in the separation of variables of §1.1.2. ∎

Exercise 1.8 (Second difference quotient)

Let F be C^2 on (a,b) and $x \in (a,b)$. Show that for $x+h \in (a,b)$

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + h^2\varphi(h), \qquad \varphi(h) \to 0 \ (h \to 0),$$

and deduce that $\big(F(x+h) + F(x-h) - 2F(x)\big)/h^2 \to F''(x)$.

Solution.

Step 1. By the fundamental theorem of calculus, $F(x+h) - F(x) = \int_x^{x+h} F'(y)\,dy$ and $F'(y) = F'(x) + \int_x^y F''(s)\,ds$. Writing $F''(s) = F''(x) + \big(F''(s) - F''(x)\big)$ and using $\int_x^{x+h}(y-x)\,dy = h^2/2$,

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + R(h), \qquad R(h) = \int_x^{x+h}\!\!\int_x^y \big(F''(s) - F''(x)\big)\,ds\,dy.$$

Step 2. Let $\omega(\delta) = \sup_{|s-x|\le\delta} |F''(s) - F''(x)|$; $\omega(\delta) \to 0$ as $\delta \to 0$ by continuity of F''. For y between x and $x+h$ the inner integral is at most $|y-x|\,\omega(|h|)$, so $|R(h)| \le \omega(|h|)\,\big|\int_x^{x+h}|y-x|\,dy\big| = \tfrac{h^2}{2}\omega(|h|)$ (for either sign of h).

Step 3. Set $\varphi(h) = R(h)/h^2$ for $h \neq 0$. Then $|\varphi(h)| \le \tfrac{1}{2}\omega(|h|) \to 0$.

Step 4. Adding the expansions at h and $-h$, the first-order terms cancel:

$$\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} = F''(x) + \varphi(h) + \varphi(-h) \to F''(x).$$

This is the limit used to pass from the particle model to the wave equation (1.1). ∎

Exercise 1.9 (Missing harmonics of the plucked string)

For the plucked string of §1.1.3, derive $A_m = \frac{2h\sin mp}{m^2 p(\pi - p)}$. For which p are the 2nd, 4th, 6th, … harmonics missing? For which p are the 3rd, 6th, 9th, … harmonics missing?

Solution.

Step 1. Since f is continuous, piecewise linear and $f(0) = f(\pi) = 0$, integrating by parts (the boundary terms vanish) gives

$$\int_0^\pi f(x)\sin mx\,dx = \frac{1}{m}\int_0^\pi f'(x)\cos mx\,dx = \frac{1}{m}\Big(\frac{h}{p}\int_0^p \cos mx\,dx - \frac{h}{\pi - p}\int_p^\pi \cos mx\,dx\Big).$$

Step 2. The two integrals are $\frac{\sin mp}{m}$ and $-\frac{\sin mp}{m}$, so the bracket equals $\frac{h\sin mp}{m}\big(\frac{1}{p} + \frac{1}{\pi - p}\big) = \frac{\pi h \sin mp}{m\,p(\pi - p)}$. Multiplying by $\frac{2}{\pi}$ gives the stated A_m.

Step 3. The m-th harmonic is missing iff $A_m = 0$ iff $\sin mp = 0$. All even harmonics vanish iff $\sin 2kp = 0$ for all $k \ge 1$; taking $k = 1$ this forces $2p \in \pi\mathbb{Z}$, i.e. $p = \pi/2$ for $p \in (0,\pi)$, and conversely $p = \pi/2$ gives $\sin(k\pi) = 0$. (The odd harmonics remain, since $\sin(m\pi/2) = \pm 1$ for odd m.)

Step 4. Similarly the multiples of 3 vanish iff $\sin 3p = 0$, i.e. $p = \pi/3$ or $p = 2\pi/3$. Plucking at the midpoint or at a third of the string removes exactly those overtones. ∎

Exercise 1.10 (Laplacian in polar coordinates)

Let u be C^2 on an open set of $\mathbb{R}^2 \setminus \{0\}$ and $v(r,\theta) = u(r\cos\theta, r\sin\theta)$. Prove

$$u_{xx} + u_{yy} = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta}, \qquad |u_x|^2 + |u_y|^2 = |v_r|^2 + \frac{1}{r^2}|v_\theta|^2.$$

Solution. Write $c = \cos\theta$, $s = \sin\theta$; all derivatives of u are evaluated at (rc, rs).

Step 1 (first derivatives). By the chain rule $v_r = cu_x + su_y$ and $v_\theta = -rsu_x + rcu_y$.

Step 2 (gradient identity). Hence $\frac{1}{r}v_\theta = -su_x + cu_y$ and

$$|v_r|^2 + \tfrac{1}{r^2}|v_\theta|^2 = |cu_x + su_y|^2 + |-su_x + cu_y|^2 = (c^2+s^2)(|u_x|^2 + |u_y|^2),$$

since the cross terms $2cs\,\mathrm{Re}(u_x\bar{u}_y)$ and $-2cs\,\mathrm{Re}(u_x\bar{u}_y)$ cancel.

Step 3 (second derivatives). Differentiating Step 1 again and using $u_{xy} = u_{yx}$ ($u \in C^2$),

$$\begin{aligned} v_{rr} &= c^2u_{xx} + 2csu_{xy} + s^2u_{yy}, \\ v_{\theta\theta} &= -rcu_x - rsu_y + r^2(s^2u_{xx} - 2csu_{xy} + c^2u_{yy}) = -rv_r + r^2(s^2u_{xx} - 2csu_{xy} + c^2u_{yy}). \end{aligned}$$

Step 4. Therefore $v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta} = (c^2+s^2)u_{xx} + (s^2+c^2)u_{yy} = u_{xx} + u_{yy}$; the mixed terms and the v_r terms cancel. ∎

Exercise 1.11 (The Euler equation $r^2F'' + rF' - n^2F = 0$)

Let $n \in \mathbb{Z}$. Show that the twice differentiable solutions on $r > 0$ of $r^2F''(r)+rF'(r)-n^2F(r) = 0$ are exactly the linear combinations of r^n, r^{-n} if $n \neq 0$, and of $1, \log r$ if $n = 0$.

Solution. The equation depends only on n^2, so assume $n \geq 0$.

Step 1 (these are solutions). For $F = r^{\pm n}$: $r^2(\pm n)(\pm n - 1)r^{\pm n-2} + r(\pm n)r^{\pm n-1} - n^2r^{\pm n} = (n^2 \mp n \pm n - n^2)r^{\pm n} = 0$. For $F = \log r$: $r^2(-r^{-2}) + r \cdot r^{-1} = 0$; constants are clearly solutions. By linearity so are their combinations.

Step 2 (reduction). Let F be a solution and $g(r) = r^{-n}F(r)$, twice differentiable. From $F = r^ng$,

$$r^2F'' + rF' - n^2F = r^n\left[r^2g'' + 2nrg' + n(n-1)g + rg' + ng - n^2g\right] = r^{n+1}\left[rg'' + (2n+1)g'\right],$$

so $rg'' + (2n+1)g' = 0$ on $(0,\infty)$.

Step 3. Then $\left(r^{2n+1}g'\right)' = r^{2n}(rg'' + (2n+1)g') = 0$, so by the mean value theorem on the interval $(0,\infty)$, $r^{2n+1}g'(r) = C$ for a constant C.

Step 4. If $n \geq 1$: $g' = Cr^{-2n-1}$ gives $g = -\frac{C}{2n}r^{-2n} + D$ and $F = Dr^n - \frac{C}{2n}r^{-n}$. If $n = 0$: $g' = C/r$ gives $F = g = C\log r + D$. This completes the classification used in §1.2.2 (and in the uniqueness part of Theorem 5.7). ∎

Chapter 2. Basic Properties of Fourier Series

2.1 Examples and formulation of the problem

Motivation. We fix what "function" and "Fourier series" mean (Riemann integrable functions on the circle) and pose the basic question: in what sense does the Fourier series of f converge back to f?

2.1.1 Main definitions and some examples

A bounded function f on $[a,b]$ is (Riemann) **integrable** if $\sup_P L(P,f) = \inf_P U(P,f)$ over partitions P; continuous and piecewise continuous functions are integrable (see the Appendix). Complex $f = u + iv$ is integrable if u and v are. A function on the circle is a 2π-periodic function on $\mathbb{R}$, equivalently a function $F(e^{i\theta}) = f(\theta)$ on $\{|z| = 1\}$; we call such f integrable on the circle if it is integrable on one (hence every) period.

Definition (Fourier coefficients and series)

For f integrable on $[a,b]$, $L = b-a$, the n-th **Fourier coefficient** is

$$\hat{f}(n) = \frac{1}{L}\int_a^b f(x)e^{-2\pi inx/L}\,dx, \qquad n \in \mathbb{Z},$$

and the **Fourier series** is $f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi inx/L}$. On the circle ($L = 2\pi$): $\hat{f}(n) = a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}\,d\theta$. The **$N$-th partial sum** is $S_N(f)(\theta) = \sum_{|n|\le N} \hat{f}(n)e^{in\theta}$. A **trigonometric polynomial** is $\sum_{|n|\le N} c_n e^{in\theta}$, of degree N if $c_{\pm N} \neq 0$.

Basic question. In what sense does $S_N(f) \to f$ as $N \to \infty$?

Example (Sawtooth)

For $f(\theta) = \frac{\pi-\theta}{2}$ on $0 < \theta < 2\pi$ (extended periodically, $f(0) = 0$): $\hat{f}(0) = 0$ and, integrating by parts, $\hat{f}(n) = \frac{1}{2in}$ for $n \neq 0$. So $f(\theta) \sim \sum_{n\neq 0} \frac{e^{in\theta}}{2in} = \sum_{n\ge 1} \frac{\sin n\theta}{n}$.

Example

For $f(\theta) = |\theta|$ on $[-\pi,\pi]$: $\hat{f}(0) = \frac{\pi}{2}$ and $\hat{f}(n) = \frac{-1+(-1)^n}{\pi n^2}$ for $n \neq 0$, so $f(\theta) \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{n \text{ odd}\ge 1} \frac{\cos n\theta}{n^2}$.

Example

For $f(\theta) = \theta$ on $(-\pi,\pi)$: $\hat{f}(0) = 0$, $\hat{f}(n) = \frac{(-1)^{n+1}}{in}$, so $f \sim 2\sum_{n\ge 1}(-1)^{n+1}\frac{\sin n\theta}{n}$.

Example (Dirichlet kernel)

$D_N(\theta) = \sum_{|n|\le N} e^{in\theta} = \frac{\sin((N+\frac{1}{2})\theta)}{\sin(\theta/2)}$; it is the trigonometric polynomial with $\hat{D}_N(n) = 1$ for $|n| \le N$ and 0 otherwise. *Proof:* with $\omega = e^{i\theta}$, $\sum_0^N \omega^n = \frac{1-\omega^{N+1}}{1-\omega}$ and $\sum_{-N}^{-1} \omega^n = \frac{\omega^{-N}-1}{1-\omega}$, so

$D_N = \frac{\omega^{-N} - \omega^{N+1}}{1-\omega}$; multiply numerator and denominator by $\omega^{-1/2}$.

Example (Poisson kernel)

For $0 \le r < 1$, $P_r(\theta) = \sum_{n\in\mathbb{Z}} r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r\cos\theta+r^2}$ (absolutely convergent series; sum the two geometric series).

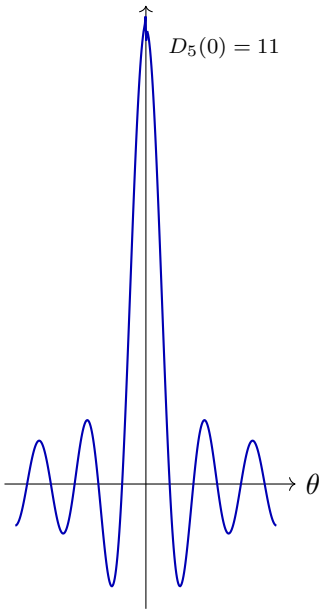


Figure 1: The Dirichlet kernel D_5: it takes negative values and $\int |D_N|$ grows like $\log N$.

2.2 Uniqueness of Fourier series

Motivation. If all Fourier coefficients of f vanish, then f vanishes at its points of continuity. Trigonometric polynomials that peak sharply at one point make this rigorous, and it already yields uniform convergence when $\sum |\hat{f}(n)| < \infty$, in particular for C^2 functions.

Theorem 2.1

Let f be integrable on the circle with $\hat{f}(n) = 0$ for all n. Then $f(\theta_0) = 0$ whenever f is continuous at θ_0.

Proof.

Step 1 (setup). First let f be real-valued. Translating (which multiplies $\hat{f}(n)$ by a unimodular factor), assume f is defined on $[-\pi, \pi]$, $\theta_0 = 0$, and suppose $f(0) > 0$. Since $\hat{f}(n) = 0$ for all n, $\int_{-\pi}^{\pi} f(\theta)p(\theta)\,d\theta = 0$ for every trigonometric polynomial p. We construct p_k with $\int f p_k \to \infty$.

Step 2 (the polynomials). Choose $0 < \delta \le \pi/2$ with $f(\theta) > f(0)/2$ for $|\theta| < \delta$. Let $p(\theta) = \epsilon + \cos\theta$ with $\epsilon > 0$ so small that $|p(\theta)| < 1 - \epsilon/2$ for $\delta \le |\theta| \le \pi$. Choose $0 < \eta < \delta$ with $p(\theta) \ge 1 + \epsilon/2$ for $|\theta| < \eta$. Let $p_k = p^k$, a trigonometric polynomial. Let B bound $|f|$.

Step 3 (estimates). On $\delta \le |\theta| \le \pi$: $|\int f p_k| \le 2\pi B(1-\epsilon/2)^k \to 0$. On $\eta \le |\theta| < \delta$: $p(\theta) \ge 0$ (as $\delta \le \pi/2$) and $f > 0$, so $\int f p_k \ge 0$ there. On $|\theta| < \eta$: $\int f p_k \ge 2\eta \frac{f(0)}{2}(1+\epsilon/2)^k \to \infty$. Hence $\int f p_k \to \infty$, contradicting $\int f p_k = 0$. So $f(0) \le 0$; applying this to $-f$, $f(0) = 0$.

Step 4 (complex f). Write $f = u + iv$ with $u = \frac{f+\bar{f}}{2}$, $v = \frac{f-\bar{f}}{2i}$. Since $\hat{\bar{f}}(n) = \overline{\hat{f}(-n)} = 0$, u and v have vanishing Fourier coefficients and are continuous at θ_0; apply the real case. ■

Corollary 2.2

If f is continuous on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$.

Proof. Theorem 2.1 at every point. ■

Corollary 2.3

If f is continuous on the circle and $\sum_n |\hat{f}(n)| < \infty$, then $S_N(f) \to f$ uniformly: $\lim_{N\to\infty} S_N(f)(\theta) = f(\theta)$ uniformly in θ.

Proof. By the Weierstrass M-test, $S_N(f)$ converges uniformly to a continuous $g(\theta) = \sum \hat{f}(n)e^{in\theta}$. Uniform convergence allows term-by-term integration, so $\hat{g}(n) = \hat{f}(n)$. By Corollary 2.2 applied to $f - g$, $f = g$. ■

Corollary 2.4

If f is twice continuously differentiable on the circle, then $\hat{f}(n) = O(1/|n|^2)$, so the Fourier series of f converges absolutely and uniformly to f.

Proof. Integrating by parts twice (boundary terms cancel by periodicity), $2\pi\hat{f}(n) = \frac{1}{in}\int f'(\theta)e^{-in\theta}d\theta = \frac{-1}{n^2}\int f''(\theta)e^{-in\theta}d\theta$, so $|n|^2|\hat{f}(n)| \le \frac{1}{2\pi}\int |f''|$. Since $\sum 1/n^2 < \infty$, Corollary 2.3 applies. ■

Remark (on the hypotheses)

In fact $f \in C^1$ is enough (Exercise in the book), and if f is C^k then $\hat{f}(n) = o(1/|n|^k)$. In general, the smoother f, the faster $\hat{f}(n)$ decays.

2.3 Convolutions

Motivation. Partial sums and the various means of a Fourier series are all convolutions $f * K$ with a kernel. Convolution turns into multiplication of Fourier coefficients, and it smooths: $f * g$ is continuous even if f, g are only integrable.

Definition (Convolution)

For 2π-periodic integrable f, g on $\mathbb{R}$, $(f * g)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)g(x-y)\,dy$. For instance $S_N(f)(x) = \sum_{|n|\le N} \hat{f}(n)e^{inx} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\sum_{|n|\le N} e^{in(x-y)}dy = (f * D_N)(x)$.

Proposition 3.1

Let f, g, h be 2π-periodic integrable functions. Then:
(i) $f * (g + h) = (f * g) + (f * h)$.
(ii) $(cf) * g = c(f * g) = f * (cg)$ for $c \in \mathbb{C}$.

(iii) $f * g = g * f$.
(iv) $(f * g) * h = f * (g * h)$.
(v) $f * g$ is continuous.
(vi) $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.

Lemma 3.2

Let f be integrable on the circle with $|f| \le B$. There are continuous functions f_k on the circle with $\sup |f_k| \le B$ and $\int_{-\pi}^{\pi} |f(x) - f_k(x)|\, dx \to 0$.

Proof. Assume f real (otherwise treat real and imaginary parts). Given $\epsilon > 0$ take a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $U(P,f) - L(P,f) < \epsilon$. Let $f^*(x) = \sup_{x_{j-1} \le y \le x_j} f(y)$ on $[x_{j-1}, x_j)$; then $f^* \ge f$ and $\int (f^* - f) \le \int f^* - \int_L = U - L < \epsilon$. Now modify the step function f^* near each x_j: on $[x_j - \delta, x_j + \delta]$ replace it by the linear segment joining $f^*(x_j - \delta)$ to $f^*(x_j + \delta)$ (and match periodically at $\pm\pi$). The resulting $\tilde{f}$ is continuous, $|\tilde{f}| \le B$, and differs from f^* on a set of total length $\le 2N\delta$, where $|\tilde{f} - f^*| \le 2B$. With δ small, $\int |f^* - \tilde{f}| < \epsilon$. Hence $\int |f - \tilde{f}| < 2\epsilon$; take $\epsilon = 1/k$. ∎

Proof of Proposition 3.1.

Step 1 (continuous f, g). (i), (ii) are linearity of the integral. For (iii), substitute $y \mapsto x - y$ and use periodicity. For (iv), interchange the order of integration (continuous integrand on a square) and substitute. For (v): f is uniformly continuous, so $|f * g(x_1) - f * g(x_2)| \le \sup_y |f(x_1 - y) - f(x_2 - y)| \cdot \sup |g| \to 0$ (using (iii)). For (vi):

$$\begin{aligned}\widehat{f * g}(n) &= \frac{1}{2\pi} \int \Big(\frac{1}{2\pi} \int f(y) g(x-y) dy \Big) e^{-inx} dx \\ &= \frac{1}{2\pi} \int f(y) e^{-iny} \Big(\frac{1}{2\pi} \int g(x-y) e^{-in(x-y)} dx \Big) dy = \hat{f}(n)\hat{g}(n).\end{aligned}$$

Step 2 (approximation). For integrable f, g, take continuous $f_k \to f$, $g_k \to g$ as in Lemma 3.2 (bounded by B). Then $f * g - f_k * g_k = (f - f_k) * g + f_k * (g - g_k)$ and $\sup_x |(f - f_k) * g(x)| \le \frac{B}{2\pi} \int |f - f_k| \to 0$, similarly for the second term. So $f_k * g_k \to f * g$ uniformly, which gives (v) (uniform limit of continuous functions) and (iii), (iv) by passing to the limit. For (vi), $\hat{f}_k(n) \to \hat{f}(n)$ since $|\hat{f}(n) - \hat{f}_k(n)| \le \frac{1}{2\pi} \int |f - f_k|$, and $\widehat{f_k * g_k}(n) \to \widehat{f * g}(n)$ by uniform convergence. ∎

2.4 Good kernels

Motivation. A family of kernels with total integral 1, bounded L^1 norms and mass concentrating at 0 reproduces f through convolution. The Dirichlet kernel fails the middle condition, which is why the partial sums need not converge, while the Fejér and Poisson kernels are good.

Definition (Good kernels)

A family $\{K_n\}$ of integrable 2π-periodic functions is a family of **good kernels** if (i) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x)\, dx = 1$ for all n;
(ii) $\int_{-\pi}^{\pi} |K_n(x)|\, dx \le M$ for all n;
(iii) for every $\delta > 0$, $\int_{\delta \le |x| \le \pi} |K_n(x)|\, dx \to 0$ as $n \to \infty$. (If $K_n \ge 0$, (ii) follows from (i).)
The same definition is used for families indexed by a continuous parameter.

Theorem 4.1

If $\{K_n\}$ is a family of good kernels and f is integrable on the circle, then $(f * K_n)(x) \to f(x)$ at every point of continuity x of f. If f is continuous, the convergence is uniform.

Proof. Let f be continuous at x and $\epsilon > 0$; choose δ with $|f(x-y) - f(x)| < \epsilon$ for $|y| < \delta$. By (i) and (iii),

$$(f * K_n)(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y)[f(x-y) - f(x)]\, dy,$$

so, with $|f| \le B$,

$$\begin{aligned} |(f * K_n)(x) - f(x)| &\le \frac{1}{2\pi} \int_{|y|<\delta} |K_n||f(x-y) - f(x)| + \frac{1}{2\pi} \int_{\delta \le |y| \le \pi} |K_n||f(x-y) - f(x)| \\ &\le \frac{\epsilon M}{2\pi} + \frac{2B}{2\pi} \int_{\delta \le |y| \le \pi} |K_n(y)| dy, \end{aligned}$$

using (ii), and the last term is $< \epsilon$ for large n by (iii). If f is continuous, it is uniformly continuous and δ is independent of x. ∎

Since $S_N(f) = f * D_N$, one might hope $\{D_N\}$ is good. It satisfies (i), but not (ii): [Supplement] $|\sin((N+\frac{1}{2})\theta)|/|\sin(\theta/2)| \ge \frac{2}{|\theta|}|\sin((N+\frac{1}{2})\theta)|$, and integrating over $[-\pi, \pi]$ gives $\int |D_N| \ge c \log N$ (sum over the $2N$ subintervals where $|\sin((N+\frac{1}{2})\theta)| \ge \frac{1}{2}$). This is the root of the difficulties of pointwise convergence.

2.5 Cesàro and Abel summability: applications to Fourier series

Motivation. Averaging the partial sums (Cesàro) or damping the terms by $r^{|n|}$ (Abel) replaces D_N by the Fejér or Poisson kernel, which are good. Hence continuous functions are uniform limits of trigonometric polynomials, and the Poisson integral solves Dirichlet's problem in the disc.

2.5.1 Cesàro means and summation

For $s_n = \sum_{k=0}^{n} c_k$, the N-th **Cesàro mean** is $\sigma_N = \frac{s_0 + \cdots + s_{N-1}}{N}$. $\sum c_k$ is **Cesàro summable** to σ if $\sigma_N \to \sigma$. Convergent series are Cesàro summable to their sum; $1 - 1 + 1 - \ldots$ is Cesàro summable to $\frac{1}{2}$ but diverges.

2.5.2 Fejér's theorem

$\sigma_N(f)(x) = \frac{S_0(f)(x) + \cdots + S_{N-1}(f)(x)}{N} = (f * F_N)(x)$, where $F_N = \frac{D_0 + \cdots + D_{N-1}}{N}$ is the **Fejér kernel**.

Lemma 5.1

$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, and $\{F_N\}$ is a family of good kernels.

Proof. With $\omega = e^{ix}$, $D_n = \frac{\omega^{-n} - \omega^{n+1}}{1-\omega}$, so

$$NF_N = \frac{1}{1-\omega}\Big(\sum_{n=0}^{N-1} \omega^{-n} - \omega \sum_{n=0}^{N-1} \omega^n\Big) = \frac{1}{1-\omega}\Big(\frac{1-\omega^{-N}}{1-\omega^{-1}} - \frac{\omega(1-\omega^N)}{1-\omega}\Big) = \frac{\omega^{-N} - 2 + \omega^N}{(1-\omega)(1-\omega^{-1})},$$

and $\omega^{-N} - 2 + \omega^N = -(2\sin(Nx/2))^2$, $(1-\omega)(1-\omega^{-1}) = 2 - 2\cos x = (2\sin(x/2))^2$ with the signs matching ($NF_N \ge 0$). Good kernel: $F_N \ge 0$; $\frac{1}{2\pi}\int F_N = 1$ since each D_n has integral 1 (average of 1's); for $\delta \le |x| \le \pi$, $\sin^2(x/2) \ge c_\delta > 0$, so $F_N(x) \le \frac{1}{Nc_\delta}$ and $\int_{\delta \le |x| \le \pi} F_N \to 0$. ∎

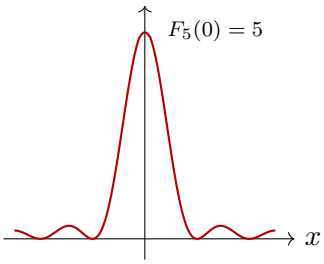


Figure 2: The Fejér kernel F_5: non-negative, integral 1, concentrating at 0.

Theorem 5.2 (Fejér)

If f is integrable on the circle, its Fourier series is Cesàro summable to f at every point of continuity. If f is continuous, the Fourier series is uniformly Cesàro summable to f.

Proof. $\sigma_N(f) = f * F_N$ with $\{F_N\}$ good (Lemma 5.1); apply Theorem 4.1. ■

Corollary 5.3

If f is integrable on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$ at all points of continuity of f.

Proof. All partial sums, hence all Cesàro means, vanish; apply Theorem 5.2. ■

Corollary 5.4

Continuous functions on the circle can be uniformly approximated by trigonometric polynomials.

Proof. $\sigma_N(f)$ is a trigonometric polynomial, and $\sigma_N(f) \to f$ uniformly (Theorem 5.2). ■

2.5.3 Abel means and summation

$\sum c_k$ is **Abel summable** to s if $A(r) = \sum c_k r^k$ converges for $0 \le r < 1$ and $A(r) \to s$ as $r \to 1^-$. Cesàro summable series are Abel summable to the same limit (not conversely: $\sum(-1)^k(k+1)$ is Abel summable to $\frac{1}{4}$ but not Cesàro summable). For $f \sim \sum a_n e^{in\theta}$, the **Abel means** are $A_r(f)(\theta) = \sum r^{|n|} a_n e^{in\theta} = (f * P_r)(\theta)$, by interchanging sum and integral (uniform convergence in the integration variable, $|a_n| \le \sup|f|$).

2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc

Lemma 5.5

If $0 \le r < 1$, then $P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}$, and $\{P_r\}_{0\le r<1}$ is a family of good kernels as $r \to 1^-$.

Proof. With $\omega = re^{i\theta}$, $P_r = \sum_{n\ge 0}\omega^n + \sum_{n\ge 1}\bar{\omega}^n = \frac{1}{1-\omega} + \frac{\bar{\omega}}{1-\bar{\omega}} = \frac{1-|\omega|^2}{|1-\omega|^2} = \frac{1-r^2}{1-2r\cos\theta+r^2}$. Good kernel: $P_r > 0$; $\frac{1}{2\pi}\int P_r = 1$ by term-by-term integration of the uniformly convergent series. For $\frac{1}{2} \le r < 1$ and $\delta \le |\theta| \le \pi$, $1 - 2r\cos\theta + r^2 = (1-r)^2 + 2r(1-\cos\theta) \ge c_\delta > 0$, so $P_r(\theta) \le \frac{1-r^2}{c_\delta} \to 0$. ■

Theorem 5.6

The Fourier series of an integrable function on the circle is Abel summable to f at every point of continuity. If f is continuous, it is uniformly Abel summable to f.

Proof. $A_r(f) = f * P_r$ with $\{P_r\}$ good; apply Theorem 4.1. ∎

Theorem 5.7

Let f be integrable on the unit circle and $u(r,\theta) = (f * P_r)(\theta)$ for $0 \le r < 1$. Then:
(i) u is C^2 in the unit disc and $\triangle u = 0$.
(ii) If θ is a point of continuity of f, then $u(r,\theta) \to f(\theta)$ as $r \to 1$. If f is continuous, $u(r,\theta) \to f(\theta)$ uniformly as $r \to 1$.
(iii) If f is continuous, then u is the unique solution of $\triangle u = 0$ in the disc that satisfies (ii).

Proof.

(i) $u(r,\theta) = \sum a_n r^{|n|} e^{in\theta}$ with $|a_n| \le B$. For $r \le \rho < 1$ the series and the series of its first and second derivatives in r and θ (with factors $|n|^2$) converge uniformly, so u is C^2 (indeed C^∞) and can be differentiated termwise; each term $r^{|n|}e^{in\theta}$ is harmonic by the polar Laplacian of §1.2.2.

(ii) Theorem 5.6.

(iii) [Supplement] (The book leaves this to an exercise.) Let v be another solution and $w = u - v$: w is harmonic in the disc and $w(r,\theta) \to 0$ uniformly as $r \to 1$ (uniform in (ii) for both). Extending w by 0 on the circle gives a continuous function on the closed disc. *Maximum principle:* for $\epsilon > 0$, $w_\epsilon = w + \epsilon(x^2+y^2)$ has $\triangle w_\epsilon = 4\epsilon > 0$, so w_ϵ has no interior maximum (there $\triangle w_\epsilon \le 0$). Hence $\max w_\epsilon$ is attained on the circle, where $w_\epsilon = \epsilon$. Letting $\epsilon \to 0$, $w \le 0$; applying this to $-w$, $w = 0$ (for complex w, take real and imaginary parts). ∎

2.6 Selected exercises

Motivation. Four exercises from the book on summation methods: Cesàro implies Abel, a Tauberian converse, the Abel means at a jump, and why Theorem 5.7(iii) needs uniform convergence.

Exercise 2.13 (Convergent ⇒ Abel, Cesàro ⇒ Abel)

Let $\sum_{k\ge0} c_k$ have partial sums $s_n = \sum_{k=0}^{n} c_k$ and Cesàro means $\sigma_N = \frac{1}{N}(s_0 + \cdots + s_{N-1})$.
(a) If $s_n \to s$, show that $\sum c_k$ is Abel summable to s.
(b) If $\sigma_N \to \sigma$, show that $\sum c_k$ is Abel summable to σ.

Solution.

(a)

Step 1. (s_n) is bounded, say $|s_n| \le M$, so $|c_k| \le 2M$ and $A(r) = \sum c_k r^k$ converges absolutely for $0 \le r < 1$. With $s_{-1} = 0$,

$$\sum_{k=0}^{N} c_k r^k = \sum_{k=0}^{N} (s_k - s_{k-1}) r^k = (1-r) \sum_{k=0}^{N-1} s_k r^k + s_N r^N.$$

Since $s_N r^N \to 0$, letting $N \to \infty$ gives $A(r) = (1-r)\sum_{k\ge0} s_k r^k$.

Step 2. As $(1-r)\sum_{k\ge0} r^k = 1$, we get $A(r) - s = (1-r)\sum_{k\ge0}(s_k - s)r^k$. Given $\epsilon > 0$ choose K with $|s_k - s| < \epsilon$ for $k \ge K$; then

$$|A(r) - s| \le (1-r)\,2MK + \epsilon(1-r)\sum_{k\ge K} r^k \le 2MK(1-r) + \epsilon,$$

so $\limsup_{r\to 1^-}|A(r)-s|\le\epsilon$ for every ϵ.

(b)

Step 1. Let $t_N=N\sigma_N=s_0+\cdots+s_{N-1}$ ($t_0=0$) and $|\sigma_N|\le M$. Then $|t_N|\le MN$, $|s_n|=|t_{n+1}-t_n|\le M(2n+1)$ and $|c_n|\le|s_n|+|s_{n-1}|\le 4M(n+1)$, so $A(r)$ converges absolutely for $r<1$, and $s_Nr^N\to 0$. Hence the identity of (a), Step 1, still gives $A(r)=(1-r)\sum_{k\ge0}s_kr^k$.

Step 2. In the same way, from $s_k=t_{k+1}-t_k$,

$$\sum_{k=0}^{N}s_kr^k=t_{N+1}r^N+(1-r)\sum_{k=1}^{N}t_kr^{k-1},$$

and $t_{N+1}r^N\to0$, so $\sum_k s_kr^k=(1-r)\sum_{k\ge1}k\sigma_kr^{k-1}$ and

$$A(r)=(1-r)^2\sum_{k\ge1}k\sigma_kr^{k-1}.$$

Step 3. Since $(1-r)^2\sum_{k\ge1}kr^{k-1}=1$, $A(r)-\sigma=(1-r)^2\sum_{k\ge1}k(\sigma_k-\sigma)r^{k-1}$. Given ϵ, choose K with $|\sigma_k-\sigma|<\epsilon$ for $k\ge K$:

$$|A(r)-\sigma|\le(1-r)^2\,2MK^2+\epsilon(1-r)^2\sum_{k\ge K}kr^{k-1}\le 2MK^2(1-r)^2+\epsilon,$$

and the conclusion follows as in (a). Together with "convergent $\Rightarrow$ Cesàro" this gives the chain convergent $\Rightarrow$ Cesàro $\Rightarrow$ Abel of §2.5.3. ∎

Exercise 2.14 (A Tauberian theorem)

If $\sum_{k\ge0}c_k$ is Cesàro summable to σ and $kc_k\to0$, prove that $\sum c_k$ converges to σ.

Solution.

Step 1. With the notation of Exercise 2.13,

$$s_{N-1}-\sigma_N=\frac{1}{N}\sum_{j=0}^{N-1}(s_{N-1}-s_j)=\frac{1}{N}\sum_{j=0}^{N-1}\sum_{k=j+1}^{N-1}c_k=\frac{1}{N}\sum_{k=1}^{N-1}kc_k,$$

since c_k occurs once for each $j<k$, i.e. k times.

Step 2. Let $b_k=kc_k\to0$, $|b_k|\le B$. Given $\epsilon>0$ choose K with $|b_k|<\epsilon$ for $k\ge K$. Then $|\frac{1}{N}\sum_{k=1}^{N-1}b_k|\le\frac{KB}{N}+\epsilon<2\epsilon$ for N large.

Step 3. Hence $s_{N-1}=\sigma_N+(s_{N-1}-\sigma_N)\to\sigma+0$, i.e. $\sum c_k=\sigma$. ∎

Exercise 2.17 (Abel means at a jump)

Let f be integrable on the circle with a jump discontinuity at θ: the limits $f(\theta^\pm)=\lim_{h\to0^+}f(\theta\pm h)$ exist. Prove that $A_r(f)(\theta)\to\frac12(f(\theta^+)+f(\theta^-))$ as $r\to1^-$.

Solution.

Step 1. P_r is even, so $\frac{1}{2\pi}\int_{-\pi}^{0}P_r=\frac{1}{2\pi}\int_0^{\pi}P_r=\frac12$ by Lemma 5.5.

Step 2. $A_r(f)(\theta)=(f*P_r)(\theta)=\frac{1}{2\pi}\int_{-\pi}^{\pi}P_r(y)f(\theta-y)\,dy$. For $y>0$, $\theta-y<\theta$, and for $y<0$, $\theta-y>\theta$. By Step 1,

$$A_r(f)(\theta)-\frac{f(\theta^+)+f(\theta^-)}{2}=\frac{1}{2\pi}\int_0^{\pi}P_r(y)[f(\theta-y)-f(\theta^-)]dy+\frac{1}{2\pi}\int_{-\pi}^{0}P_r(y)[f(\theta-y)-f(\theta^+)]dy.$$

Step 3. Let $|f| \le B$ and $\epsilon > 0$; choose δ with $|f(\theta - y) - f(\theta^-)| < \epsilon$ for $0 < y < \delta$ and $|f(\theta - y) - f(\theta^+)| < \epsilon$ for $-\delta < y < 0$. Since $P_r > 0$,

$$\left| A_r(f)(\theta) - \frac{f(\theta^+) + f(\theta^-)}{2} \right| \le \frac{\epsilon}{2\pi} \int_{-\pi}^{\pi} P_r + \frac{2B}{2\pi} \int_{\delta \le |y| \le \pi} P_r(y)\, dy = \epsilon + o(1) \quad (r \to 1^-),$$

by the good-kernel property (iii) of $\{P_r\}$. ∎

Exercise 2.18 (Pointwise boundary values do not determine u)

Let $u(r, \theta) = \partial_\theta P_r(\theta)$ for $0 \le r < 1$. Show that $\triangle u = 0$ in the disc and $u(r, \theta) \to 0$ as $r \to 1$ for each θ, but $u \not\equiv 0$.

Solution.

Step 1 (P is harmonic). With $z = x + iy = re^{i\theta}$,

$$\frac{1+z}{1-z} = \frac{(1+z)(1-\bar z)}{|1-z|^2} = \frac{1 - |z|^2 + (z - \bar z)}{|1-z|^2}, \qquad \text{so} \qquad P_r(\theta) = \operatorname{Re} \frac{1+z}{1-z}.$$

The real part U of a holomorphic $G = U + iV$ on $|z| < 1$ is C^∞ and harmonic: by Cauchy–Riemann $U_x = V_y$, $U_y = -V_x$, so $U_{xx} + U_{yy} = V_{yx} - V_{xy} = 0$.

Step 2 (u is harmonic). By the chain rule $\partial_\theta = x\partial_y - y\partial_x$, so $u = xP_y - yP_x$. Using $\triangle(xg) = x\triangle g + 2g_x$,

$$\triangle u = x\,\triangle P_y + 2P_{xy} - y\,\triangle P_x - 2P_{xy} = x\,\partial_y(\triangle P) - y\,\partial_x(\triangle P) = 0.$$

Step 3 (boundary limits). Differentiating the closed form,

$$u(r, \theta) = -\frac{2r(1 - r^2)\sin\theta}{(1 - 2r\cos\theta + r^2)^2}.$$

If $\theta \in 2\pi\mathbb{Z}$, then $u(r, \theta) = 0$ for all r. Otherwise the denominator tends to $(2 - 2\cos\theta)^2 > 0$ while the numerator tends to 0, so $u(r, \theta) \to 0$.

Step 4. But $u(\tfrac12, \tfrac{\pi}{2}) = -\frac{3/4}{(5/4)^2} = -\frac{12}{25} \neq 0$. The convergence in Step 3 cannot be uniform, since otherwise Theorem 5.7(iii) with $f = 0$ would force $u \equiv 0$. ∎

Chapter 3. Convergence of Fourier Series

3.1 Mean-square convergence of Fourier series

Motivation. Pointwise convergence is delicate, but in the mean-square sense the Fourier series of every integrable function converges. The reason is geometric: $\{e^{in\theta}\}$ is an orthonormal family, $S_N(f)$ is the orthogonal projection onto trigonometric polynomials of degree $\le N$, and trigonometric polynomials are dense (Fejér).

Theorem 1.1 (Mean-square convergence)

If f is integrable on the circle, then

$$\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2\, d\theta \to 0 \quad \text{as } N \to \infty.$$

3.1.1 Vector spaces and inner products

An **inner product** $(\cdot,\cdot)$ on a complex vector space V is linear in the first variable, $(Y,X) = \overline{(X,Y)}$, and $(X,X) \ge 0$; it is **strictly positive definite** if $(X,X) = 0$ only for $X = 0$. $\|X\| = (X,X)^{1/2}$. Basic facts, for any inner product:

- **Pythagoras:** if $X \perp Y$ ($(X,Y) = 0$), then $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$.
- **Cauchy–Schwarz:** $|(X,Y)| \le \|X\|\|Y\|$.
- **Triangle inequality:** $\|X+Y\| \le \|X\| + \|Y\|$.

Proofs [Supplement]. Pythagoras: expand. Cauchy–Schwarz: if $\|Y\| = 0$, then $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\,\mathrm{Re}(X,Y)$ for all real t forces $\mathrm{Re}(X,Y) = 0$, and similarly $\mathrm{Im}(X,Y) = 0$. Otherwise, with $c = (X,Y)/\|Y\|^2$, $X - cY \perp Y$ and $\|X\|^2 = \|X - cY\|^2 + |c|^2\|Y\|^2 \ge |(X,Y)|^2/\|Y\|^2$. Triangle: $\|X+Y\|^2 = \|X\|^2 + 2\,\mathrm{Re}(X,Y) + \|Y\|^2 \le (\|X\| + \|Y\|)^2$.

Examples. (1) $\mathbb{R}^d$, $\mathbb{C}^d$ with the usual inner products.

(2) $\ell^2(\mathbb{Z})$: complex sequences $(a_n)_{n\in\mathbb{Z}}$ with $\sum |a_n|^2 < \infty$ and $(A,B) = \sum a_n \bar{b}_n$ (finite by Cauchy–Schwarz in $\mathbb{C}^d$ and a limit). It is a vector space (by $|a+b|^2 \le 2|a|^2 + 2|b|^2$) and **complete**: Cauchy sequences converge.

(3) $\mathcal{R}$: integrable functions on $[0,2\pi]$ with $(f,g) = \frac{1}{2\pi}\int_0^{2\pi} f\bar{g}$. This inner product is not strictly positive definite ($f = 0$ except at one point has $\|f\| = 0$), and $\mathcal{R}$ is not complete. Completing it leads to Lebesgue's L^2.

3.1.2 Proof of mean-square convergence

Let $e_n(\theta) = e^{in\theta}$; then $\{e_n\}$ is orthonormal in $\mathcal{R}$: $(e_n, e_m) = \frac{1}{2\pi}\int e^{i(n-m)\theta} d\theta = \delta_{nm}$. With $a_n = \hat{f}(n) = (f, e_n)$, $S_N(f) = \sum_{|n|\le N} a_n e_n$.

Key orthogonality. $f - S_N(f) \perp e_n$ for $|n| \le N$, since $(f - S_N(f), e_n) = a_n - a_n = 0$. Hence $f - S_N(f) \perp \sum_{|n|\le N} b_n e_n$ for all b_n, and in particular (Pythagoras)

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{|n|\le N} |a_n|^2. \tag{3.1}$$

Lemma 1.2 (Best approximation)

If f is integrable on the circle with Fourier coefficients a_n, then $\|f - S_N(f)\| \le \|f - \sum_{|n|\le N} c_n e_n\|$ for all complex c_n, with equality iff $c_n = a_n$ for all $|n| \le N$.

Proof. $f - \sum c_n e_n = (f - S_N(f)) + \sum b_n e_n$ with $b_n = a_n - c_n$. The two summands are orthogonal, so $\|f - \sum c_n e_n\|^2 = \|f - S_N(f)\|^2 + \sum |b_n|^2$. ∎

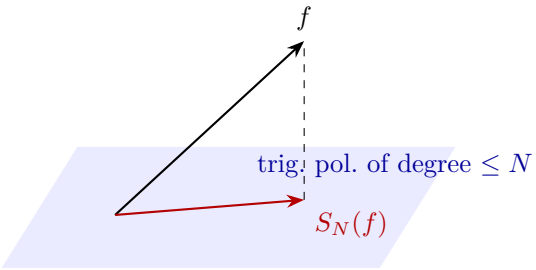


Figure. *$S_N(f)$ is the orthogonal projection of f, hence the best approximation (Lemma 1.2).*

Proof of Theorem 1.1.

Step 1 (continuous f). Given $\epsilon > 0$, Corollary 5.4 of Chapter 2 gives a trigonometric polynomial P of some degree M with $|f - P| < \epsilon$ everywhere, so $\|f - P\| < \epsilon$. By Lemma 1.2, $\|f - S_N(f)\| \le \|f - P\| < \epsilon$ for all $N \ge M$.

Step 2 (integrable f). Let $|f| \le B$. By Lemma 3.2 of Chapter 2 there is a continuous g with $|g| \le B$ and $\int |f - g| < \epsilon^2$, so $\|f - g\|^2 = \frac{1}{2\pi}\int |f-g|^2 \le \frac{2B}{2\pi}\int |f-g| < C\epsilon^2$. Choose a trigonometric polynomial P with $\|g - P\| < \epsilon$ (Step 1). Then $\|f - P\| < C'\epsilon$, and again Lemma 1.2 gives $\|f - S_N(f)\| < C'\epsilon$ for N large. ∎

Theorem 1.3 (Parseval)

Let $f \sim \sum a_n e^{in\theta}$ be integrable on the circle. Then:
(i) Mean-square convergence: $\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \to 0$.
(ii) Parseval's identity: $\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^2 d\theta$.
(iii) The map $f \mapsto (a_n)_{n\in\mathbb{Z}}$ is "unitary": $(f, g) = \sum a_n \bar{b}_n$ for $g \sim \sum b_n e^{in\theta}$ (Lemma 1.5).

Proof. (i) is Theorem 1.1.

(ii) Let $N \to \infty$ in (3.1).

(iii) is Lemma 1.5 below. ∎

Bessel's inequality $\sum |a_n|^2 \le \|f\|^2$ already follows from (3.1).

Theorem 1.4 (Riemann–Lebesgue lemma)

If f is integrable on the circle, then $\hat{f}(n) \to 0$ as $|n| \to \infty$. Equivalently, $\int_0^{2\pi} f(\theta) \sin N\theta \, d\theta \to 0$ and $\int_0^{2\pi} f(\theta) \cos N\theta \, d\theta \to 0$ as $N \to \infty$.

Proof. By Parseval $\sum |\hat{f}(n)|^2 < \infty$, so the terms tend to 0. The reformulation follows from $\cos N\theta = \frac{e^{iN\theta} + e^{-iN\theta}}{2}$, $\sin N\theta = \frac{e^{iN\theta} - e^{-iN\theta}}{2i}$. ∎

Lemma 1.5

If $F \sim \sum a_n e^{in\theta}$ and $G \sim \sum b_n e^{in\theta}$ are integrable on the circle, then $\frac{1}{2\pi}\int_0^{2\pi} F(\theta)\overline{G(\theta)}\,d\theta = \sum_{n=-\infty}^{\infty} a_n \bar{b}_n$, the series converging absolutely.

Proof. [Supplement] Absolute convergence: $|a_n\bar{b}_n| \le \frac{1}{2}(|a_n|^2 + |b_n|^2)$ and Parseval. Polarization: $(F,G) = \frac{1}{4}\sum_{k=0}^{3} i^k \|F + i^k G\|^2$ and the same identity in $\ell^2(\mathbb{Z})$, i.e. $\sum a_n\bar{b}_n = \frac{1}{4}\sum_k i^k \sum_n |a_n + i^k b_n|^2$; the Fourier coefficients of $F + i^k G$ are $a_n + i^k b_n$, and Parseval applies to each term. ∎

3.2 Return to pointwise convergence

Motivation. Differentiability at a point suffices for convergence there, and convergence depends only on the behaviour of f near the point. Mere continuity does not suffice: breaking the symmetry between $e^{in\theta}$ and $e^{-in\theta}$ produces a continuous function whose Fourier series diverges at 0.

3.2.1 A local result

Theorem 2.1

If f is integrable on the circle and differentiable at θ_0, then $S_N(f)(\theta_0) \to f(\theta_0)$.

Proof. Define, for $|t| \le \pi$,

$$F(t) = \begin{cases} \dfrac{f(\theta_0 - t) - f(\theta_0)}{t} & t \neq 0, \\ -f'(\theta_0) & t = 0. \end{cases}$$

F is bounded near 0 (differentiability) and integrable on $[-\pi, \pi]$ (Appendix, Proposition 1.4). Since $\frac{1}{2\pi}\int D_N = 1$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(\theta_0 - t) - f(\theta_0)) D_N(t)\,dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(t)\, t\, D_N(t)\,dt.$$

With $D_N(t) = \frac{\sin((N+1/2)t)}{\sin(t/2)}$ and $\sin((N+\frac{1}{2})t) = \sin Nt \cos\frac{t}{2} + \cos Nt \sin\frac{t}{2}$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int F(t)\frac{t}{\sin(t/2)}\cos\tfrac{t}{2}\,\sin Nt\,dt + \frac{1}{2\pi}\int F(t)\,t\cos Nt\,dt.$$

The functions $F(t)\frac{t\cos(t/2)}{\sin(t/2)}$ and $tF(t)$ are integrable ($t/\sin(t/2)$ is bounded on $[-\pi,\pi]$), so both integrals tend to 0 by the Riemann–Lebesgue lemma (Theorem 1.4). ∎

(The same proof works if f is only Lipschitz at θ_0: $|f(\theta) - f(\theta_0)| \le M|\theta - \theta_0|$.)

Theorem 2.2 (Riemann localization principle)

If f, g are integrable on the circle and $f = g$ on an open interval I containing θ_0, then $S_N(f)(\theta_0) - S_N(g)(\theta_0) \to 0$.

Proof. $f - g$ vanishes in I, so it is differentiable at θ_0 with value 0 there; apply Theorem 2.1 to $f - g$. ∎

3.2.2 A continuous function with diverging Fourier series

Symmetry breaking. For the sawtooth $f(\theta) = i(\pi - \theta)$ on $0 < \theta < \pi$, odd, we have $f \sim \sum_{n \neq 0} \frac{e^{in\theta}}{n}$. The "half" $\sum_{n \leq -1} \frac{e^{in\theta}}{n}$ is not the Fourier series of any integrable (in particular bounded) $\tilde{f}$: otherwise $|A_r(\tilde{f})(0)| = \sum_{n \geq 1} \frac{r^n}{n} \to \infty$ as $r \to 1$, whereas $|A_r(\tilde{f})(0)| \leq \frac{1}{2\pi} \int |\tilde{f}| P_r \leq \sup |\tilde{f}|$. For $N \geq 1$ let

$$f_N(\theta) = \sum_{1 \leq |n| \leq N} \frac{e^{in\theta}}{n}, \qquad \tilde{f}_N(\theta) = \sum_{-N \leq n \leq -1} \frac{e^{in\theta}}{n}.$$

Then (i) $|\tilde{f}_N(0)| = \sum_{n=1}^{N} \frac{1}{n} \geq \log N$ (compare with $\int_1^N dx/x$), and (ii) $f_N(\theta)$ is bounded uniformly in N and θ, by the following lemma.

Lemma 2.3

Suppose the Abel means $A_r = \sum_{n \geq 1} r^n c_n$ of $\sum c_n$ are bounded as $r \to 1^-$, and $c_n = O(1/n)$. Then the partial sums $S_N = \sum_{n=1}^{N} c_n$ are bounded.

Proof. Let $r = 1 - 1/N$ and $n|c_n| \leq M$. Then

$$|S_N - A_r| \leq \sum_{n=1}^{N} |c_n|(1-r^n) + \sum_{n>N} r^n |c_n| \leq M \sum_{n=1}^{N} (1-r) + \frac{M}{N} \sum_{n>N} r^n \leq MN(1-r) + \frac{M}{N} \frac{1}{1-r} = 2M,$$

using $1 - r^n = (1-r)(1 + r + \cdots + r^{n-1}) \leq n(1-r)$. So if also $|A_r| \leq M$, then $|S_N| \leq 3M$. ∎

Apply it to $\sum_{n \geq 1} c_n$ with $c_n = \frac{e^{in\theta}}{n} - \frac{e^{-in\theta}}{n}$, whose partial sums are $f_N(\theta)$: $|c_n| \leq 2/n$, and the Abel means are $(f * P_r)(\theta)$, bounded by $\sup |f|$ since P_r is a good kernel. Hence $|f_N(\theta)| \leq C$ for all N, θ.

Now shift frequencies by $2N$: $P_N(\theta) = e^{i2N\theta} f_N(\theta)$, $\tilde{P}_N(\theta) = e^{i2N\theta} \tilde{f}_N(\theta)$, trigonometric polynomials whose nonzero coefficients sit at $N \leq n \leq 3N$, $n \neq 2N$ (resp. $N \leq n \leq 2N - 1$).

Lemma 2.4

$S_M(P_N) = P_N$ if $M \geq 3N$; $S_M(P_N) = \tilde{P}_N$ if $M = 2N$; $S_M(P_N) = 0$ if $M < N$.

Proof. S_M keeps exactly the frequencies $|n| \leq M$. The frequencies of P_N lie in $[N, 3N]$; those $\leq 2N$ are exactly the frequencies of $\tilde{P}_N$. ∎

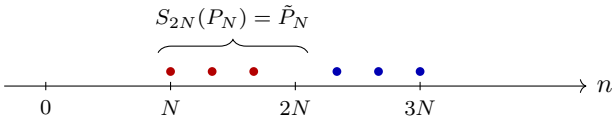


Figure. *Frequencies of P_N (centre $2N$); S_{2N} keeps only the left half and breaks the symmetry.*

The construction. Choose $\alpha_k > 0$ with $\sum \alpha_k < \infty$ and integers N_k with $N_{k+1} > 3N_k$ and $\alpha_k \log N_k \to \infty$; e.g. $\alpha_k = 1/k^2$, $N_k = 3^{2^k}$ (then $\alpha_k \log N_k = \frac{2^k \log 3}{k^2} \to \infty$). Let

$$f(\theta) = \sum_{k=1}^{\infty} \alpha_k P_{N_k}(\theta).$$

Since $|P_N(\theta)| = |f_N(\theta)| \leq C$, the series converges uniformly, so f is continuous, and its Fourier coefficients are obtained termwise. By Lemma 2.4 (the frequency ranges $[N_k, 3N_k]$ are disjoint and $3N_k < N_{k+1}$, so $2N_m < N_k$ for $k > m$ and $2N_m \geq 3N_k$ for $k < m$),

$$S_{2N_m}(f) = \sum_{k<m} \alpha_k P_{N_k} + \alpha_m \tilde{P}_{N_m}.$$

At $\theta = 0$: $|\sum_{k<m} \alpha_k P_{N_k}(0)| \le C \sum \alpha_k$, while $|\tilde{P}_{N_m}(0)| = |\tilde{f}_{N_m}(0)| \ge \log N_m$. So $|S_{2N_m}(f)(0)| \ge \alpha_m \log N_m - C' \to \infty$: *the Fourier series of the continuous function f diverges at 0.*

3.3 Selected exercises

Motivation. Four exercises from the book showing what Parseval's identity and the Riemann–Lebesgue lemma buy: sharp Poincaré-type inequalities, absolute convergence for C^1 and Lipschitz functions, and the Dirichlet integral.

Exercise 3.11 (Wirtinger and Poincaré inequalities)

(a) Let f be T-periodic, continuous and piecewise C^1 with $\int_0^T f = 0$. Show that $\int_0^T |f|^2 \le \frac{T^2}{4\pi^2} \int_0^T |f'|^2$, with equality iff $f(t) = A\sin(2\pi t/T) + B\cos(2\pi t/T)$.
(b) Let f be C^1 on $[a,b]$ with $f(a) = f(b) = 0$. Show that $\int_a^b |f|^2 \le \frac{(b-a)^2}{\pi^2} \int_a^b |f'|^2$, with equality iff $f(t) = A\sin(\pi \frac{t-a}{b-a})$; in particular the constant is optimal.

Solution.

(a)

Step 1 (reduction to $T = 2\pi$). Let $g(s) = f(Ts/2\pi)$. Then $g'(s) = \frac{T}{2\pi} f'(Ts/2\pi)$, $\int_0^{2\pi} |g|^2 = \frac{2\pi}{T} \int_0^T |f|^2$ and $\int_0^{2\pi} |g'|^2 = \frac{T}{2\pi} \int_0^T |f'|^2$. So the claim for f is the claim $\int_0^{2\pi} |g|^2 \le \int_0^{2\pi} |g'|^2$ for g, and the extremals correspond.

Step 2 (coefficients of g'). Let $T = 2\pi$. Integrating by parts on each interval where f is C^1, the boundary terms telescope (continuity) and the end terms cancel (periodicity), so $\hat{f}'(n) = in\hat{f}(n)$; also $\hat{f}(0) = 0$.

Step 3. Parseval for f and for the piecewise continuous f' gives

$$\frac{1}{2\pi}\int_0^{2\pi} |f|^2 = \sum_{n\neq 0} |\hat{f}(n)|^2 \le \sum_{n\neq 0} n^2 |\hat{f}(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f'|^2.$$

Step 4 (equality). Equality holds iff $(n^2-1)|\hat{f}(n)|^2 = 0$ for all n, i.e. $\hat{f}(n) = 0$ for $|n| \ge 2$. Then f and $\hat{f}(-1)e^{-it} + \hat{f}(1)e^{it}$ are continuous with the same coefficients, so they coincide (Corollary 2.2); this is $A\sin t + B\cos t$. Conversely such f give equality.

(b)

Step 1. Translate so that $a = 0$, and let $L = b$. Extend f to $[-L, L]$ oddly, $\tilde{f}(t) = -f(-t)$ for $t < 0$, and then $2L$-periodically. Since $f(0) = f(L) = 0$, $\tilde{f}$ is continuous, and it is piecewise C^1; being odd, $\int_{-L}^{L} \tilde{f} = 0$.

Step 2. By (a) with $T = 2L$, $\int_{-L}^{L} |\tilde{f}|^2 \le \frac{L^2}{\pi^2} \int_{-L}^{L} |\tilde{f}'|^2$. Both integrals are twice the corresponding integrals over $[0, L]$ (as $|\tilde{f}(-t)| = |f(t)|$, $|\tilde{f}'(-t)| = |f'(t)|$), which gives the inequality.

Step 3. Equality forces $\tilde{f} = A\sin(\pi t/L) + B\cos(\pi t/L)$ by (a); oddness gives $B = 0$. Conversely $f(t) = \sin(\pi t/L)$ gives equality, so L^2/π^2 cannot be improved. ∎

Exercise 3.12 (The Dirichlet integral)

Prove that $\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}$ (as an improper Riemann integral).

Solution.

Step 1 (existence). For $R>1$, integrating by parts, $\int_1^R \frac{\sin x}{x}\,dx = \cos 1 - \frac{\cos R}{R} - \int_1^R \frac{\cos x}{x^2}\,dx$, which converges as $R\to\infty$ since $|\cos x|/x^2 \le x^{-2}$. The integrand is continuous on $(0,1]$ and tends to 1 at 0. Hence $I = \lim_{R\to\infty}\int_0^R \frac{\sin x}{x}\,dx$ exists, and it suffices to compute the limit along $R_N = (N+\frac12)\pi$.

Step 2 (splitting D_N). $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N = 1$ and $D_N(\theta) = \sin((N+\frac12)\theta)/\sin(\theta/2)$. Let $h(\theta) = \frac{1}{\sin(\theta/2)} - \frac{2}{\theta}$ for $0<|\theta|\le\pi$, $h(0)=0$. Since

$$h(\theta) = \frac{\theta - 2\sin(\theta/2)}{\theta\sin(\theta/2)} = \frac{\theta^3/24 + O(\theta^5)}{\theta^2/2 + O(\theta^4)} \to 0,$$

h is continuous on $[-\pi,\pi]$.

Step 3 (Riemann–Lebesgue). $\sin((N+\frac12)\theta) = \sin N\theta\cos\frac{\theta}{2} + \cos N\theta\sin\frac{\theta}{2}$, and $h\cos\frac{\theta}{2}$, $h\sin\frac{\theta}{2}$ are integrable, so Theorem 1.4 gives $\int_{-\pi}^{\pi} h(\theta)\sin((N+\frac12)\theta)\,d\theta \to 0$. Therefore

$$\int_{-\pi}^{\pi} \frac{2\sin((N+\frac12)\theta)}{\theta}\,d\theta = 2\pi - \int_{-\pi}^{\pi} h(\theta)\sin\big((N+\tfrac12)\theta\big)d\theta \to 2\pi.$$

Step 4. The integrand is even, so $\int_0^{\pi}\frac{\sin((N+\frac12)\theta)}{\theta}d\theta \to \frac{\pi}{2}$. Substituting $x = (N+\frac12)\theta$ this is $\int_0^{R_N}\frac{\sin x}{x}\,dx$, so $I = \frac{\pi}{2}$. ∎

Exercise 3.14 (C^1 functions have absolutely convergent Fourier series)

Prove that the Fourier series of a C^1 function f on the circle converges absolutely, hence uniformly to f.

Solution.

Step 1. As in Exercise 3.11, $\hat{f'}(n) = in\hat f(n)$, and Parseval for f' gives $\sum_n n^2|\hat f(n)|^2 = \frac{1}{2\pi}\int|f'|^2 < \infty$.

Step 2. By the Cauchy–Schwarz inequality, for every N,

$$\sum_{0<|n|\le N}|\hat f(n)| = \sum_{0<|n|\le N}|n\hat f(n)|\cdot\frac{1}{|n|} \le \Big(\sum_n n^2|\hat f(n)|^2\Big)^{1/2}\Big(\sum_{n\neq0}\frac{1}{n^2}\Big)^{1/2}.$$

The right side is finite and independent of N, so $\sum|\hat f(n)|<\infty$.

Step 3. By Corollary 2.3, $S_N(f)\to f$ uniformly. This improves Corollary 2.4, which assumed $f\in C^2$. ∎

Exercise 3.16 (Lipschitz functions: Bernstein's argument)

Let f be 2π-periodic with $|f(x)-f(y)|\le K|x-y|$. Prove that $\sum_n|\hat f(n)|<\infty$, so the Fourier series of f converges absolutely and uniformly to f.

Solution.

Step 1. For $h>0$ let $g_h(x) = f(x+h)-f(x-h)$. Then $\hat g_h(n) = (e^{inh}-e^{-inh})\hat f(n) = 2i\sin(nh)\hat f(n)$. By Parseval and $|g_h|\le 2Kh$,

$$\sum_n 4\sin^2(nh)|\hat f(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi}|g_h|^2 \le 4K^2h^2, \qquad\text{so}\qquad \sum_n \sin^2(nh)|\hat f(n)|^2 \le K^2h^2.$$

Step 2. Fix $p\ge 0$ and take $h = \pi/2^{p+1}$. For $2^{p-1}<|n|\le 2^p$ we have $\frac{\pi}{4}<|n|h\le\frac{\pi}{2}$, hence $\sin^2(nh)\ge\frac12$, and Step 1 gives

$$\sum_{2^{p-1}<|n|\le 2^p}|\hat f(n)|^2 \le 2K^2h^2 = \frac{K^2\pi^2}{2^{2p+1}}.$$

Step 3. The block contains at most 2^{p+1} integers, so by Cauchy–Schwarz

$$\sum_{2^{p-1}<|n|\le 2^p} |\hat f(n)| \le (2^{p+1})^{1/2}\Big(\frac{K^2\pi^2}{2^{2p+1}}\Big)^{1/2} = \frac{K\pi}{2^{p/2}}.$$

Step 4. The blocks $p = 0, 1, 2, \dots$ cover all $n \neq 0$, and $\sum_p 2^{-p/2} < \infty$, so $\sum_n |\hat f(n)| < \infty$. Since f is continuous, Corollary 2.3 gives uniform convergence. (The same argument works for Hölder exponent $\alpha > \tfrac12$.) ∎

Chapter 4. Some Applications of Fourier Series

4.1 The isoperimetric inequality

Motivation. Among closed curves of given length the circle encloses the largest area. Writing the curve's coordinates as Fourier series turns length and area into sums of squares of coefficients, and Parseval's identity reduces the inequality to $|n| \le n^2$.

Curves. A **parametrized curve** is a map $\gamma : [a,b] \to \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$; it is **simple** if injective on $[a,b)$, **closed** if $\gamma(a) = \gamma(b)$, and of class C^1 (with closed curves $(b-a)$-periodic). The **curve** Γ is the image. Its **length** is $\ell = \int_a^b |\gamma'(s)|\, ds$, independent of C^1 reparametrizations. Every curve with $\gamma' \neq 0$ can be parametrized by **arc-length**, $|\gamma'(s)| = 1$, and then $\ell = b - a$. For a simple closed C^1 curve enclosing the region Ω, Green's theorem gives the **area**

$$\mathcal{A} = \frac{1}{2}\Big|\int_\Gamma (x\,dy - y\,dx)\Big| = \frac{1}{2}\Big|\int_a^b (x(s)y'(s) - y(s)x'(s))ds\Big|,$$

which we take as the definition of the enclosed area.

Theorem 1.1 (Isoperimetric inequality)

If Γ is a simple closed C^1 curve in $\mathbb{R}^2$ of length ℓ enclosing area $\mathcal{A}$, then $\mathcal{A} \le \frac{\ell^2}{4\pi}$, with equality iff Γ is a circle.

Proof.

Step 1 (normalization). Rescaling $(x,y) \mapsto \delta(x,y)$ multiplies ℓ by δ and $\mathcal{A}$ by δ^2, so we may take $\ell = 2\pi$ and show $\mathcal{A} \le \pi$. Parametrize by arc-length on $[0, 2\pi]$, so $x'(s)^2 + y'(s)^2 = 1$ and

$$\frac{1}{2\pi}\int_0^{2\pi} (|x'(s)|^2 + |y'(s)|^2)ds = 1.$$

Step 2 (Fourier coefficients). Let $x(s) \sim \sum a_n e^{ins}$, $y(s) \sim \sum b_n e^{ins}$. Then $x'(s) \sim \sum a_n in e^{ins}$, $y' \sim \sum b_n in e^{ins}$ (integration by parts, periodic C^1). Parseval gives

$$\sum_n |n|^2(|a_n|^2 + |b_n|^2) = 1.$$

Since x, y are real, $\overline{a_n} = a_{-n}$, $\overline{b_n} = b_{-n}$. By Lemma 1.5 of Chapter 3,

$$\mathcal{A} = \frac{1}{2}\Big|\int_0^{2\pi} (xy' - yx')\, ds\Big| = \pi\Big|\sum_n n(a_n\overline{b_n} - b_n\overline{a_n})\Big|.$$

Step 3 (the inequality). $|a_n\overline{b_n} - b_n\overline{a_n}| \le 2|a_n||b_n| \le |a_n|^2 + |b_n|^2$ and $|n| \le |n|^2$, so

$$\mathcal{A} \le \pi\sum_n |n|(|a_n|^2 + |b_n|^2) \le \pi\sum_n |n|^2(|a_n|^2 + |b_n|^2) = \pi.$$

Step 4 (equality). If $\mathcal{A} = \pi$, then $|n| = |n|^2$ wherever $|a_n|^2 + |b_n|^2 \neq 0$, so $a_n = b_n = 0$ for $|n| \ge 2$, and $x(s) = a_{-1}e^{-is} + a_0 + a_1 e^{is}$, $y(s) = b_{-1}e^{-is} + b_0 + b_1 e^{is}$. With $\overline{a_1} = a_{-1}$, $\overline{b_1} = b_{-1}$, the identity $2(|a_1|^2 + |b_1|^2) = 1$ gives $|a_1|^2 + |b_1|^2 = \frac{1}{2}$, and equality in $2|a_1\overline{b_1} - b_1\overline{a_1}| \le \dots$ requires $2|a_1||b_1| = |a_1|^2 + |b_1|^2$, i.e. $|a_1| = |b_1| = \frac{1}{2}$, and $|a_1\overline{b_1} - \overline{a_1}b_1| = 2|a_1||b_1|$. Write $a_1 = \frac{1}{2}e^{i\alpha}$, $b_1 = \frac{1}{2}e^{i\beta}$; then $|e^{i(\alpha-\beta)} - e^{-i(\alpha-\beta)}| = 2$, so $|\sin(\alpha-\beta)| = 1$, $\alpha - \beta = k\pi/2$ with k odd. Hence $x(s) = a_0 + \cos(\alpha + s)$, $y(s) = b_0 \pm \sin(\alpha + s)$: a circle of radius 1. Conversely the circle gives equality. ∎

4.2 Weyl's equidistribution theorem

Motivation. For irrational γ the fractional parts $\langle n\gamma\rangle$ fill $[0,1)$ uniformly. Testing against exponentials, the average of $e^{2\pi ikn\gamma}$ over $n \le N$ is a geometric sum that tends to 0; trigonometric polynomials approximate continuous and then Riemann integrable functions.

For $x \in \mathbb{R}$, $[x]$ is the integer part and $\langle x\rangle = x - [x] \in [0,1)$ the **fractional part**; $x = y \bmod \mathbb{Z}$ if $x - y \in \mathbb{Z}$. For γ rational, $\langle n\gamma\rangle$ takes finitely many values; for γ irrational the $\langle n\gamma\rangle$ are distinct (if $\langle n_1\gamma\rangle = \langle n_2\gamma\rangle$, then $(n_1 - n_2)\gamma \in \mathbb{Z}$).

Definition (Equidistribution)

A sequence $\xi_1, \xi_2, \dots$ in $[0,1)$ is **equidistributed** if for every interval $(a,b) \subseteq [0,1)$,

$$\lim_{N\to\infty} \frac{\#\{1 \le n \le N : \xi_n \in (a,b)\}}{N} = b - a.$$

Theorem 2.1 (Weyl)

If γ is irrational, then $\langle\gamma\rangle, \langle 2\gamma\rangle, \langle 3\gamma\rangle, \dots$ is equidistributed in $[0,1)$. In particular $\{\langle n\gamma\rangle\}$ is dense in $[0,1)$ (**Kronecker's theorem**).

Let $\chi_{(a,b)}$ be the 1-periodic extension of the characteristic function of $(a,b) \subseteq [0,1)$. Then $\#\{n \le N : \langle n\gamma\rangle \in (a,b)\} = \sum_{n=1}^{N} \chi_{(a,b)}(n\gamma)$, and the theorem says

$$\frac{1}{N}\sum_{n=1}^{N} \chi_{(a,b)}(n\gamma) \to \int_0^1 \chi_{(a,b)}(x)\,dx. \tag{4.1}$$

Lemma 2.2

If f is continuous, 1-periodic, and γ is irrational, then $\frac{1}{N}\sum_{n=1}^{N} f(n\gamma) \to \int_0^1 f(x)\,dx$.

Proof.

Step 1 (exponentials). For $f = 1$ both sides are 1. For $f(x) = e^{2\pi ikx}$, $k \neq 0$: $\int_0^1 f = 0$, and with $e^{2\pi ik\gamma} \neq 1$ (γ irrational),

$$\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma} = \frac{e^{2\pi ik\gamma}}{N}\frac{1 - e^{2\pi ikN\gamma}}{1 - e^{2\pi ik\gamma}}, \qquad \left|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma}\right| \le \frac{2}{N|1 - e^{2\pi ik\gamma}|} \to 0.$$

By linearity the lemma holds for trigonometric polynomials.

Step 2 (approximation). Given $\epsilon > 0$, Corollary 5.4 of Chapter 2 gives a trigonometric polynomial P with $|f - P| < \epsilon/3$. Then $\left|\frac{1}{N}\sum f(n\gamma) - \frac{1}{N}\sum P(n\gamma)\right| < \epsilon/3$, $\left|\int f - \int P\right| < \epsilon/3$, and $\left|\frac{1}{N}\sum P(n\gamma) - \int P\right| < \epsilon/3$ for N large. ∎

Corollary 2.3

The conclusion of Lemma 2.2 holds for every 1-periodic f that is Riemann integrable on $[0,1]$.

Proof. Let f be real. Given $\epsilon > 0$, choose a partition with upper and lower step functions $f_U \ge f \ge f_L$ and $\int(f_U - f_L) < \epsilon$. Modifying near the partition points (as in Lemma 3.2 of Chapter 2, keeping $f_\pm$ above/below), we get continuous 1-periodic $f_+ \ge f \ge f_-$ with $\int(f_+ - f_-) < 2\epsilon$. By Lemma 2.2,

$$\limsup \frac{1}{N}\sum f(n\gamma) \le \lim \frac{1}{N}\sum f_+(n\gamma) = \int f_+ \le \int f + 2\epsilon,$$

and similarly $\liminf \geq \int f - 2\epsilon$. ∎

Proof of Theorem 2.1. Apply Corollary 2.3 to $f = \chi_{(a,b)}$, which gives (4.1). ∎

Weyl's criterion (one direction proved above, the other an exercise in the book): $\{\xi_n\} \subseteq [0,1)$ is equidistributed iff $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$. For example $\langle n^2\gamma \rangle$ is equidistributed for irrational γ. *Geometric interpretation:* a light ray reflected inside a square mirror box traces a closed periodic path if the slope is rational and a dense (indeed equidistributed) path if it is irrational.

4.3 A continuous but nowhere differentiable function

Motivation. Lacunary Fourier series $\sum 2^{-n\alpha} e^{i2^n x}$ converge uniformly but oscillate too wildly to be differentiable anywhere. Delayed Cesàro means isolate a single lacunary term, whose derivative is huge, whereas differentiability would force all such derivatives to be $O(\log N)$.

Theorem 3.1 (Weierstrass)

For $0 < \alpha < 1$, $f_\alpha(x) = f(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i2^n x}$ is continuous but nowhere differentiable.

Continuity is clear: the series converges absolutely and uniformly since $\sum 2^{-n\alpha} < \infty$. For nowhere differentiability we use the **delayed means**

$$\triangle_N(f) = 2\sigma_{2N}(f) - \sigma_N(f).$$

Facts. (a) Since $\sigma_M(f) = \sum_{|n|<M} (1 - \frac{|n|}{M}) a_n e^{inx}$, the coefficient of $a_n e^{inx}$ in $\triangle_N(f)$ is $2(1 - \frac{|n|}{2N}) - (1 - \frac{|n|}{N}) = 1$ for $|n| \leq N$ and $2 - \frac{|n|}{N}$ for $N < |n| < 2N$:

$$\triangle_N(f) = S_N(f) + \sum_{N<|n|<2N} \Big(2 - \frac{|n|}{N}\Big) a_n e^{inx}.$$

(b) For lacunary f (frequencies 2^k only) and $N = 2^{n-1}$, no frequency lies strictly between N and $2N$, so $\triangle_N(f) = S_N(f)$.

Lemma 3.2

If g is continuous and differentiable at x_0, then $\sigma_N(g)'(x_0) = O(\log N)$, hence $\triangle_N(g)'(x_0) = O(\log N)$.

Proof.

Step 1. $\sigma_N(g)'(x_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N'(t) g(x_0 - t)\, dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N'(t)(g(x_0 - t) - g(x_0))dt$, since $\int F_N' = 0$ (F_N periodic). Differentiability at x_0 and boundedness of g give $|g(x_0 - t) - g(x_0)| \leq C|t|$.

Step 2 (bounds for F_N'). $F_N = \sum_{|n|<N}(1 - \frac{|n|}{N}) e^{int}$ gives $|F_N'(t)| \leq \sum_{|n|<N} |n| \leq N^2$. And from $F_N(t) = \frac{\sin^2(Nt/2)}{N \sin^2(t/2)}$, [Supplement] differentiating,

$$|F_N'(t)| \leq \frac{|\sin(Nt/2)\cos(Nt/2)|}{\sin^2(t/2)} + \frac{\sin^2(Nt/2)|\cos(t/2)|}{N|\sin(t/2)|^3} \leq \frac{A}{t^2} + \frac{A}{N|t|^3} \leq \frac{2A}{t^2} \quad (|t| \geq 1/N).$$

Step 3. Hence $|\sigma_N(g)'(x_0)| \leq C\int_{|t|\leq 1/N} N^2|t|\, dt + C\int_{1/N\leq|t|\leq\pi} \frac{2A}{|t|}\, dt = O(1) + O(\log N)$. Finally $\triangle_N(g)' = 2\sigma_{2N}(g)' - \sigma_N(g)'$. ∎

Lemma 3.3

If $2N = 2^n$, then $\triangle_{2N}(f) - \triangle_N(f) = 2^{-n\alpha} e^{i2^n x}$.

Proof. By Fact (b), $\triangle_{2N}(f) = S_{2N}(f)$ and $\triangle_N(f) = S_N(f)$, and the only frequency in $(N, 2N]$ is $2N = 2^n$. ∎

Proof of Theorem 3.1. Suppose f is differentiable at x_0. By Lemma 3.3, $\triangle_{2N}(f)'(x_0) - \triangle_N(f)'(x_0) = i2^{n(1-\alpha)} e^{i2^n x_0}$, of modulus $2^{n(1-\alpha)} = (2N)^{1-\alpha}$. By Lemma 3.2 it is $O(\log N)$. But $N^{1-\alpha}/\log N \to \infty$ since $\alpha < 1$: a contradiction. ∎

4.4 The heat equation on the circle

Motivation. On a ring the temperature evolves by $\partial_t u = \partial_x^2 u$. Separating variables gives $u = \sum a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = f * H_t$: high frequencies are damped extremely fast, and the heat kernel is a good kernel.

Model the ring by the circle with angle $\theta = 2\pi x$, $x \in [0, 1)$. After rescaling time the temperature satisfies $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with $u(x, 0) = f(x)$. Separating $u = A(x)B(t)$ gives $\frac{B'}{B} = \frac{A''}{A} = \lambda$, and periodicity forces $\lambda = -4\pi^2 n^2$, $A = e^{\pm 2\pi i n x}$, $B = e^{-4\pi^2 n^2 t}$. Superposing,

$$u(x, t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}, \qquad a_n = \hat{f}(n).$$

For integrable f the a_n are bounded, the series converges with all derivatives for $t > 0$, and u solves the equation. Moreover $u(\cdot, t) \to f$ in mean square as $t \to 0$ (Parseval). In convolution form $u(x, t) = (f * H_t)(x)$ with the **heat kernel** of the circle

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x}.$$

$\{H_t\}_{t>0}$ is a family of good kernels as $t \to 0$ (Chapter 5, Corollary 3.4), so $u(x, t) \to f(x)$ at every point of continuity, uniformly if f is continuous. Unlike the Poisson kernel, H_t is not given by an elementary closed formula; its positivity and concentration follow from Poisson summation (Chapter 5).

4.5 Selected exercises

Motivation. Four exercises from the book: the converse half of Weyl's criterion and a uniform version of it, a natural sequence that fails to be equidistributed, and the mean-square boundary behaviour of the heat equation stated above.

Exercise 4.5 (Powers of the golden ratio)

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\gamma_n = \langle \varphi^n \rangle$. Prove that $\{\gamma_n\}$ is not equidistributed in $[0, 1)$.

Solution.

Step 1. Let $\psi = \frac{1-\sqrt{5}}{2}$. Both φ and ψ are roots of $x^2 = x + 1$, so $U_n = \varphi^n + \psi^n$ satisfies $U_{n+1} = U_n + U_{n-1}$, with $U_0 = 2$, $U_1 = 1$. By induction every U_n is an integer.

Step 2. $\psi \approx -0.618$, so $0 < |\psi| < 1$ and $\varphi^n = U_n - \psi^n$. If n is odd, $-\psi^n = |\psi|^n \in (0, 1)$, so $\gamma_n = |\psi|^n$. If n is even, $-\psi^n \in (-1, 0)$, so $\gamma_n = 1 - |\psi|^n$.

Step 3. Choose n_0 with $|\psi|^{n_0} < \frac{1}{4}$. For $n \geq n_0$, $\gamma_n \in [0, \frac{1}{4}) \cup (\frac{3}{4}, 1)$. Hence

$$\frac{\#\{1 \leq n \leq N : \gamma_n \in (\frac{1}{4}, \frac{3}{4})\}}{N} \leq \frac{n_0}{N} \to 0 \neq \frac{1}{2},$$

so the sequence is not equidistributed. ∎

Exercise 4.7 (Weyl's criterion, converse direction)

Let $\{\xi_n\} \subseteq [0, 1)$ be equidistributed. Prove that $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$.

Solution. Write $\#_N(E) = \#\{1 \leq n \leq N : \xi_n \in E\}$.

Step 1 (single points are negligible). Let $c \in [0, 1)$. If $c = 0$: $\#_N(\{0\}) \leq N - \#_N((0, 1))$, and $\#_N((0, 1))/N \to 1$. If $c > 0$: for small $\delta > 0$, $\#_N(\{c\}) \leq \#_N((c - \delta, c + \delta))$, whose ratio to N tends to 2δ. In both cases $\#_N(\{c\})/N \to 0$.

Step 2 (continuous f). Let f be real and continuous on $[0, 1]$, $|f| \leq B$, and $\epsilon > 0$. By uniform continuity choose $0 = x_0 < x_1 < \cdots < x_M = 1$ with $|f(x) - f(x_j)| < \epsilon$ on $[x_{j-1}, x_j]$. Every ξ_n lies in some (x_{j-1}, x_j) or equals some x_j with $j < M$, so

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) - \sum_{j=1}^{M} f(x_j)\frac{\#_N((x_{j-1}, x_j))}{N}\Big| \leq \epsilon + B\sum_{j=0}^{M-1}\frac{\#_N(\{x_j\})}{N}.$$

Step 3. As $N \to \infty$ the second sum tends to $\sum_j f(x_j)(x_j - x_{j-1})$, a Riemann sum within ϵ of $\int_0^1 f$, and the error term tends to ϵ by Step 1. So $\limsup_N \left|\frac{1}{N}\sum f(\xi_n) - \int_0^1 f\right| \leq 2\epsilon$ for every ϵ, i.e. $\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f$.

Step 4. Apply Step 3 to $\cos(2\pi k x)$ and $\sin(2\pi k x)$, whose integrals over $[0, 1]$ vanish for $k \neq 0$. ∎

Exercise 4.10 (Uniform averages along an equidistributed sequence)

Let $\{\xi_n\}$ be equidistributed in $[0, 1)$ and f continuous and 1-periodic with $\int_0^1 f = 0$. Prove that $\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n) \to 0$ uniformly in x.

Solution.

Step 1 (exponentials). For $k \neq 0$,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k(x+\xi_n)}\Big| = \Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n}\Big| \to 0$$

by Exercise 4.7, and the left side does not depend on x. By linearity the claim holds uniformly in x for every trigonometric polynomial Q with zero constant term.

Step 2 (approximation). Let $\epsilon > 0$. By Corollary 5.4 (rescaled to period 1) there is a trigonometric polynomial P with $\sup|f - P| < \epsilon$. Its constant term $c_0 = \int_0^1 P$ satisfies $|c_0| = |\int_0^1 (P - f)| < \epsilon$, so $Q = P - c_0$ has zero constant term and $\sup|f - Q| < 2\epsilon$.

Step 3. For all x,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n)\Big| \leq 2\epsilon + \Big|\frac{1}{N}\sum_{n=1}^{N} Q(x + \xi_n)\Big| < 3\epsilon$$

once $N \geq N_0$, with N_0 independent of x by Step 1. ∎

Exercise 4.11 (Mean-square initial values for the heat equation)

Let f be integrable on $[0,1]$ (1-periodic) and $u(x,t) = (f * H_t)(x)$. Prove that $\int_0^1 |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0^+$.

Solution.

Step 1. For $t > 0$, $u(\cdot,t) = \sum_n a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}$ converges uniformly ($|a_n| \le \sup|f|$), so its Fourier coefficients are $a_n e^{-4\pi^2 n^2 t}$ and those of $u(\cdot,t) - f$ are $a_n(e^{-4\pi^2 n^2 t} - 1)$.

Step 2. Parseval (Theorem 1.3, rescaled to period 1) for the integrable function $u(\cdot,t) - f$ gives

$$\int_0^1 |u(x,t) - f(x)|^2 dx = \sum_n |a_n|^2 (1 - e^{-4\pi^2 n^2 t})^2.$$

Step 3. Each factor $(1 - e^{-4\pi^2 n^2 t})^2$ lies in $[0,1]$ and tends to 0 as $t \to 0$. Given $\epsilon > 0$, choose M with $\sum_{|n|>M} |a_n|^2 < \epsilon$ (possible since $\sum |a_n|^2 = \int_0^1 |f|^2 < \infty$). Then the sum is at most $\sum_{|n|\le M} |a_n|^2 (1 - e^{-4\pi^2 M^2 t})^2 + \epsilon$, which is $< 2\epsilon$ for t small. ∎

Chapter 5. The Fourier Transform on $\mathbb{R}$

5.1 Elementary theory of the Fourier transform

Motivation. On the real line Fourier series become integrals: $\hat{f}(\xi) = \int f(x)e^{-2\pi ix\xi}dx$. On the Schwartz space $\mathcal{S}(\mathbb{R})$ of smooth rapidly decreasing functions the Fourier transform interchanges differentiation and multiplication by x, fixes the Gaussian, and is a bijection that preserves L^2 norms.

5.1.1 Integration of functions on the real line

f is of **moderate decrease** if it is continuous and $|f(x)| \le \frac{A}{1+x^2}$ for some A. For such f, $\int_{-N}^{N} f$ is Cauchy as $N \to \infty$, and we set $\int_{-\infty}^{\infty} f = \lim_{N\to\infty} \int_{-N}^{N} f$. These functions form a vector space $\mathcal{M}(\mathbb{R})$.

Proposition 1.1

For f, g of moderate decrease:
(i) Linearity: $\int(af + bg) = a\int f + b\int g$.
(ii) Translation invariance: $\int f(x-h)\,dx = \int f(x)\,dx$ for $h \in \mathbb{R}$.
(iii) Scaling: $\delta \int f(\delta x)\,dx = \int f(x)\,dx$ for $\delta > 0$.
(iv) Continuity: $\int |f(x-h) - f(x)|\,dx \to 0$ as $h \to 0$.

Proof. (i) Clear from finite intervals.

(ii) $\int_{-N}^{N} f(x-h)dx = \int_{-N-h}^{N-h} f$, and $\int_{-N-h}^{-N} f$, $\int_{N-h}^{N} f \to 0$ by the decay.

(iii) $\delta \int_{-N}^{N} f(\delta x)dx = \int_{-\delta N}^{\delta N} f$.

(iv) Given ϵ, choose N with $\int_{|x|\ge N} |f(x)|$, $\int_{|x|\ge N} |f(x-h)| < \epsilon$ for $|h| \le 1$; on $[-N, N]$, f is uniformly continuous, so $\int_{-N}^{N} |f(x-h) - f(x)| < \epsilon$ for small h. ∎

5.1.2 Definition of the Fourier transform

For $f \in \mathcal{M}(\mathbb{R})$, $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}\,dx$; $\hat{f}$ is bounded and continuous. (If $\hat{f}$ is also of moderate decrease, inversion and Plancherel hold, §5.1.7.)

5.1.3 The Schwartz space

$\mathcal{S}(\mathbb{R})$ consists of the C^∞ functions f with $\sup_x |x|^k |f^{(\ell)}(x)| < \infty$ for all $k, \ell \ge 0$ (**rapidly decreasing**). It is a vector space closed under differentiation and multiplication by polynomials. The **Gaussian** e^{-ax^2} $(a > 0)$ belongs to $\mathcal{S}(\mathbb{R})$: its derivatives are $P(x)e^{-ax^2}$ with polynomials P. Smooth compactly supported functions also belong to it; the **bump function** $e^{-1/(1-x^2)}$ on $|x| < 1$ (0 elsewhere) is an example.

5.1.4 The Fourier transform on $\mathcal{S}$

Proposition 1.2

If $f \in \mathcal{S}(\mathbb{R})$, writing $f(x) \to \hat{f}(\xi)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi ih\xi}$ for $h \in \mathbb{R}$.
(ii) $f(x)e^{-2\pi ixh} \to \hat{f}(\xi+h)$ for $h \in \mathbb{R}$.
(iii) $f(\delta x) \to \delta^{-1}\hat{f}(\delta^{-1}\xi)$ for $\delta > 0$.
(iv) $f'(x) \to 2\pi i\xi\hat{f}(\xi)$.
(v) $-2\pi ixf(x) \to \frac{d}{d\xi}\hat{f}(\xi)$.

Proof. (i)–(iii) by Proposition 1.1(ii), (iii).

(iv) Integrate by parts on $[-N, N]$: $\int_{-N}^{N} f'(x)e^{-2\pi ix\xi}dx = [f(x)e^{-2\pi ix\xi}]_{-N}^{N} + 2\pi i\xi \int_{-N}^{N} f(x)e^{-2\pi ix\xi}dx$, and the boundary term $\to 0$.

(v) Let $g(x) = -2\pi ixf(x) \in \mathcal{S}$. For $\epsilon > 0$,

$$\frac{\hat{f}(\xi+h) - \hat{f}(\xi)}{h} - \hat{g}(\xi) = \int f(x)e^{-2\pi ix\xi}\Big[\frac{e^{-2\pi ixh} - 1}{h} + 2\pi ix\Big]dx.$$

Choose N with $\int_{|x|\geq N} |f|, \int_{|x|\geq N} |x||f| < \epsilon$. For $|x| \leq N$, $|\frac{e^{-2\pi ixh}-1}{h} + 2\pi ix| \leq C|h|$ uniformly (Taylor). Also $|\frac{e^{-2\pi ixh}-1}{h}| \leq 2\pi|x|$. So the difference is $\leq C'\epsilon + C''N|h| < C\epsilon$ for $|h|$ small. ∎

Theorem 1.3

If $f \in \mathcal{S}(\mathbb{R})$, then $\hat{f} \in \mathcal{S}(\mathbb{R})$.

Proof. $\hat{f}$ is bounded. By Proposition 1.2(iv), (v), $\xi^k(\frac{d}{d\xi})^\ell \hat{f}(\xi)$ is the Fourier transform of $\frac{1}{(2\pi i)^k}(\frac{d}{dx})^k[(-2\pi ix)^\ell f(x)] \in \mathcal{S}$, hence bounded. And $\hat{f}$ is C^∞ by repeated (v). ∎

Theorem 1.4

If $f(x) = e^{-\pi x^2}$, then $\hat{f}(\xi) = f(\xi)$.

Proof. $F(\xi) = \hat{f}(\xi)$ has $F(0) = \int e^{-\pi x^2}dx = 1$. By Proposition 1.2(v) and $f'(x) = -2\pi xf(x)$, then (iv),

$$F'(\xi) = \int f(x)(-2\pi ix)e^{-2\pi ix\xi}dx = i\int f'(x)e^{-2\pi ix\xi}dx = i(2\pi i\xi)F(\xi) = -2\pi\xi F(\xi).$$

So $G(\xi) = F(\xi)e^{\pi\xi^2}$ has $G' = 0$ and $G = G(0) = 1$. ∎

($\int e^{-\pi x^2}dx = 1$: square it and use polar coordinates, $\int_0^{2\pi}\int_0^\infty e^{-\pi r^2} r\,dr\,d\theta = 1$.)

Corollary 1.5

If $\delta > 0$ and $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}$.

Proof. Proposition 1.2(iii) with δ replaced by $\delta^{-1/2}$ and Theorem 1.4. ∎

As $\delta \to 0$, K_δ concentrates at 0 while $\widehat{K_\delta}$ flattens out: the uncertainty principle in miniature (§5.4).

Definition (Good kernels on $\mathbb{R}$)

$\{K_\delta\}_{\delta>0}$ is a family of **good kernels** as $\delta \to 0$ if (i) $\int K_\delta = 1$;
(ii) $\int |K_\delta| \le M$;
(iii) for every $\eta > 0$, $\int_{|x|>\eta} |K_\delta| \to 0$.

Theorem 1.6

$\{K_\delta\}_{\delta>0}$, $K_\delta(x) = \delta^{-1/2} e^{-\pi x^2/\delta}$, is a family of good kernels as $\delta \to 0$.

Proof. $K_\delta > 0$ and $\int K_\delta = \widehat{K_\delta}(0) = 1$. Substituting $x = \delta^{1/2} y$, $\int_{|x|>\eta} K_\delta = \int_{|y|>\eta/\sqrt{\delta}} e^{-\pi y^2} dy \to 0$. ■

For $f, g \in \mathcal{S}(\mathbb{R})$, $(f * g)(x) = \int f(x-t) g(t)\, dt$.

Corollary 1.7

If $f \in \mathcal{S}(\mathbb{R})$, then $(f * K_\delta)(x) \to f(x)$ uniformly in x as $\delta \to 0$.

Proof. f is uniformly continuous: given ϵ, take R with $|f| < \epsilon/4$ for $|x| \ge R$, and use uniform continuity on $[-R-1, R+1]$. Choose η with $|f(x-t) - f(x)| < \epsilon$ for $|t| < \eta$. Then

$$|(f * K_\delta)(x) - f(x)| \le \int_{|t|<\eta} K_\delta(t) |f(x-t) - f(x)| dt + \int_{|t|\ge\eta} K_\delta(t)\, 2 \sup |f|\, dt \le \epsilon + C \int_{|t|\ge\eta} K_\delta < 2\epsilon$$

for δ small, uniformly in x. ■

5.1.5 The Fourier inversion

Proposition 1.8 (Multiplication formula)

If $f, g \in \mathcal{S}(\mathbb{R})$, then $\int f(x) \hat{g}(x)\, dx = \int \hat{f}(y) g(y)\, dy$.

Proof. $F(x,y) = f(x) g(y) e^{-2\pi i x y}$ has $|F| \le \frac{C}{(1+x^2)(1+y^2)}$, so (Appendix, Theorem 3.1) the iterated integrals may be interchanged: $\int \Big(\int F\, dy \Big) dx = \int f(x) \hat{g}(x) dx$ and $\int \Big(\int F\, dx \Big) dy = \int \hat{f}(y) g(y) dy$. ■

Theorem 1.9 (Fourier inversion)

If $f \in \mathcal{S}(\mathbb{R})$, then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi$.

Proof.

Step 1 ($x = 0$). Let $G_\delta(x) = e^{-\pi \delta x^2}$, so $\widehat{G_\delta} = K_\delta$ (Corollary 1.5). By Proposition 1.8, $\int f(x) K_\delta(x) dx = \int \hat{f}(\xi) G_\delta(\xi) d\xi$. As $\delta \to 0$, the left side tends to $f(0)$ (Corollary 1.7, K_δ even), and the right side to $\int \hat{f}$ (dominated by $|\hat{f}|$, and $G_\delta \to 1$ uniformly on compacts). So $f(0) = \int \hat{f}(\xi)\, d\xi$.

Step 2 (general x). Apply Step 1 to $F(y) = f(y + x)$, whose transform is $\hat{f}(\xi) e^{2\pi i x \xi}$ (Proposition 1.2(i)). ■

Write $\mathcal{F}(f) = \hat{f}$ and $\mathcal{F}^*(g)(x) = \int g(\xi) e^{2\pi i x \xi} d\xi$; then $\mathcal{F}^* \mathcal{F} = I$ on $\mathcal{S}$, and $\mathcal{F}^*(g)(x) = \mathcal{F}(g)(-x)$.

Corollary 1.10

The Fourier transform is a bijection of $\mathcal{S}(\mathbb{R})$ onto itself.

Proof. $\mathcal{F}$ maps $\mathcal{S}$ to $\mathcal{S}$ (Theorem 1.3), $\mathcal{F}^* \circ \mathcal{F} = I$ (Theorem 1.9), and similarly $\mathcal{F} \circ \mathcal{F}^* = I$ (since $\mathcal{F}^*(g) = \mathcal{F}(g)(-\cdot)$). ■

5.1.6 The Plancherel formula

Proposition 1.11

If $f, g \in \mathcal{S}(\mathbb{R})$:
(i) $f * g \in \mathcal{S}(\mathbb{R})$;
(ii) $f * g = g * f$;
(iii) $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Proof.

(ii) Substitute $t \mapsto x - t$.

(i) For $\ell \geq 0$: $\sup |x|^\ell |g(x-y)| \leq A_\ell (1+|y|)^\ell$ since $|x|^\ell \leq 2^\ell(|x-y|^\ell + |y|^\ell)$. So $\sup_x |x^\ell (f*g)(x)| \leq A_\ell \int |f(y)|(1+|y|)^\ell dy < \infty$. Derivatives: $\frac{d}{dx}(f*g) = f * g'$ (differentiate under the integral), and repeat.

(iii) $F(x,y) = f(y)g(x-y)e^{-2\pi i x\xi}$ is integrable on $\mathbb{R}^2$ (rapid decay), so by Fubini (Theorem 3.1), $\int (f*g)(x)e^{-2\pi i x\xi}dx = \int f(y)e^{-2\pi i y\xi}\Big(\int g(x-y)e^{-2\pi i(x-y)\xi}dx\Big)dy = \hat{f}(\xi)\hat{g}(\xi)$. ■

Theorem 1.12 (Plancherel)

If $f \in \mathcal{S}(\mathbb{R})$, then $\|\hat{f}\| = \|f\|$, where $\|f\| = \left(\int |f|^2\right)^{1/2}$.

Proof. Let $f^\flat(x) = \overline{f(-x)}$; then $\widehat{f^\flat}(\xi) = \overline{\hat{f}(\xi)}$. With $h = f * f^\flat$, $\hat{h} = \hat{f}\overline{\hat{f}} = |\hat{f}|^2$ (Proposition 1.11) and $h(0) = \int |f|^2$. By Fourier inversion at 0, $h(0) = \int \hat{h}(\xi)d\xi = \int |\hat{f}|^2$. ■

5.1.7 Extension to functions of moderate decrease

For f, g of moderate decrease, $f * g$ is of moderate decrease and $\widehat{f*g} = \hat{f}\hat{g}$; the multiplication formula still holds. Hence Fourier inversion and Plancherel hold whenever f and $\hat{f}$ are both of moderate decrease.

5.1.8 The Weierstrass approximation theorem

Theorem 1.13 (Weierstrass)

If f is continuous on $[a,b]$ and $\epsilon > 0$, there is a polynomial P with $\sup_{[a,b]} |f - P| < \epsilon$.

Proof. Choose $[-M, M] \supset [a,b]$ in its interior, and extend f to a continuous g on $\mathbb{R}$, linear from $f(b)$ to 0 on $[b, M]$, from $f(a)$ to 0 on $[-M, a]$, and 0 outside $[-M, M]$; $|g| \leq B$. g is uniformly continuous, so as in Corollary 1.7, $g * K_\delta \to g$ uniformly; fix δ with $|g * K_\delta - g| < \epsilon/2$. Expand $e^{-\pi x^2/\delta} = \sum_n \frac{(-\pi x^2/\delta)^n}{n!}$, uniformly convergent on $|x| \leq 2M$, and truncate: a polynomial R with

$|K_\delta(x) - R(x)| < \frac{\epsilon}{4MB}$ for $|x| \le 2M$. Then $P(x) = \int_{-M}^{M} g(t)R(x-t)\,dt$ is a polynomial in x, and for $|x| \le M$ (so $|x-t| \le 2M$),

$$|(g * K_\delta)(x) - P(x)| = \Big| \int_{-M}^{M} g(t)(K_\delta(x-t) - R(x-t))dt \Big| \le 2MB\frac{\epsilon}{4MB} = \frac{\epsilon}{2}.$$

■

5.2 Applications to some partial differential equations

Motivation. The Fourier transform turns ∂_x^2 into multiplication by $-4\pi^2\xi^2$, so the heat equation on $\mathbb{R}$ is solved by convolving with a Gaussian, and Laplace's equation in the half-plane by convolving with the Poisson kernel. Uniqueness needs some control at infinity.

5.2.1 The time-dependent heat equation on the real line

Seek $u(x,t)$ with $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ $(t>0)$, $u(x,0) = f(x)$. Transforming in x: $\partial_t \hat{u}(\xi,t) = -4\pi^2\xi^2\hat{u}(\xi,t)$, so $\hat{u}(\xi,t) = \hat{f}(\xi)e^{-4\pi^2\xi^2 t}$, and $u = f * \mathcal{H}_t$ with the **heat kernel**

$$\mathcal{H}_t(x) = K_\delta(x) \text{ with } \delta = 4\pi t, \qquad \mathcal{H}_t(x) = \frac{1}{(4\pi t)^{1/2}}e^{-x^2/4t}, \qquad \widehat{\mathcal{H}_t}(\xi) = e^{-4\pi^2 t\xi^2}.$$

Theorem 2.1

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,t) = (f * \mathcal{H}_t)(x)$ for $t > 0$. Then:
(i) u is C^2 for $x \in \mathbb{R}$, $t > 0$, and solves the heat equation.
(ii) $u(x,t) \to f(x)$ uniformly in x as $t \to 0$; hence, setting $u(x,0) = f(x)$, u is continuous on $\{t \ge 0\}$.
(iii) $\int |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0$.

Proof.

(i) $u(x,t) = \int \hat{f}(\xi)e^{-4\pi^2\xi^2 t}e^{2\pi i x\xi}d\xi$ (inversion, Proposition 1.11). Differentiating under the integral (justified by the rapid decay of $\hat{f}$) gives $\partial_t u = \int \hat{f}(-4\pi^2\xi^2)e^{\cdots} = \partial_x^2 u$.

(ii) Corollary 1.7 with $\delta = 4\pi t$.

(iii) By Plancherel, $\int |u-f|^2 = \int |\hat{f}(\xi)|^2 |e^{-4\pi^2\xi^2 t} - 1|^2 d\xi$. Split at $|\xi| \le N$ (where $|e^{-4\pi^2\xi^2 t} - 1| \le 4\pi^2 N^2 t$) and $|\xi| > N$ (where it is ≤ 1, and $\int_{|\xi|>N} |\hat{f}|^2 < \epsilon$). ■

Corollary 2.2

$u(\cdot,t) \in \mathcal{S}(\mathbb{R})$ uniformly in t: for every $T > 0$ and $k, \ell \ge 0$,

$$\sup_{x\in\mathbb{R},\ 0<t<T} |x|^k |\partial_x^\ell u(x,t)| < \infty. \tag{5.1}$$

Proof. $\partial_x^\ell u = (\partial_x^\ell f) * \mathcal{H}_t$, so it suffices to treat $\ell = 0$; let $|x| \ge 1$. Split $\int f(x-y)\mathcal{H}_t(y)dy$ into $|y| \le |x|/2$ and $|y| \ge |x|/2$. On the first part $|x-y| \ge |x|/2$, so $|f(x-y)| \le C_k|x|^{-k}$, and $\int \mathcal{H}_t = 1$. On the second part $\mathcal{H}_t(y) \le (4\pi t)^{-1/2}e^{-x^2/16t}$, and [Supplement] $\sup_{0<t<T} t^{-1/2}e^{-x^2/16t} \le C_{k,T}|x|^{-k}$ (put $s = x^2/t \ge x^2/T$: $t^{-1/2}e^{-x^2/16t} = |x|^{-1}s^{1/2}e^{-s/16} \le C|x|^{-1}e^{-s/32} \le C|x|^{-1}e^{-x^2/32T}$); together with $\int |f| < \infty$ this gives $|u(x,t)| \le C|x|^{-k}$. ■

Theorem 2.3 (Uniqueness)

Suppose (i) u is continuous on $\{t \geq 0\}$;
(ii) u satisfies the heat equation for $t > 0$;
(iii) $u(x,0) = 0$;
(iv) $u(\cdot,t) \in \mathcal{S}(\mathbb{R})$ uniformly in t as in (5.1). Then $u = 0$.

Proof. Let $E(t) = \int |u(x,t)|^2 dx \geq 0$, $E(0) = 0$. Differentiating under the integral and integrating by parts (the decay kills boundary terms),

$$\frac{dE}{dt} = \int (\partial_t u\, \bar{u} + u\, \overline{\partial_t u}) = \int (\partial_x^2 u\, \bar{u} + u\, \overline{\partial_x^2 u}) = -2 \int |\partial_x u|^2 \leq 0.$$

So E is decreasing, $E \equiv 0$ (by continuity at $t = 0$), and $u = 0$. ∎

($\mathcal{H}_t(x)$ itself, or u with very fast growth, shows the role of (iv): without decay conditions uniqueness fails.)

5.2.2 The steady-state heat equation in the upper half-plane

Find u on $\mathbb{R}^2_+ = \{(x,y) : y > 0\}$ with $\triangle u = 0$ and $u(x,0) = f(x)$. Transforming in x: $\partial_y^2 \hat{u} = 4\pi^2 \xi^2 \hat{u}$, so $\hat{u} = A(\xi) e^{2\pi|\xi|y} + B(\xi) e^{-2\pi|\xi|y}$; boundedness gives $\hat{u}(\xi, y) = \hat{f}(\xi) e^{-2\pi|\xi|y}$, and $u = f * \mathcal{P}_y$ with the **Poisson kernel** $\mathcal{P}_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$.

Lemma 2.4

$\int_{-\infty}^{\infty} e^{-2\pi|\xi|y} e^{2\pi i \xi x} d\xi = \mathcal{P}_y(x)$ and $\int_{-\infty}^{\infty} \mathcal{P}_y(x) e^{-2\pi i x \xi} dx = e^{-2\pi|\xi|y}$.

Proof. Split the first integral at 0: $\int_0^\infty e^{-2\pi\xi(y-ix)} d\xi + \int_0^\infty e^{-2\pi\xi(y+ix)} d\xi = \frac{1}{2\pi(y-ix)} + \frac{1}{2\pi(y+ix)} = \frac{1}{\pi} \frac{y}{x^2+y^2}$. Both $\mathcal{P}_y$ and $e^{-2\pi|\xi|y}$ are of moderate decrease, so inversion (§5.1.7) gives the second identity. ∎

Lemma 2.5

$\{\mathcal{P}_y\}$ is a family of good kernels on $\mathbb{R}$ as $y \to 0$.

Proof. $\mathcal{P}_y > 0$; $\int \mathcal{P}_y = \widehat{\mathcal{P}_y}(0) = 1$ (or $\frac{1}{\pi} \int \frac{dx}{1+x^2} = 1$ after $x \mapsto yx$); $\int_{|x|\geq\delta} \mathcal{P}_y = \frac{1}{\pi} \int_{|x|\geq\delta/y} \frac{dx}{1+x^2} \to 0$. ∎

Theorem 2.6

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,y) = (f * \mathcal{P}_y)(x)$. Then:
(i) u is C^2 in $\mathbb{R}^2_+$ and $\triangle u = 0$;
(ii) $u(x,y) \to f(x)$ uniformly as $y \to 0$;
(iii) $\int |u(x,y) - f(x)|^2 dx \to 0$ as $y \to 0$;
(iv) with $u(x,0) = f(x)$, u is continuous on $\overline{\mathbb{R}^2_+}$ and vanishes at infinity: $u(x,y) \to 0$ as $|x| + y \to \infty$.

Proof.

(i)–(iii) As for Theorem 2.1: $u = \int \hat{f}(\xi) e^{-2\pi|\xi|y} e^{2\pi i x \xi} d\xi$ and $(\partial_x^2 + \partial_y^2) e^{-2\pi|\xi|y} e^{2\pi i x \xi} = 0$; Lemma 2.5; Plancherel.

(iv) Splitting the integral at $|t| \leq |x|/2$ and $|t| \geq |x|/2$: $|u(x,y)| \leq C(\frac{1}{1+x^2} + \frac{y}{x^2+y^2})$. Also $|u| \leq \sup \mathcal{P}_y \int |f| \leq C/y$. Use the first bound for $|x| \geq y$ and the second for $|x| \leq y$. ∎

Theorem 2.7

If u is continuous on $\overline{\mathbb{R}^2_+}$, $\triangle u = 0$ in $\mathbb{R}^2_+$, $u(x,0) = 0$, and u vanishes at infinity, then $u = 0$.

Lemma 2.8 (Mean-value property)

Let u be C^2 with $\triangle u = 0$ in an open $\Omega \subseteq \mathbb{R}^2$. If the closed disc of radius R centred at (x,y) lies in Ω, then $u(x,y) = \frac{1}{2\pi}\int_0^{2\pi} u(x + r\cos\theta, y + r\sin\theta)\, d\theta$ for $0 \le r \le R$.

Proof. Let $U(r,\theta) = u(x + r\cos\theta, y + r\sin\theta)$; in polar coordinates $0 = \partial_\theta^2 U + r\partial_r(r\partial_r U)$. With $F(r) = \frac{1}{2\pi}\int_0^{2\pi} U(r,\theta)d\theta$, $r\partial_r(r\partial_r F) = -\frac{1}{2\pi}\int \partial_\theta^2 U\, d\theta = 0$ (periodicity). So $rF'(r)$ is constant, equal to its value 0 at $r = 0$; F is constant, $F(r) = F(0) = u(x,y)$. ∎

Proof of Theorem 2.7. (The example $u = y$ shows the decay is needed.) Take u real (real and imaginary parts) and suppose $u(x_0,y_0) > 0$. Choose R so large that $u \le \frac{1}{2}u(x_0,y_0)$ outside the half-disc $D_R^+ = \{x^2 + y^2 \le R^2,\ y \ge 0\}$. u attains its maximum M on D_R^+ at some (x_1,y_1), and $u \le M$ on all of $\overline{\mathbb{R}^2_+}$; $y_1 > 0$ since $u = 0$ on $y = 0$ and $M > 0$. By the mean-value property on discs of radius $r < y_1$, $M = u(x_1,y_1) = \frac{1}{2\pi}\int u(x_1 + r\cos\theta, y_1 + r\sin\theta)d\theta \le M$, so $u = M$ on all such circles (continuity), hence on the disc of radius y_1 around (x_1,y_1). Its boundary touches the real axis, where $u = 0$: contradiction with $M > 0$. Similarly $u \ge 0$. ∎

5.3 The Poisson summation formula

Motivation. Periodizing a function on $\mathbb{R}$ gives a function on the circle whose Fourier coefficients are samples of $\hat{f}$ at the integers. This links Fourier analysis on $\mathbb{R}$ and on the circle: it gives the functional equation of the theta function and identifies the heat and Poisson kernels of the circle as periodizations.

For $f \in \mathcal{S}(\mathbb{R})$ the **periodization** $F_1(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ is 1-periodic and continuous (uniform convergence on compacts); another periodic function attached to f is $F_2(x) = \sum_n \hat{f}(n)e^{2\pi inx}$.

Theorem 3.1 (Poisson summation formula)

If $f \in \mathcal{S}(\mathbb{R})$, then $\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}$. In particular $\sum_n f(n) = \sum_n \hat{f}(n)$.

Proof. Both sides are continuous and 1-periodic, and the right side converges absolutely. The m-th Fourier coefficient of the left side is

$$\int_0^1 \sum_n f(x+n)e^{-2\pi imx}dx = \sum_n \int_n^{n+1} f(y)e^{-2\pi imy}dy = \int_{-\infty}^{\infty} f(y)e^{-2\pi imy}dy = \hat{f}(m),$$

interchanging sum and integral by uniform convergence. That of the right side is also $\hat{f}(m)$. By uniqueness of Fourier coefficients for continuous functions (Corollary 2.2 of Chapter 2, after rescaling to period 1), the two sides agree. ∎

(The formula also holds when f and $\hat{f}$ are of moderate decrease, with the same proof.)

5.3.1 Theta and zeta functions

The **theta function** is $\vartheta(s) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 s}$ for $s > 0$.

Theorem 3.2

$s^{-1/2}\vartheta(1/s) = \vartheta(s)$ for $s > 0$.

Proof. Apply Poisson summation (at $x = 0$) to $f(x) = e^{-\pi s x^2}$, whose transform is $\hat{f}(\xi) = s^{-1/2}e^{-\pi\xi^2/s}$ (Corollary 1.5 with $\delta = 1/s$). ∎

This functional equation of ϑ yields, via $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty t^{s/2-1}\frac{\vartheta(t)-1}{2}dt$, the functional equation of the Riemann zeta function $\zeta(s) = \sum n^{-s}$ (Book II).

5.3.2 Heat kernels

Theorem 3.3

The heat kernel of the circle is the periodization of the heat kernel of the line: $H_t(x) = \sum_{n\in\mathbb{Z}} \mathcal{H}_t(x+n)$.

Proof. Poisson summation for $f = \mathcal{H}_t$, $\hat{f}(\xi) = e^{-4\pi^2 t\xi^2}$: $\sum_n \mathcal{H}_t(x+n) = \sum_n e^{-4\pi^2 tn^2}e^{2\pi inx} = H_t(x)$. ∎

Corollary 3.4

H_t is a good kernel as $t \to 0$ (on the circle $\mathbb{R}/\mathbb{Z}$, with integrals over $|x| \le \frac{1}{2}$).

Proof. $H_t > 0$ by Theorem 3.3, and $\int_{|x|\le 1/2} H_t = \hat{H}_t(0) = 1$. For $|x| \le \frac{1}{2}$, $H_t(x) = \mathcal{H}_t(x) + E_t(x)$ with $E_t(x) = \sum_{n\neq 0}\mathcal{H}_t(x+n) \le Ct^{-1/2}\sum_{n\ge 1} e^{-cn^2/t} \to 0$ uniformly as $t \to 0$ ($|x+n| \ge |n|/2$). For $\delta \le |x| \le \frac{1}{2}$, $\mathcal{H}_t(x) \le (4\pi t)^{-1/2}e^{-\delta^2/4t} \to 0$. So $\int_{\delta\le|x|\le 1/2} H_t \to 0$. ∎

5.3.3 Poisson kernels

Theorem 3.5

$P_r(2\pi x) = \sum_{n\in\mathbb{Z}} \mathcal{P}_y(x+n)$ with $r = e^{-2\pi y}$.

Proof. Poisson summation for $f = \mathcal{P}_y$, $\hat{f}(\xi) = e^{-2\pi|\xi|y}$ (both of moderate decrease): $\sum_n \mathcal{P}_y(x+n) = \sum_n e^{-2\pi|n|y}e^{2\pi inx} = \sum r^{|n|}e^{in(2\pi x)} = P_r(2\pi x)$. ∎

5.4 The Heisenberg uncertainty principle

Motivation. A function and its Fourier transform cannot both be sharply localized: the product of their spreads is at least $1/16\pi^2$, with equality only for Gaussians. In quantum mechanics this is the uncertainty relation between position and momentum.

Theorem 4.1 (Heisenberg)

If $\psi \in \mathcal{S}(\mathbb{R})$ with $\int |\psi|^2 = 1$, then

$$\Big(\int x^2|\psi(x)|^2dx\Big)\Big(\int \xi^2|\hat{\psi}(\xi)|^2d\xi\Big) \ge \frac{1}{16\pi^2},$$

with equality iff $\psi(x) = Ae^{-Bx^2}$ with $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$. More generally $\big(\int (x - x_0)^2|\psi|^2\big)\big(\int (\xi - \xi_0)^2|\hat{\psi}|^2\big) \geq \frac{1}{16\pi^2}$ for all $x_0, \xi_0 \in \mathbb{R}$.

Proof.

Step 1. Integrating by parts (ψ, ψ' rapidly decreasing),

$$1 = \int |\psi|^2 = -\int x \frac{d}{dx}|\psi|^2 = -\int (x\psi'\bar{\psi} + x\psi\overline{\psi'}),$$

so by Cauchy–Schwarz $1 \leq 2\int |x||\psi||\psi'| \leq 2\big(\int x^2|\psi|^2\big)^{1/2}\big(\int |\psi'|^2\big)^{1/2}$.

Step 2. By Proposition 1.2(iv) and Plancherel, $\int |\psi'|^2 = 4\pi^2 \int \xi^2|\hat{\psi}|^2$. Squaring gives the inequality.

Step 3 (equality). Equality in Cauchy–Schwarz forces $\psi'(x) = \beta x\psi(x)$ (and the real-part step to be equality), so $\psi = Ae^{\beta x^2/2}$; $\psi \in \mathcal{S}$ forces $\beta = -2B < 0$, and $\int |\psi|^2 = |A|^2\sqrt{\pi/2B} = 1$.

Step 4. The general form follows by applying the inequality to $e^{-2\pi i x\xi_0}\psi(x + x_0)$ and changing variables. ∎

Quantum mechanics. For a state function ψ with $\int |\psi|^2 = 1$, the probability that the particle lies in (a, b) is $\int_a^b |\psi|^2$; the expected position is $\bar{x} = \int x|\psi|^2$ and the variance $(\Delta x)^2 = \int (x - \bar{x})^2|\psi|^2$. The momentum is governed in the same way by $\hat{\psi}$ (frequency ξ corresponds to momentum $h\xi$, h Planck's constant), and Theorem 4.1 becomes $(\Delta x)(\Delta\xi) \geq \frac{1}{4\pi}$, i.e. $(\Delta\,\text{position})(\Delta\,\text{momentum}) \geq \frac{h}{4\pi}$.

Chapter 6. The Fourier Transform on $\mathbb{R}^d$

6.1 Preliminaries

Motivation. In higher dimensions rotations enter alongside translations and dilations, and integration in polar coordinates splits $\mathbb{R}^d$ into radii and directions. These are the tools for radial functions, spherical means and the Radon transform.

6.1.1 Symmetries

Analysis on $\mathbb{R}^d$ is shaped by **translations** $x \mapsto x+h$, **dilations** $x \mapsto \delta x$ ($\delta > 0$) and **rotations**: linear R with $R(x)\cdot R(y) = x\cdot y$, equivalently $|Rx| = |x|$, or $R^t = R^{-1}$; then $\det R = \pm 1$ (**proper** if $+1$). On $\mathbb{R}$ the rotations are $\pm I$; on $\mathbb{R}^2$ they are $z \mapsto e^{i\varphi}z$ and $z \mapsto e^{i\varphi}\bar{z}$ (identifying $\mathbb{R}^2 = \mathbb{C}$). For $d = 3$, Euler angles parametrize the proper rotations.

6.1.2 Integration on $\mathbb{R}^d$

A continuous f is **rapidly decreasing** if $\sup_x |x|^k |f(x)| < \infty$ for all k; then $\int_{\mathbb{R}^d} f = \lim_{N\to\infty} \int_{Q_N} f$ over the cubes $Q_N = \{|x_i| \le N/2\}$ (or balls). It suffices that $\sup |x|^{d+\epsilon}|f(x)| < \infty$; **moderate decrease** means $\epsilon = 1$. For such f:

$$\int f(x+h)\,dx = \int f(x)\,dx, \qquad \delta^d \int f(\delta x)\,dx = \int f(x)\,dx, \qquad \int f(Rx)\,dx = \int f(x)\,dx.$$

Polar coordinates. Writing $x = r\gamma$, $r \ge 0$, $\gamma \in S^{d-1}$, with the surface measure $d\sigma$ on the sphere,

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

For $d = 2$: $\int f = \int_0^{2\pi}\int_0^\infty f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$. For $d = 3$: $x = (r\sin\theta\cos\varphi, r\sin\theta\sin\varphi, r\cos\theta)$, $d\sigma = \sin\theta\, d\theta\, d\varphi$. The area of S^{d-1} is $\omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ (computed from $\int e^{-\pi|x|^2}dx = 1$).

6.2 Elementary theory of the Fourier transform

Motivation. Everything from Chapter 5 extends to $\mathbb{R}^d$ with $x\xi$ replaced by $x\cdot\xi$. The new feature is the interaction with rotations: the Fourier transform commutes with them, so radial functions have radial transforms.

Multi-index notation: $x^\alpha = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$, $(\frac{\partial}{\partial x})^\alpha$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$. $\mathcal{S}(\mathbb{R}^d)$ consists of smooth f with $\sup_x |x^\alpha (\frac{\partial}{\partial x})^\beta f(x)| < \infty$ for all α, β. The Fourier transform is $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi} dx$.

Proposition 2.1

For $f \in \mathcal{S}(\mathbb{R}^d)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi i\xi\cdot h}$;
(ii) $f(x)e^{-2\pi i x\cdot h} \to \hat{f}(\xi+h)$;
(iii) $f(\delta x) \to \delta^{-d}\hat{f}(\delta^{-1}\xi)$, $\delta > 0$;
(iv) $(\frac{\partial}{\partial x})^\alpha f(x) \to (2\pi i\xi)^\alpha \hat{f}(\xi)$;
(v) $(-2\pi i x)^\alpha f(x) \to (\frac{\partial}{\partial \xi})^\alpha \hat{f}(\xi)$;

(vi) $f(Rx) \to \hat{f}(R\xi)$ for every rotation R.

Proof. (i)–(v) as in Proposition 1.2 of Chapter 5 (one variable at a time).

(vi) Substitute $y = Rx$: $\int f(Rx)e^{-2\pi ix\cdot\xi}dx = \int f(y)e^{-2\pi iR^{-1}y\cdot\xi}dy = \int f(y)e^{-2\pi iy\cdot R\xi}dy = \hat{f}(R\xi)$, since $R^{-1}y\cdot\xi = y\cdot R\xi$ ($R^{-1} = R^t$). ■

Corollary 2.2

The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ to itself.

Proof. By (iv), (v), $\xi^\alpha(\partial_\xi)^\beta\hat{f}$ is the transform of a Schwartz function, hence bounded. ■

Corollary 2.3

The Fourier transform of a radial function ($f(Rx) = f(x)$ for all rotations R) is radial.

Proof. By (vi), $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R. ■

The Gaussian $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$ is radial and its own Fourier transform (product of the one-dimensional transforms); hence $K_\delta(x) = \delta^{-d/2}e^{-\pi|x|^2/\delta}$ has $\widehat{K_\delta}(\xi) = e^{-\pi\delta|\xi|^2}$ and is a good kernel as $\delta \to 0$.

Theorem 2.4 (Fourier inversion and Plancherel)

If $f \in \mathcal{S}(\mathbb{R}^d)$, then $f(x) = \int_{\mathbb{R}^d}\hat{f}(\xi)e^{2\pi ix\cdot\xi}d\xi$ and $\int|\hat{f}(\xi)|^2d\xi = \int|f(x)|^2dx$. In particular $\mathcal{F}$ is a bijection of $\mathcal{S}(\mathbb{R}^d)$.

Proof. Exactly as in Chapter 5: the multiplication formula $\int f\hat{g} = \int\hat{f}g$ (Fubini), applied with $g = e^{-\pi\delta|x|^2}$, gives inversion at 0 via the good kernels K_δ; translate for general x. For Plancherel, use $h = f * f^\flat$, $f^\flat(x) = \overline{f(-x)}$, with $\widehat{f*g} = \hat{f}\hat{g}$. ■

6.3 The wave equation in $\mathbb{R}^d \times \mathbb{R}$

Motivation. The Fourier transform solves the wave equation explicitly and shows that energy is conserved. In three dimensions the solution is an average over spheres, which yields Huygens' principle: signals propagate sharply. Two dimensions are handled by viewing them as three-dimensional data independent of x_3 ("descent").

Cauchy problem. Find $u(x,t)$ with $\triangle u = \frac{\partial^2 u}{\partial t^2}$ (speed $c = 1$), $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$, $f, g \in \mathcal{S}(\mathbb{R}^d)$.

6.3.1 Solution in terms of Fourier transforms

Transforming in x: $\partial_t^2\hat{u}(\xi,t) = -4\pi^2|\xi|^2\hat{u}(\xi,t)$, so $\hat{u} = \hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$.

Theorem 3.1

A solution of the Cauchy problem for the wave equation is

$$u(x,t) = \int_{\mathbb{R}^d} \Big[\hat f(\xi)\cos(2\pi|\xi|t) + \hat g(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}\Big] e^{2\pi i x\cdot\xi} d\xi. \tag{6.1}$$

Proof. $\cos(2\pi|\xi|t)$ and $\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$ are smooth in ξ (even power series in $|\xi|$) and, with all their t-derivatives, bounded by polynomials in $|\xi|$ for bounded t. Since $\hat f, \hat g$ are rapidly decreasing, we may differentiate under the integral twice in x and t: each term $[\dots]e^{2\pi i x\cdot\xi}$ satisfies $\triangle_x = -4\pi^2|\xi|^2 = \partial_t^2$. At $t=0$, Fourier inversion gives $u = f$, $\partial_t u = g$. ∎

The **energy** of u is $E(t) = \int_{\mathbb{R}^d} \Big(\big|\frac{\partial u}{\partial t}\big|^2 + \sum_j \big|\frac{\partial u}{\partial x_j}\big|^2\Big) dx$.

Theorem 3.2

If u is given by (6.1), then $E(t) = E(0)$ for all t.

Lemma 3.3

For $a, b \in \mathbb{C}$ and $\alpha \in \mathbb{R}$: $|a\cos\alpha + b\sin\alpha|^2 + |-a\sin\alpha + b\cos\alpha|^2 = |a|^2 + |b|^2$.

Proof. The vectors $e_1 = (\cos\alpha, \sin\alpha)$, $e_2 = (-\sin\alpha, \cos\alpha)$ form an orthonormal basis of $\mathbb{R}^2 \subseteq \mathbb{C}^2$, and $(a,b)\cdot e_1$, $(a,b)\cdot e_2$ are the coordinates of (a,b) in it; the identity is Pythagoras (expand: cross terms cancel). ∎

Proof of Theorem 3.2. By Plancherel and Proposition 2.1(iv), $\int |\partial_t u|^2 = \int \big|-2\pi|\xi|\hat f\sin(2\pi|\xi|t) + \hat g\cos(2\pi|\xi|t)\big|^2 d\xi$ and $\sum_j \int |\partial_{x_j} u|^2 = \int \big|2\pi|\xi|\hat f\cos(2\pi|\xi|t) + \hat g\sin(2\pi|\xi|t)\big|^2 d\xi$ (since $\sum_j |2\pi i\xi_j|^2 = 4\pi^2|\xi|^2$). By Lemma 3.3 with $a = 2\pi|\xi|\hat f$, $b = \hat g$, $\alpha = 2\pi|\xi|t$, the sum of the integrands is $4\pi^2|\xi|^2|\hat f|^2 + |\hat g|^2$, independent of t. ∎

Uniqueness [Supplement]

Conservation of energy (for any solution with Schwartz-type decay, by integrating by parts as in Theorem 2.3 of Chapter 5: $E'(t) = 2\,\mathrm{Re}\int(\partial_t u\,\overline{\partial_t^2 u} + \sum \partial_j u\,\overline{\partial_j \partial_t u}) = 2\,\mathrm{Re}\int \partial_t u\,\overline{(\partial_t^2 u - \triangle u)} = 0$) shows that a solution with zero data has $E \equiv 0$, so u is constant in x and t, hence 0. Thus (6.1) is the only such solution.

6.3.2 The wave equation in $\mathbb{R}^3 \times \mathbb{R}$

The **spherical mean** of f is $M_t(f)(x) = \frac{1}{4\pi}\int_{S^2} f(x - t\gamma)\, d\sigma(\gamma)$, the average of f over the sphere of radius t centred at x.

Lemma 3.4

If $f \in \mathcal{S}(\mathbb{R}^3)$ and t is fixed, then $M_t(f) \in \mathcal{S}(\mathbb{R}^3)$; moreover $M_t(f)$ is C^∞ in t and each t-derivative lies in $\mathcal{S}(\mathbb{R}^3)$.

Proof. For $|x| \geq 2|t|$ and $\gamma \in S^2$, $|x - t\gamma| \geq |x|/2$, so $|f(x - t\gamma)| \leq C_N(1 + |x|)^{-N}$ and $M_t(f)$ is rapidly decreasing. Derivatives in x and t pass under the integral and are spherical means of derivatives of f (times polynomials in γ), so they are rapidly decreasing too. ∎

Lemma 3.5

$\frac{1}{4\pi}\int_{S^2} e^{-2\pi i\xi\cdot\gamma}d\sigma(\gamma) = \frac{\sin(2\pi|\xi|)}{2\pi|\xi|}$.

Proof. The left side is radial in ξ (rotation invariance of $d\sigma$), so take $\xi = (0,0,\rho)$, $\rho = |\xi|$. In spherical coordinates

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} e^{-2\pi i\rho\cos\theta}\sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{-1}^{1} e^{-2\pi i\rho u}du = \frac{\sin(2\pi\rho)}{2\pi\rho}.$$

■

Consequently (Fubini),

$$\widehat{M_t(f)}(\xi) = \hat{f}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}. \tag{6.2}$$

Theorem 3.6 (Kirchhoff)

For $d = 3$, the solution of the Cauchy problem is $u(x,t) = \frac{\partial}{\partial t}(tM_t(f)(x)) + tM_t(g)(x)$.

Proof. By Theorem 3.1 with $f = 0$, the solution with data $(0, g)$ is

$$u_1 = \int \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}e^{2\pi ix\cdot\xi}d\xi = t\int \hat{g}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi = tM_t(g)(x),$$

by (6.2) and inversion. With data $(f, 0)$, $u_2 = \int \hat{f}\cos(2\pi|\xi|t)e^{2\pi ix\cdot\xi}d\xi = \frac{\partial}{\partial t}\left(t\int \hat{f}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi\right) = \partial_t(tM_t(f))$. Superpose: $u = u_1 + u_2$. ■

Huygens' principle. In $d = 1$, $u(x,t)$ (d'Alembert) depends on the data in the whole interval $[x-t, x+t]$, but only on the endpoints if $g = 0$. In $d = 3$, $u(x,t)$ always depends only on the data on the *sphere* $|y - x| = t$: a disturbance localized near a point is felt at distance t exactly at time t and then passes (sharp signals; light and sound propagate cleanly).

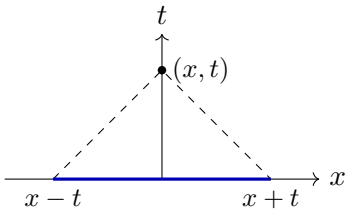


Figure. *Domain of dependence in $d = 1$: $u(x,t)$ depends on the data in $[x-t, x+t]$; in $d = 3$ only on the sphere of radius t.*

6.3.3 The wave equation in $\mathbb{R}^2 \times \mathbb{R}$: descent

For $f \in \mathcal{S}(\mathbb{R}^2)$ let $\tilde{M}_t(f)(x) = \frac{1}{2\pi}\int_{|y|\le 1} f(x - ty)(1 - |y|^2)^{-1/2}dy$.

Theorem 3.7

A solution of the Cauchy problem in two dimensions with data $f, g \in \mathcal{S}(\mathbb{R}^2)$ is

$$u(x,t) = \frac{\partial}{\partial t}(t\tilde{M}_t(f)(x)) + t\tilde{M}_t(g)(x). \tag{6.3}$$

Proof.

Step 1 (descent idea). Let $\tilde{f}(x_1,x_2,x_3)=f(x_1,x_2)$; a solution of the 3-dimensional problem with data independent of x_3 is independent of x_3 and solves the 2-dimensional problem. Projecting the upper and lower hemispheres $\gamma=(y,\pm\sqrt{1-|y|^2})$, $|y|\le 1$, onto the unit disc, $d\sigma=(1-|y|^2)^{-1/2}dy$ on each, so $M_t(\tilde{f})(x,x_3)=\frac{2}{4\pi}\int_{|y|\le1} f(x-ty)(1-|y|^2)^{-1/2}dy=\tilde{M}_t(f)(x)$.

Step 2 (truncation). Since $\tilde{f}\notin\mathcal{S}(\mathbb{R}^3)$, fix $T>0$ and $\eta\in\mathcal{S}(\mathbb{R})$ with $\eta(x_3)=1$ for $|x_3|\le 3T$, and let $\tilde{f}^\flat=f(x_1,x_2)\eta(x_3)$, $\tilde{g}^\flat$ likewise. By Theorem 3.6, $u^\flat=\partial_t(tM_t(\tilde{f}^\flat))+tM_t(\tilde{g}^\flat)$ solves the 3-dimensional problem with these data. For $|x_3|\le T$ and $|t|\le T$, the spheres of radius t around (x,x_3) stay in $|x_3|\le 2T$, where $\tilde{f}^\flat=\tilde{f}$, so $u^\flat(x,x_3,t)$ equals (6.3). It is independent of x_3 there, so $\partial_{x_3}^2u^\flat=0$ and (6.3) solves the 2-dimensional equation for $|t|<T$. T is arbitrary. ∎

In two dimensions $u(x,t)$ depends on the data on the whole disc of radius t, so Huygens' principle fails (ripples on water linger).

6.4 Radial symmetry and Bessel functions

Motivation. For radial $f(x)=f_0(|x|)$ the transform is radial, $\hat{f}(\xi)=F_0(|\xi|)$. In dimensions 1 and 3 the relation between f_0 and F_0 is elementary; in dimension 2 it involves the Bessel function J_0.

- $d=1$ (radial = even): $F_0(\rho)=\int f(x)e^{-2\pi ix\rho}dx=2\int_0^\infty\cos(2\pi\rho r)f_0(r)\,dr$.
- $d=3$: by polar coordinates and Lemma 3.5,

$$\begin{aligned}F_0(\rho)&=\int_0^\infty f_0(r)\Big(\int_{S^2}e^{-2\pi ir\gamma\cdot\xi}d\sigma(\gamma)\Big)r^2dr\\&=\int_0^\infty f_0(r)\frac{2\sin(2\pi\rho r)}{\rho r}r^2dr=2\rho^{-1}\int_0^\infty\sin(2\pi\rho r)f_0(r)\,r\,dr.\end{aligned}$$

- $d=2$: the **Bessel function** of order $n\in\mathbb{Z}$ is $J_n(\rho)=\frac{1}{2\pi}\int_0^{2\pi}e^{i\rho\sin\theta}e^{-in\theta}d\theta$, the n-th Fourier coefficient of $e^{i\rho\sin\theta}$, so $e^{i\rho\sin\theta}=\sum_n J_n(\rho)e^{in\theta}$. Taking $\xi=(0,-\rho)$,

$$F_0(\rho)=\int_0^{2\pi}\int_0^\infty f_0(r)e^{2\pi ir\rho\sin\theta}r\,dr\,d\theta=2\pi\int_0^\infty J_0(2\pi r\rho)f_0(r)\,r\,dr.$$

In general F_0 is given by a Bessel function of order $d/2-1$; for odd d these are elementary, for even d not.

6.5 The Radon transform and some of its applications

Motivation. X-ray scans measure integrals of the attenuation coefficient along lines. Reconstructing a function from its integrals over lines (in $\mathbb{R}^2$) or planes (in $\mathbb{R}^3$) is possible because the one-dimensional Fourier transform of the Radon transform is a slice of the full Fourier transform.

6.5.1 The X-ray transform in $\mathbb{R}^2$

A beam of intensity I_0 traversing an object with attenuation coefficient ρ along a line L emerges with intensity $I=I_0e^{-\int_L\rho}$. So measuring I for all lines gives the **X-ray transform** $X(\rho)(L)=\int_L\rho$. **Reconstruction problem:** recover ρ from $X(\rho)$ (uniqueness and an inversion formula). In $\mathbb{R}^2$ the inversion formula involves a Hilbert transform; we treat the simpler 3-dimensional analogue.

6.5.2 The Radon transform in $\mathbb{R}^3$

Planes in $\mathbb{R}^3$ are $\mathcal{P}_{t,\gamma} = \{x : x \cdot \gamma = t\}$, $t \in \mathbb{R}$, $\gamma \in S^2$ (and $\mathcal{P}_{t,\gamma} = \mathcal{P}_{-t,-\gamma}$). For $f \in \mathcal{S}(\mathbb{R}^3)$ choose e_1, e_2 with $\{e_1, e_2, \gamma\}$ orthonormal and set

$$\mathcal{R}(f)(t,\gamma) = \int_{\mathcal{P}_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du_1\, du_2.$$

Proposition 5.1

For $f \in \mathcal{S}(\mathbb{R}^3)$ and each γ, $\mathcal{R}(f)(t,\gamma)$ does not depend on the choice of e_1, e_2, and $\int_{-\infty}^{\infty} \mathcal{R}(f)(t,\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\, dx$.

Proof. Another choice e_1', e_2' differs by a rotation (or reflection) of $\mathbb{R}^2$, under which $\int_{\mathbb{R}^2}$ is invariant. For the identity, the linear change of variables $x = t\gamma + u_1 e_1 + u_2 e_2$ is a rotation of $\mathbb{R}^3$, so $\int_{\mathbb{R}^3} f = \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du\, dt$ (Fubini for rapidly decreasing functions). ∎

Lemma 5.2

For $f \in \mathcal{S}(\mathbb{R}^3)$ and fixed γ, $\mathcal{R}(f)(\cdot,\gamma) \in \mathcal{S}(\mathbb{R})$, and $\widehat{\mathcal{R}(f)}(s,\gamma) = \hat{f}(s\gamma)$ (one-dimensional transform in t on the left, three-dimensional on the right).

Proof. $(1+|t|)^N |f(t\gamma + u)| \le C(1+|u|)^{-3}$ since $|t\gamma + u|^2 = t^2 + |u|^2$, so $\mathcal{R}(f)$ decays rapidly in t; derivatives in t are Radon transforms of directional derivatives of f. By Proposition 5.1 applied to $f(x)e^{-2\pi i s x\cdot\gamma}$ (which equals $e^{-2\pi i s t} f(x)$ on $\mathcal{P}_{t,\gamma}$), $\int \mathcal{R}(f)(t,\gamma) e^{-2\pi i s t} dt = \int_{\mathbb{R}^3} f(x) e^{-2\pi i s x\cdot\gamma} dx = \hat{f}(s\gamma)$. ∎

Corollary 5.3

If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Proof. By Lemma 5.2, $\hat{f}(s\gamma) = \hat{g}(s\gamma)$ for all s, γ, i.e. $\hat{f} = \hat{g}$; apply Fourier inversion. ∎

The **dual Radon transform** of $F(t,\gamma)$ is $\mathcal{R}^*(F)(x) = \int_{S^2} F(x\cdot\gamma, \gamma)\, d\sigma(\gamma)$: it averages F over all planes through x.

Theorem 5.4 (Radon inversion)

If $f \in \mathcal{S}(\mathbb{R}^3)$, then $\triangle(\mathcal{R}^*\mathcal{R}(f)) = -8\pi^2 f$.

Proof. By Lemma 5.2 and one-dimensional inversion, $\mathcal{R}(f)(t,\gamma) = \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i t s} ds$, so $\mathcal{R}^*\mathcal{R}(f)(x) = \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i x\cdot\gamma s} ds\, d\sigma(\gamma)$. Applying $\triangle$ (under the integral) multiplies the integrand by $-4\pi^2 s^2 |\gamma|^2 = -4\pi^2 s^2$. The change $(s,\gamma) \mapsto (-s,-\gamma)$ shows that the parts $s<0$ and $s>0$ are equal, so

$$\triangle(\mathcal{R}^*\mathcal{R}(f))(x) = -8\pi^2 \int_{S^2} \int_0^{\infty} \hat{f}(s\gamma) e^{2\pi i x\cdot s\gamma} s^2 ds\, d\sigma(\gamma) = -8\pi^2 f(x),$$

the last step being Fourier inversion written in polar coordinates. ∎

So $f = -\frac{1}{8\pi^2} \triangle \mathcal{R}^*\mathcal{R}(f)$: integrals over planes determine f explicitly, the mathematical basis of tomography.

6.6 Selected exercises

Motivation. Three exercises from the book that use the d-dimensional tools of this chapter: rotation invariance of integrals, Plancherel with differentiation, and Fubini on $\mathbb{R}^3 \times S^2$.

Exercise 6.5 (Gaussian integrals with a matrix)

Let A be a real symmetric positive definite $d \times d$ matrix. Show that

$$\int_{\mathbb{R}^d} e^{-\pi (x, Ax)}\, dx = (\det A)^{-1/2}.$$

Solution.

Step 1. By the spectral theorem there is an orthogonal R ($R^t = R^{-1}$) with $R^t A R = D = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$, where $\lambda_i > 0$ are the eigenvalues of A. With $\lambda = \min_i \lambda_i$ we get $(x, Ax) = (R^t x, D R^t x) \geq \lambda |R^t x|^2 = \lambda |x|^2$, so $f(x) = e^{-\pi (x, Ax)}$ is continuous and rapidly decreasing, and the integral exists.

Step 2. R is a rotation in the sense of Section 6.1.1, so $\int f(x)\, dx = \int f(Ry)\, dy$ (Section 6.1.2). Since $(Ry, ARy) = (y, R^t A R y) = (y, Dy) = \sum_i \lambda_i y_i^2$,

$$\int_{\mathbb{R}^d} e^{-\pi (x, Ax)}\, dx = \int_{\mathbb{R}^d} \prod_{i=1}^{d} e^{-\pi \lambda_i y_i^2}\, dy = \prod_{i=1}^{d} \int_{\mathbb{R}} e^{-\pi \lambda_i t^2}\, dt,$$

where the integral of the product over the cube Q_N factors, and we let $N \to \infty$.

Step 3. Substituting $s = \sqrt{\lambda_i}\, t$, $\int_{\mathbb{R}} e^{-\pi \lambda_i t^2} dt = \lambda_i^{-1/2} \int_{\mathbb{R}} e^{-\pi s^2} ds = \lambda_i^{-1/2}$. Hence the integral is $(\lambda_1 \cdots \lambda_d)^{-1/2} = (\det A)^{-1/2}$. ∎

Exercise 6.6 (Heisenberg uncertainty principle in $\mathbb{R}^d$)

Let $\psi \in \mathcal{S}(\mathbb{R}^d)$ with $\int |\psi|^2 = 1$. Show that

$$\Big(\int_{\mathbb{R}^d} |x|^2 |\psi(x)|^2 dx \Big) \Big(\int_{\mathbb{R}^d} |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi \Big) \geq \frac{d^2}{16 \pi^2}.$$

Solution.

Step 1. Fix j and the other coordinates. Integrating by parts in x_j,

$$\int_{\mathbb{R}} |\psi|^2\, dx_j = \big[x_j |\psi|^2 \big]_{x_j = -\infty}^{\infty} - \int_{\mathbb{R}} x_j\, \partial_j (|\psi|^2)\, dx_j = - \int_{\mathbb{R}} x_j\, \partial_j (|\psi|^2)\, dx_j,$$

the boundary term vanishing by rapid decay. All functions involved are rapidly decreasing on $\mathbb{R}^d$, so integrating over the remaining variables (Fubini) gives $1 = - \int x_j \partial_j (|\psi|^2)\, dx$.

Step 2. Since $\partial_j |\psi|^2 = \partial_j \psi\, \bar{\psi} + \psi\, \partial_j \bar{\psi} = 2 \operatorname{Re}(\bar{\psi}\, \partial_j \psi)$, summing over j gives

$$d = -2 \operatorname{Re} \int_{\mathbb{R}^d} \bar{\psi}(x) \sum_{j=1}^{d} x_j\, \partial_j \psi(x)\, dx \leq 2 \int_{\mathbb{R}^d} |x|\, |\psi(x)|\, |\nabla \psi(x)|\, dx,$$

by the Cauchy–Schwarz inequality in $\mathbb{R}^d$: $|\sum_j x_j \partial_j \psi| \leq |x|\, |\nabla \psi|$.

Step 3. The Cauchy–Schwarz inequality for integrals gives

$$d^2 \leq 4 \int |x|^2 |\psi|^2 dx \cdot \int |\nabla \psi|^2 dx.$$

Step 4. By Proposition 2.1(iv), $\widehat{\partial_j \psi}(\xi) = 2\pi i \xi_j \hat{\psi}(\xi)$, and by Plancherel (Theorem 2.4),

$$\int |\nabla \psi|^2 dx = \sum_{j=1}^{d} \int 4\pi^2 \xi_j^2 |\hat{\psi}(\xi)|^2 d\xi = 4\pi^2 \int |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi.$$

Substituting into Step 3, $d^2 \le 16\pi^2 \int |x|^2|\psi|^2 \int |\xi|^2|\hat{\psi}|^2$, which is the claim. ∎

Exercise 6.12 (The dual Radon transform is the adjoint)

For $f \in \mathcal{S}(\mathbb{R}^3)$ and $F \in \mathcal{S}(\mathbb{R} \times S^2)$ prove

$$\int_{\mathbb{R}} \int_{S^2} \mathcal{R}(f)(t,\gamma)\, \overline{F(t,\gamma)}\, d\sigma(\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\, \overline{\mathcal{R}^*(F)(x)}\, dx.$$

Solution.

Step 1. Fix $\gamma \in S^2$ and put $h_\gamma(x) = f(x)\overline{F(x\cdot\gamma,\gamma)}$. It is continuous and rapidly decreasing (F is bounded), and on the plane $\mathcal{P}_{t,\gamma}$ it equals $f(x)\overline{F(t,\gamma)}$. Hence $\mathcal{R}(h_\gamma)(t,\gamma) = \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)}$.

Step 2. The proof of Proposition 5.1 (a rotation of coordinates and Fubini) uses only continuity and rapid decrease, so it applies to h_γ:

$$\int_{\mathbb{R}} \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)}\, dt = \int_{\mathbb{R}^3} f(x)\overline{F(x\cdot\gamma,\gamma)}\, dx.$$

Step 3. Integrate over $\gamma \in S^2$. The function $(x,\gamma) \mapsto f(x)\overline{F(x\cdot\gamma,\gamma)}$ is continuous and $\int_{S^2}\int_{\mathbb{R}^3} |f(x)|\,|F(x\cdot\gamma,\gamma)|\, dx\, d\sigma(\gamma) \le 4\pi \sup|F| \int|f| < \infty$, so Fubini allows us to exchange the integrals:

$$\int_{S^2}\int_{\mathbb{R}} \mathcal{R}(f)\overline{F}\, dt\, d\sigma(\gamma) = \int_{\mathbb{R}^3} f(x)\, \overline{\int_{S^2} F(x\cdot\gamma,\gamma)\, d\sigma(\gamma)}\, dx = \int_{\mathbb{R}^3} f\, \overline{\mathcal{R}^*(F)}\, dx.$$

Step 4. On the left, $|\mathcal{R}(f)(t,\gamma)F(t,\gamma)| \le \sup|F| \cdot \mathcal{R}(|f|)(t,\gamma)$, and $\int_{\mathbb{R}} \mathcal{R}(|f|)(t,\gamma)\, dt = \int |f|$ for every γ (Step 2 with f, F replaced by $|f|, 1$). So the integrand is integrable on $\mathbb{R}\times S^2$, and Fubini lets us write the iterated integral in the stated order $\int_{\mathbb{R}}\int_{S^2}$. Thus $(\mathcal{R}f, F)_2 = (f, \mathcal{R}^*F)_1$. ∎

Chapter 7. Finite Fourier Analysis

7.1 Fourier analysis on $\mathbb{Z}(N)$

Motivation. Replacing the circle by the N-th roots of unity, Fourier series become finite sums: inversion and Parseval are plain linear algebra with an orthogonal basis of exponentials. Splitting into even and odd indices computes all N coefficients in $O(N \log N)$ operations (the fast Fourier transform).

7.1.1 The group $\mathbb{Z}(N)$

For $N \geq 1$ let $\zeta = e^{2\pi i/N}$. The N-th roots of unity $\{1, \zeta, \ldots, \zeta^{N-1}\}$ form a group under multiplication, and $\zeta^k \mapsto k$ identifies it with $\mathbb{Z}(N) = \mathbb{Z}/N\mathbb{Z}$ under addition mod N. Functions on $\mathbb{Z}(N)$ are N-periodic functions on $\mathbb{Z}$; one can view $\mathbb{Z}(N)$ as N equally spaced points on the circle (*discretization*).

7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$

V, the complex functions on $\mathbb{Z}(N)$, is an N-dimensional vector space with inner product $(F, G) = \sum_{k=0}^{N-1} F(k)\overline{G(k)}$ and norm $\|F\|^2 = \sum |F(k)|^2$. For $\ell = 0, \ldots, N-1$ let $e_\ell(k) = \zeta^{\ell k}$.

Lemma 1.1

The family $\{e_0, \ldots, e_{N-1}\}$ is orthogonal: $(e_m, e_\ell) = N$ if $m = \ell$ and 0 if $m \neq \ell$. Hence $\{e_\ell/\sqrt{N}\}$ is an orthonormal basis of V.

Proof. $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$. If $m = \ell$ this is N. Otherwise $q = \zeta^{m-\ell} \neq 1$ and $q^N = 1$, so the geometric sum is $\frac{1-q^N}{1-q} = 0$. N orthogonal nonzero vectors in an N-dimensional space form a basis. ■

The n-th **Fourier coefficient** of F is $a_n = \frac{1}{N}(F, e_n) = \frac{1}{N}\sum_{k=0}^{N-1} F(k)\zeta^{-kn}$.

Theorem 1.2

If F is a function on $\mathbb{Z}(N)$, then $F(k) = \sum_{n=0}^{N-1} a_n \zeta^{nk}$ (inversion) and $\sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |F(k)|^2$ (Parseval).

Proof. Expand F in the orthonormal basis $e_n^* = e_n/\sqrt{N}$: $F = \sum (F, e_n^*)e_n^* = \sum \frac{1}{N}(F, e_n)e_n = \sum a_n e_n$. Pythagoras gives $\|F\|^2 = \sum |(F, e_n^*)|^2 = \sum N|a_n|^2$. ■

(Discrete analogue of Fourier series: if $F(k) = f(k/N)$ for a 1-periodic f, then a_n is a Riemann sum for $\hat{f}(n)$.)

7.1.3 The fast Fourier transform

Computing all a_n naively takes about N^2 operations. Let $\#(M)$ be the minimal number of operations (additions and multiplications) needed to compute all Fourier coefficients of any function on $\mathbb{Z}(M)$, here with $\omega_M = e^{-2\pi i/M}$, so that $a_k^M(F) = \frac{1}{M}\sum_r F(r)\omega_M^{kr}$.

Theorem 1.3

For $N = 2^n$, the Fourier coefficients of a function on $\mathbb{Z}(N)$ can be computed with at most $4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$ operations, given ω_N.

Lemma 1.4

Given ω_{2M}, $\#(2M) \le 2\#(M) + 8M$.

Proof. Computing $\omega_{2M}, \omega_{2M}^2, \dots, \omega_{2M}^{2M}$ takes $\le 2M$ operations, and gives $\omega_M = \omega_{2M}^2$. For F on $\mathbb{Z}(2M)$ put $F_0(r) = F(2r)$, $F_1(r) = F(2r+1)$, functions on $\mathbb{Z}(M)$. Splitting the sum into even and odd indices,

$$a_k^{2M}(F) = \frac{1}{2M} \sum_{r=0}^{2M-1} F(r)\omega_{2M}^{kr} = \frac{1}{2}\Big(\frac{1}{M}\sum_{\ell} F_0(\ell)\omega_M^{k\ell} + \frac{1}{M}\sum_{m} F_1(m)\omega_M^{km}\omega_{2M}^{k}\Big)$$
$$= \frac{1}{2}\big(a_k^M(F_0) + a_k^M(F_1)\omega_{2M}^k\big)$$

(with a_k^M read M-periodically in k). Computing all $a^M(F_0)$, $a^M(F_1)$ costs $2\#(M)$; then each of the $2M$ coefficients costs 3 operations. So $\#(2M) \le 2M + 2\#(M) + 6M$. ∎

Proof of Theorem 1.3. Induction on n. For $N = 2$: $a_0 = \frac{1}{2}(F(0) + F(1))$, $a_1 = \frac{1}{2}(F(0) - F(1))$, at most $5 \le 8$ operations. If $\#(2^{n-1}) \le 4 \cdot 2^{n-1}(n-1)$, then by Lemma 1.4, $\#(2^n) \le 2 \cdot 4 \cdot 2^{n-1}(n-1) + 8 \cdot 2^{n-1} = 4 \cdot 2^n n$. ∎

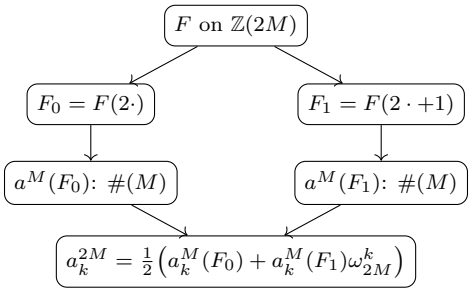


Figure. *One step of the fast Fourier transform (Lemma 1.4).*

7.2 Fourier analysis on finite abelian groups

Motivation. On a general finite abelian group the role of the exponentials is played by characters, homomorphisms into the unit circle. They are orthonormal, and there are enough of them to form a basis (simultaneous diagonalization of the translation operators). This gives Fourier inversion and Plancherel on every finite abelian group, the tool needed for Dirichlet's theorem.

7.2.1 Abelian groups

An **abelian group** $(G, \cdot)$ is associative and commutative with identity 1 and inverses. Examples: $\mathbb{Z}(N)$; the circle $\mathbb{R}/\mathbb{Z}$; $\mathbb{R}$, $\mathbb{R}^+ = (0, \infty)$ under multiplication; $\mathbb{Z}^*(q)$, the units of $\mathbb{Z}(q)$, i.e. residues coprime to q, under multiplication. A **homomorphism** $f : G \to H$ satisfies $f(ab) = f(a)f(b)$; an **isomorphism** is a bijective homomorphism. The **direct product** $G_1 \times G_2$ has componentwise multiplication. (E.g. $\mathbb{Z}^*(8) \cong \mathbb{Z}(2) \times \mathbb{Z}(2)$, while $\mathbb{Z}^*(5) \cong \mathbb{Z}(4)$.) Every finite abelian group is a direct product of cyclic groups; we will not need this.

7.2.2 Characters

A **character** of G is a function $e : G \to S^1 = \{|z| = 1\}$ with $e(ab) = e(a)e(b)$, i.e. a homomorphism $G \to S^1$. The **trivial character** is $e \equiv 1$. $\hat{G}$ denotes the set of characters. On $\mathbb{Z}(N)$ the characters are exactly $e_\ell(k) = \zeta^{\ell k}$ (a character is determined by $e(1)$, an N-th root of unity).

Lemma 2.1

$\hat{G}$ is an abelian group under $(e_1 e_2)(a) = e_1(a)e_2(a)$.

Proof. Products of characters are characters; the trivial character is the identity; the inverse of e is $\bar{e} = 1/e$ (as $|e| = 1$); associativity and commutativity are pointwise. ∎

Lemma 2.2

If G is finite and $e : G \to \mathbb{C} - \{0\}$ satisfies $e(ab) = e(a)e(b)$, then e is a character.

Proof. G is finite, so $|e(a)|$ is bounded above and below by positive numbers. If $|e(a)| > 1$, then $|e(a^n)| = |e(a)|^n \to \infty$, impossible; if $|e(a)| < 1$, $|e(a^n)| \to 0$, also impossible. So $|e(a)| = 1$. ∎

7.2.3 The orthogonality relations

On the space V of complex functions on finite G use $(f, g) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{g(a)}$.

Theorem 2.3

The characters of G form an orthonormal family in V.

Lemma 2.4

If e is a nontrivial character, then $\sum_{a \in G} e(a) = 0$.

Proof. Choose b with $e(b) \neq 1$. Since $a \mapsto ba$ permutes G, $e(b) \sum_a e(a) = \sum_a e(ba) = \sum_a e(a)$; so $\sum_a e(a) = 0$. ∎

Proof of Theorem 2.3. $(e, e) = \frac{1}{|G|} \sum |e(a)|^2 = 1$. For $e \neq e'$, $e\overline{e'} = e(e')^{-1}$ is a nontrivial character, so $(e, e') = \frac{1}{|G|} \sum_a e(a)\overline{e'(a)} = 0$ by Lemma 2.4. ∎

Consequently $|\hat{G}| \leq |G|$ (orthonormal vectors are independent).

7.2.4 Characters as a total family

Theorem 2.5

The characters of a finite abelian group G form a basis of the vector space V of functions on G; in particular $|\hat{G}| = |G|$.

Lemma 2.6

A commuting family $\{T_1, \ldots, T_k\}$ of unitary transformations of a finite-dimensional inner product space V ($T_i T_j = T_j T_i$) is simultaneously diagonalizable: V has a basis of common eigenvectors.

Proof. Induction on k. For $k = 1$ this is the spectral theorem for unitary operators. Write $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$, the eigenspaces of T_k. Each T_j, $j < k$, maps V_{λ_i} into itself: if $T_k v = \lambda_i v$, then $T_k T_j v = T_j T_k v = \lambda_i T_j v$. The restrictions of $T_1, \dots, T_{k-1}$ to V_{λ_i} are commuting unitary operators, so by induction V_{λ_i} has a basis of common eigenvectors. Their union is the desired basis of V. ∎

Proof of Theorem 2.5. For $a \in G$ define T_a on V by $(T_a f)(x) = f(ax)$. $T_a T_b = T_b T_a$ (G abelian), and T_a is unitary since $x \mapsto ax$ permutes G. By Lemma 2.6 there is a basis $\{v_b\}$ of common eigenvectors, $T_a v = \lambda_a v$. For such v: $v(1) \neq 0$, since otherwise $v(a) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a. Let $w(x) = v(x)/v(1)$. Then $w(x) = \lambda_x$ (as $v(x) = (T_x v)(1) = \lambda_x v(1)$), so $w(ab) = \lambda_{ab}$, and $\lambda_{ab} v = T_{ab} v = T_a T_b v = \lambda_a \lambda_b v$ gives $w(ab) = w(a)w(b)$; also $w(x) \neq 0$ (as T_x is invertible). By Lemma 2.2, w is a character. The basis $\{v_b\}$ consists of multiples of characters, so the characters span V; with Theorem 2.3 they form a basis. ∎

7.2.5 Fourier inversion and Plancherel formula

For f on G and $e \in \hat{G}$, the **Fourier coefficient** is $\hat{f}(e) = (f, e) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{e(a)}$, and the **Fourier series** is $f \sim \sum_{e \in \hat{G}} \hat{f}(e) e$.

Theorem 2.7

The characters of a finite abelian group G form an orthonormal basis of V with $(f, g) = \frac{1}{|G|} \sum_a f(a)\overline{g(a)}$. For every f on G, $f = \sum_{e \in \hat{G}} \hat{f}(e) e$ (inversion) and $\|f\|^2 = \sum_{e \in \hat{G}} |\hat{f}(e)|^2$ (Plancherel).

Proof. Theorems 2.3 and 2.5; then expand in the orthonormal basis and apply Pythagoras. ∎

For $G = \mathbb{Z}(N)$ this recovers Theorem 1.2 (up to the normalization of the inner product).

7.3 Selected exercises

Motivation. Three exercises from the book linking the finite theory to the circle (decay of discrete coefficients, sampling of trigonometric polynomials) and recasting finite Fourier series as convolution with a "delta" kernel.

Exercise 7.2 (Decay of discrete Fourier coefficients)

Let f be a C^1 function on the circle and, for $N \geq 1$, $a_N(n) = \frac{1}{N} \sum_{k=1}^{N} f(e^{2\pi i k/N}) e^{-2\pi i k n/N}$. Prove that there is c, independent of N and n, with $|a_N(n)| \leq c/|n|$ whenever $0 < |n| \leq N/2$.

Solution. Write $g(x) = f(e^{2\pi i x})$, a 1-periodic C^1 function, and $M = \sup |g'|$.

Step 1. For an integer ℓ, reindexing $k \mapsto k - \ell$ and using that $k \mapsto g(k/N) e^{-2\pi i k n/N}$ is N-periodic in k,

$$\frac{1}{N} \sum_{k=1}^{N} g\Big(\frac{k+\ell}{N}\Big) e^{-2\pi i k n/N} = e^{2\pi i \ell n/N} a_N(n).$$

Subtracting this from the definition of $a_N(n)$,

$$a_N(n)\big(1 - e^{2\pi i \ell n/N}\big) = \frac{1}{N} \sum_{k=1}^{N} \Big[g\Big(\frac{k}{N}\Big) - g\Big(\frac{k+\ell}{N}\Big)\Big] e^{-2\pi i k n/N}.$$

Step 2. By the mean value theorem each bracket is at most $M\ell/N$ in modulus, so the right side is at most $M\ell/N$.

Step 3. Choose $\ell = \lfloor N/(2|n|)\rfloor$, which is ≥ 1 since $|n| \leq N/2$. Then $\ell|n|/N \leq \frac{1}{2}$, and also $\ell|n|/N \geq |n|/N$ and $\ell|n|/N > \frac{1}{2} - |n|/N$; the larger of these two lower bounds is at least $\frac{1}{4}$. So $s = \ell n/N$ has $\frac{1}{4} \leq |s| \leq \frac{1}{2}$ and

$$|1 - e^{2\pi i s}| = 2|\sin(\pi s)| \geq 2\sin(\pi/4) = \sqrt{2}.$$

Step 4. Combining, $|a_N(n)| \leq \frac{M\ell}{\sqrt{2}N} \leq \frac{M}{2\sqrt{2}\,|n|}$, using $\ell \leq N/(2|n|)$. Take $c = M/(2\sqrt{2})$. ∎

Exercise 7.8 (Sampling a trigonometric polynomial)

Let $P(x) = \sum_{n=1}^{N} a_n e^{2\pi i n x}$.
(a) Show that $\int_0^1 |P(x)|^2 dx = \frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2$.
(b) Prove that $P(x) = \sum_{j=1}^{N} P(j/N) K(x - j/N)$, where

$$K(x) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n x} = \frac{e^{2\pi i x}}{N}\,\frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} \quad (x \notin \mathbb{Z}),$$

and that $K(0) = 1$, $K(j/N) = 0$ for $j \not\equiv 0 \pmod N$.

Solution. Let $\zeta = e^{2\pi i/N}$, so $P(j/N) = \sum_{n=1}^{N} a_n \zeta^{nj}$.

Step 1. For integers $n, m \in \{1, \dots, N\}$, $\frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \mathbb{1}\{n = m\}$: the sum over $j = 1, \dots, N$ equals the sum over $j = 0, \dots, N-1$ ($\zeta^N = \zeta^0$), and $n - m \in (-N, N)$ is $\equiv 0 \pmod N$ only if $n = m$; apply Lemma 1.1.

Step 2. Hence

$$\frac{1}{N}\sum_{j=1}^{N} P(j/N)\zeta^{-mj} = \sum_{n=1}^{N} a_n \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = a_m, \qquad 1 \leq m \leq N.$$

(a) On the circle, $\int_0^1 e^{2\pi i(n-m)x} dx = \mathbb{1}\{n = m\}$, so $\int_0^1 |P|^2 = \sum_{n,m} a_n \bar{a}_m \int_0^1 e^{2\pi i(n-m)x} dx = \sum_n |a_n|^2$. On $\mathbb{Z}(N)$, by Step 1,

$$\frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2 = \sum_{n,m} a_n \bar{a}_m \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \sum_{n} |a_n|^2.$$

(These are the Parseval identities on the circle and, by Step 2, on $\mathbb{Z}(N)$, Theorem 1.2.) The two sides agree.

(b) Insert Step 2 into the definition of P:

$$P(x) = \sum_{n=1}^{N} \Big(\frac{1}{N}\sum_{j=1}^{N} P(j/N) e^{-2\pi i n j/N}\Big) e^{2\pi i n x} = \sum_{j=1}^{N} P(j/N)\,\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n (x - j/N)},$$

which is $\sum_j P(j/N) K(x - j/N)$. For $x \notin \mathbb{Z}$, $q = e^{2\pi i x} \neq 1$ and $\sum_{n=1}^{N} q^n = q\frac{1-q^N}{1-q}$, giving the closed form. Finally $K(0) = \frac{1}{N}\cdot N = 1$, and for $j \not\equiv 0$ the number $q = \zeta^j$ satisfies $q \neq 1$, $q^N = 1$, so $K(j/N) = \frac{1}{N}\sum_{n=1}^{N} q^n = \frac{q}{N}\frac{1-q^N}{1-q} = 0$. Thus P is determined by its N samples $P(j/N)$. ∎

Exercise 7.13 (Convolution and the delta kernel on G)

Let G be a finite abelian group with unit 1_G and $(f*g)(a) = \frac{1}{|G|}\sum_{b\in G} f(b)g(ab^{-1})$.
(a) Show that $\widehat{f*g}(e) = \hat{f}(e)\hat{g}(e)$ for all $e \in \hat{G}$.
(b) Show that $\sum_{e\in\hat{G}} e(c) = 0$ whenever $c \neq 1_G$.
(c) Let $D(c) = \sum_{e\in\hat{G}} e(c)$. Show that the Fourier series $Sf = \sum_e \hat{f}(e)e$ equals $f * D$, and deduce $Sf = f$.

Solution.

(a) Since $\overline{e(bc)} = \overline{e(b)}\,\overline{e(c)}$ and $c = ab^{-1}$ runs over G as a does (for fixed b),

$$\begin{aligned}\widehat{f*g}(e) &= \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{a\in G} g(ab^{-1})\overline{e(a)} = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{c\in G} g(c)\overline{e(bc)}\\ &= \Big(\frac{1}{|G|}\sum_b f(b)\overline{e(b)}\Big)\Big(\frac{1}{|G|}\sum_c g(c)\overline{e(c)}\Big) = \hat{f}(e)\hat{g}(e).\end{aligned}$$

(b)

Step 1. There is $e_0 \in \hat{G}$ with $e_0(c) \neq 1$. Otherwise every character takes the same value at c and at 1_G, hence so does every linear combination of characters; by Theorem 2.5 every function is such a combination, but the indicator of $\{1_G\}$ takes the values 1 and 0 there.

Step 2. By Lemma 2.1, $e \mapsto e_0 e$ is a bijection of $\hat{G}$. Hence $\sum_e e(c) = \sum_e (e_0 e)(c) = e_0(c)\sum_e e(c)$, and $e_0(c) \neq 1$ forces $\sum_e e(c) = 0$.

(c)

Step 1. By (b) and $|\hat{G}| = |G|$ (Theorem 2.5), $D(1_G) = |G|$ and $D(c) = 0$ for $c \neq 1_G$.

Step 2. Using $\overline{e(b)} = e(b)^{-1} = e(b^{-1})$,

$$Sf(a) = \sum_{e\in\hat{G}} \frac{1}{|G|}\sum_{b\in G} f(b)\overline{e(b)}e(a) = \frac{1}{|G|}\sum_{b\in G} f(b)\sum_{e\in\hat{G}} e(ab^{-1}) = (f*D)(a).$$

Step 3. By Step 1 only $b = a$ contributes, so $(f*D)(a) = \frac{1}{|G|}f(a)|G| = f(a)$. Hence $Sf = f$, a second proof of the inversion formula in Theorem 2.7. ∎

Chapter 8. Dirichlet's Theorem

8.1 A little elementary number theory

Motivation. Dirichlet's theorem on primes in arithmetic progressions is proved by Fourier analysis on the finite abelian group $\mathbb{Z}^*(q)$. This section collects the arithmetic needed: unique factorization, and Euler's analytic proof that there are infinitely many primes via $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ and the divergence of $\sum 1/p$.

8.1.1 The fundamental theorem of arithmetic

For integers a, b, $a \mid b$ means $b = ac$ for some $c \in \mathbb{Z}$. A prime is an integer $p > 1$ whose only positive divisors are 1 and p. The greatest common divisor $\gcd(a, b)$ of positive integers is the largest positive common divisor; a, b are relatively prime if $\gcd(a, b) = 1$.

Theorem 1.1 (Euclid's algorithm)

For any integers a and b with $b > 0$ there exist unique integers q and r with $0 \le r < b$ such that $a = qb + r$.

Proof.

Step Existence. The set $\{a - qb : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty (take q very negative), so it has a least element $r = a - qb \ge 0$. If $r \ge b$ then $r - b = a - (q+1)b \ge 0$ is smaller, a contradiction; so $0 \le r < b$.

Step Uniqueness. If $a = qb + r = q'b + r'$ with $0 \le r, r' < b$, then $b \mid r - r'$ and $|r - r'| < b$, so $r = r'$ and then $q = q'$. ∎

Theorem 1.2

If $\gcd(a, b) = d$, then there exist integers x and y such that $ax + by = d$.

Proof. Let S be the set of positive integers of the form $ax + by$, and $s = ax + by$ its least element. Every common divisor of a, b divides s, so $d \le s$. Write $a = qs + r$ with $0 \le r < s$; then $r = a(1 - qx) + b(-qy)$ is of the same form, so by minimality $r = 0$, i.e. $s \mid a$. Likewise $s \mid b$, so $s \le d$. Hence $s = d$. ∎

Corollary 1.3

Positive integers a and b are relatively prime if and only if $ax + by = 1$ for some integers x, y.

Proof. One direction is Theorem 1.2. Conversely, if $ax + by = 1$ then any common positive divisor of a, b divides 1. ∎

Corollary 1.4

If a and c are relatively prime and $c \mid ab$, then $c \mid b$. In particular, if a prime p does not divide a and $p \mid ab$, then $p \mid b$.

Proof. Write $1 = ax + cy$; then $b = abx + cby$, and c divides both terms. For a prime $p \nmid a$ we have $\gcd(a, p) = 1$. ∎

Corollary 1.5

If a prime p divides $a_1 \cdots a_r$, then $p \mid a_i$ for some i.

Proof. If $p \nmid a_1$ then $p \mid a_2 \cdots a_r$ by Corollary 1.4; induct on r. ∎

Theorem 1.6

Every integer greater than 1 factors uniquely into a product of primes (up to order).

Proof.

Step Existence. Suppose not, and let n be the smallest integer > 1 without a factorization. Then n is not prime, so $n = ab$ with $1 < a, b < n$. Both a and b factor, and hence so does n, a contradiction.

Step Uniqueness. Suppose $n = p_1 \cdots p_r = q_1 \cdots q_s$. By Corollary 1.5, $p_1 \mid q_j$ for some j, so $p_1 = q_j$. Cancel and induct on r. ∎

8.1.2 The infinitude of primes

Theorem 1.7

There are infinitely many primes.

Proof. If $p_1, \ldots, p_n$ were all the primes, $N = p_1 \cdots p_n + 1$ would have a prime factor (Theorem 1.6), which must be some p_i. But $p_i \mid N - p_1 \cdots p_n = 1$, which is absurd. ∎

Euler's analytic argument proves more ($\sum 1/p = \infty$) and is the model for Dirichlet's proof. For $s > 1$ let $\zeta(s) = \sum_{n \geq 1} n^{-s}$. The series converges, and comparing with $\int x^{-s} dx$,

$$\frac{1}{s-1} \leq \zeta(s) \leq 1 + \frac{1}{s-1}, \qquad \text{so } \zeta(s) \to \infty \text{ as } s \to 1^+. \tag{8.1}$$

(Indeed $\int_n^{n+1} x^{-s} dx \leq n^{-s} \leq \int_{n-1}^{n} x^{-s} dx$ for $n \geq 2$.) An infinite product $\prod A_n$ **converges** if $\prod_{n \leq N} A_n$ has a limit as $N \to \infty$.

Lemma 1.8

The exponential and logarithm functions satisfy:
(i) $e^{\log x} = x$ for $x > 0$.
(ii) $\log(1 + x) = x + E(x)$ with $|E(x)| \leq x^2$ if $|x| < 1/2$.
(iii) If $\log(1 + x) = y$ and $|x| < 1/2$, then $|y| \leq 2|x|$.

Proof. (i) is the definition of log as the inverse of exp.

(ii) For $|x| < 1$, $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so $|E(x)| \leq \frac{x^2}{2}(1 + |x| + |x|^2 + \cdots) = \frac{x^2}{2(1-|x|)} \leq x^2$ when $|x| < 1/2$.

(iii) By (ii), $|y| \leq |x| + x^2 \leq |x| + \frac{1}{2}|x| \leq 2|x|$. ∎

Proposition 1.9

If $A_n = 1 + a_n$ and $\sum |a_n|$ converges, then $\prod_n A_n$ converges, and the product vanishes if and only if some factor A_n vanishes. Also, if $a_n \neq 1$ for all n, then $\prod_n 1/(1 - a_n)$ converges.

Proof. Real a_n here (the complex case is Proposition 3.2). Discard the finitely many n with $|a_n| \geq 1/2$; they only multiply the limit by a fixed finite factor (zero exactly when some $A_n =$

0). For the rest, $\prod_{n\le N} A_n = e^{B_N}$ with $B_N = \sum_{n\le N}\log(1+a_n)$, and $|\log(1+a_n)| \le 2|a_n|$ by Lemma 1.8(iii). So $B_N \to B$, and by continuity of exp, $\prod A_n \to e^B \neq 0$. For the last statement, $1/(1-a_n) = 1 + a_n/(1-a_n)$ and $|a_n/(1-a_n)| \le 2|a_n|$ once $|a_n| < 1/2$. ∎

Theorem 1.10

For every $s > 1$, $\zeta(s) = \prod_p \dfrac{1}{1-p^{-s}}$, the product being over all primes.

Proof. The product converges by Proposition 1.9 with $a_n = p_n^{-s}$, since $\sum_p p^{-s} \le \sum n^{-s} < \infty$. Let $p_1 < \cdots < p_N$ be the primes $\le M$, and expand each factor as a geometric series:

$$\prod_{j\le N} \frac{1}{1-p_j^{-s}} = \prod_{j\le N}\left(1 + p_j^{-s} + p_j^{-2s} + \cdots\right) = \sum_{n\in\mathcal{N}_M} n^{-s},$$

where by unique factorization (Theorem 1.6) $\mathcal{N}_M$ is the set of integers all of whose prime factors are $\le M$, each appearing once (absolutely convergent series may be multiplied out and rearranged). Since $\{1,\dots,M\} \subset \mathcal{N}_M$,

$$\sum_{n\le M} n^{-s} \le \prod_{p\le M} \frac{1}{1-p^{-s}} \le \zeta(s).$$

Let $M \to \infty$. ∎

Proposition 1.11

The series $\sum_p 1/p$ over all primes diverges.

Proof. Take logarithms in Theorem 1.10 (legitimate since the partial products converge to a positive limit and log is continuous): by Lemma 1.8(ii),

$$\log\zeta(s) = -\sum_p \log(1-p^{-s}) = \sum_p \left(p^{-s} + O(p^{-2s})\right) = \sum_p p^{-s} + O(1),$$

since $\sum_p p^{-2s} \le \sum n^{-2}$ for $s \ge 1$. As $s \to 1^+$, $\zeta(s) \to \infty$ by (8.1), so $\sum_p p^{-s} \to \infty$. Since $p^{-s} \le p^{-1}$, $\sum_p 1/p = \infty$. ∎

8.2 Dirichlet's theorem

Motivation. To count primes $p \equiv \ell \pmod q$ we must isolate the progression inside $\sum_p p^{-s}$. The indicator δ_ℓ of the class ℓ is a function on the group $\mathbb{Z}^*(q)$, so it expands in characters; each character gives a series $\sum \chi(p)p^{-s}$, and the theorem reduces to showing that these stay bounded for $\chi \neq \chi_0$, i.e. to the behaviour of the L-functions at $s = 1$.

Theorem 2.1 (Dirichlet)

If q and ℓ are relatively prime positive integers, then there are infinitely many primes of the form $\ell + kq$ with $k \in \mathbb{Z}$.

Following Euler, we shall show that $\sum_{p\equiv\ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$; the proof is completed in Section 8.3.

8.2.1 Fourier analysis, Dirichlet characters, and reduction of the theorem

$\mathbb{Z}^*(q)$ denotes the multiplicative group of residues mod q that are relatively prime to q (a group by Corollary 1.3); its order is Euler's $\varphi(q)$. Let δ_ℓ be the indicator of ℓ on $\mathbb{Z}^*(q)$. Since $\mathbb{Z}^*(q)$ is finite abelian, Theorem 2.7 gives

$$\delta_\ell(m) = \sum_{e \in \widehat{\mathbb{Z}^*(q)}} \hat{\delta}_\ell(e) e(m), \qquad \hat{\delta}_\ell(e) = \frac{1}{\varphi(q)} \sum_{m \in \mathbb{Z}^*(q)} \delta_\ell(m) \overline{e(m)} = \frac{1}{\varphi(q)} \overline{e(\ell)}.$$

For a character e of $\mathbb{Z}^*(q)$, the **Dirichlet character** $\chi = \chi_e$ is the function on $\mathbb{Z}$ with $\chi(m) = e(m \bmod q)$ if $\gcd(m,q) = 1$ and $\chi(m) = 0$ otherwise. The trivial Dirichlet character is χ_0 (from $e \equiv 1$). Extend δ_ℓ to $\mathbb{Z}$ by $\delta_\ell(m) = 1$ if $m \equiv \ell \pmod q$, 0 otherwise.

Lemma 2.2

The Dirichlet characters are multiplicative: $\chi(mn) = \chi(m)\chi(n)$ for all $m, n \in \mathbb{Z}$. Moreover

$$\delta_\ell(m) = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \chi(m),$$

where the sum is over all Dirichlet characters mod q.

Proof. If m, n are both prime to q, so is mn, and $\chi(mn) = e(mn) = e(m)e(n)$. If either is not, then neither is mn, and both sides vanish. The formula holds for m prime to q by the expansion above. If $\gcd(m,q) > 1$, both sides are 0: the left because ℓ is prime to q and so $m \not\equiv \ell$, the right because $\chi(m) = 0$. ∎

Hence, for $s > 1$,

$$\sum_{p \equiv \ell\,(q)} \frac{1}{p^s} = \sum_p \frac{\delta_\ell(p)}{p^s} = \frac{1}{\varphi(q)} \sum_{p \nmid q} \frac{1}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \frac{\chi(p)}{p^s}. \tag{8.2}$$

The first term on the right tends to ∞ as $s \to 1^+$ (Proposition 1.11; only finitely many primes divide q). So Dirichlet's theorem follows from:

Theorem 2.3

If χ is a nontrivial Dirichlet character, then $\sum_p \chi(p) p^{-s}$ remains bounded as $s \to 1^+$.

The proof occupies Section 8.3.

8.2.2 Dirichlet L-functions

For a Dirichlet character χ and $s > 1$ define

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Theorem 2.4 (Dirichlet product formula)

If $s > 1$, then

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \chi(p) p^{-s}}.$$

Proof. Identical to Theorem 1.10, using $|\chi(p)p^{-s}| \le p^{-s}$ (so the product converges by Proposition 3.2) and the multiplicativity of χ (Lemma 2.2): expanding $\prod_{p\le M}(1+\chi(p)p^{-s}+\chi(p)^2p^{-2s}+\cdots)$ gives $\sum_{n\in\mathcal{N}_M}\chi(n)n^{-s}$, since $\chi(p_1^{a_1}\cdots p_k^{a_k}) = \chi(p_1)^{a_1}\cdots\chi(p_k)^{a_k}$. The difference from $\sum_n \chi(n)n^{-s}$ is bounded by $\sum_{n>M} n^{-s} \to 0$. ∎

Formally, taking logarithms and using $\log(1+x) = x + O(x^2)$,

$$\log L(s,\chi) = -\sum_p \log\left(1-\chi(p)p^{-s}\right) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So if $L(1,\chi)$ is finite and nonzero, $\sum_p \chi(p)p^{-s}$ is bounded as $s \to 1^+$. Making this rigorous requires:

(a) a logarithm of complex numbers, since χ may be complex-valued;

(b) continuity of $L(s,\chi)$ down to $s=1$; and (c) the key fact $L(1,\chi) \neq 0$.

8.3 Proof of the theorem

Motivation. Three ingredients make the formal argument rigorous: a power-series logarithm $\log_1$ on the disc and a logarithm $\log_2$ of $L(s,\chi)$ defined by integrating L'/L; the extension of $L(s,\chi)$ to $s>0$ via the bounded partial sums of a nontrivial character; and the non-vanishing $L(1,\chi)\neq 0$, proved for complex χ by a positivity argument and for real χ by summation along hyperbolas.

8.3.1 Logarithms

For $|z|<1$ define

$$\log_1\left(\frac{1}{1-z}\right) = \sum_{k=1}^{\infty}\frac{z^k}{k}.$$

Proposition 3.1

The function $\log_1$ satisfies:
(i) If $|z|<1$, then $e^{\log_1(\frac{1}{1-z})} = \frac{1}{1-z}$.
(ii) If $|z|<1$, then $\log_1\left(\frac{1}{1-z}\right) = z + E_1(z)$, where $|E_1(z)| \le |z|^2$ if $|z| < 1/2$.
(iii) If $|z|<1/2$, then $\left|\log_1\left(\frac{1}{1-z}\right)\right| \le 2|z|$.

Proof. (i) Write $z = re^{i\theta}$, $0\le r<1$; it suffices to show $(1-re^{i\theta})\,e^{\sum_k (re^{i\theta})^k/k} = 1$. Differentiating in r gives

$$\Big[-e^{i\theta} + (1-re^{i\theta})\,e^{i\theta}\sum_{k\ge 1}(re^{i\theta})^{k-1}\Big]e^{\sum_k (re^{i\theta})^k/k} = \Big[-e^{i\theta} + (1-re^{i\theta})\frac{e^{i\theta}}{1-re^{i\theta}}\Big]e^{\cdots} = 0.$$

So the left side is constant in r, and at $r=0$ it equals 1.

(ii) $|E_1(z)| = \left|\sum_{k\ge 2} z^k/k\right| \le \frac{|z|^2}{2}\sum_{j\ge 0}|z|^j = \frac{|z|^2}{2(1-|z|)} \le |z|^2$ for $|z|<1/2$.

(iii) By (ii), $\left|\log_1(\frac{1}{1-z})\right| \le |z| + |z|^2 \le 2|z|$. ∎

Proposition 3.2

If $\sum|a_n|$ converges and $a_n \neq 1$ for all n, then $\prod_{n=1}^{\infty}\frac{1}{1-a_n}$ converges and is nonzero.

Proof. As in Proposition 1.9: for $n \geq n_0$ we have $|a_n| < 1/2$, and by Proposition 3.1(i),

$$\prod_{n_0 \leq n \leq N} \frac{1}{1-a_n} = \exp\Big(\sum_{n_0 \leq n \leq N} \log_1 \Big(\frac{1}{1-a_n} \Big) \Big).$$

The series in the exponent converges absolutely by (iii), so the partial products converge to $e^B \neq 0$. The finitely many factors with $n < n_0$ are finite and nonzero. ∎

8.3.2 L-functions

Proposition 3.3

Let χ_0 be the trivial Dirichlet character ($\chi_0(n) = 1$ if $\gcd(n,q) = 1$, 0 otherwise) and $q = p_1^{a_1} \cdots p_N^{a_N}$. Then

$$L(s,\chi_0) = (1-p_1^{-s})(1-p_2^{-s}) \cdots (1-p_N^{-s})\, \zeta(s).$$

Therefore $L(s,\chi_0) \to \infty$ as $s \to 1^+$.

Proof. By Theorems 2.4 and 1.10, $L(s,\chi_0) = \prod_{p \nmid q} (1-p^{-s})^{-1}$ is the Euler product for ζ with the factors for $p_1, \ldots, p_N$ removed. The last statement follows from (8.1). ∎

Proposition 3.4

If χ is a nontrivial Dirichlet character, then $\sum_{n \geq 1} \chi(n)/n^s$ converges for $s > 0$; we denote its sum by $L(s,\chi)$. Moreover:
(i) $L(s,\chi)$ is continuously differentiable for $0 < s < \infty$.
(ii) There are constants $c, c' > 0$ with $L(s,\chi) = 1 + O(e^{-cs})$ and $L'(s,\chi) = O(e^{-c's})$ as $s \to \infty$.

Lemma 3.5

If χ is a nontrivial Dirichlet character, then $\big|\sum_{n=1}^{k} \chi(n)\big| \leq q$ for all k.

Proof. First $S = \sum_{n=1}^{q} \chi(n) = 0$: for $a \in \mathbb{Z}^*(q)$, multiplicativity gives $\chi(a)S = \sum_n \chi(an) = S$, because $n \mapsto an$ permutes the residues mod q; and $\chi(a) \neq 1$ for some a. Since χ is q-periodic, writing $k = aq + b$ with $0 \leq b < q$ leaves $\sum_{n=1}^{k} \chi(n) = \sum_{aq < n \leq aq+b} \chi(n)$, at most q terms each of modulus ≤ 1. ∎

Proof of Proposition 3.4. Let $s_k = \sum_{n=1}^{k} \chi(n)$, $s_0 = 0$, so $|s_k| \leq q$. Summation by parts gives

$$\sum_{n=1}^{N} \frac{\chi(n)}{n^s} = \sum_{n=1}^{N} \frac{s_n - s_{n-1}}{n^s} = \sum_{n=1}^{N-1} s_n \Big[\frac{1}{n^s} - \frac{1}{(n+1)^s} \Big] + \frac{s_N}{N^s}.$$

Let $f(x) = x^{-s}$; by the mean value theorem $|n^{-s} - (n+1)^{-s}| \leq \sup_{[n,n+1]} |f'| \leq s\, n^{-s-1}$. For $s \geq \delta > 0$ the boundary term tends to 0, and the sum converges absolutely and uniformly on compact subsets of $[\delta, \infty)$, since $|s_n|\, s\, n^{-s-1} \leq qs\, n^{-\delta-1}$. So $L(s,\chi) = \sum_n s_n[n^{-s} - (n+1)^{-s}]$ is continuous for $s > 0$.

(i) The term-wise derivative is $\sum_n s_n[-\log n\, n^{-s} + \log(n+1)(n+1)^{-s}]$; with $g(x) = x^{-s} \log x$, $|g'(x)| \leq C_\delta x^{-1-\delta/2}$ for $s \geq \delta$, so this series also converges uniformly on compacts in $(0,\infty)$. Hence $L(\cdot,\chi) \in C^1(0,\infty)$.

(ii) $\chi(1) = 1$, so $|L(s,\chi) - 1| \leq \sum_{n \geq 2} n^{-s} \leq 2^{-s} + \int_2^\infty x^{-s} dx = 2^{-s}(1 + \frac{2}{s-1}) = O(e^{-cs})$ for $s \geq 2$, with any $c < \log 2$. Similarly $|L'(s,\chi)| \leq \sum_{n \geq 2} (\log n) n^{-s} = O(e^{-c's})$. ∎

We now define a logarithm of $L(s,\chi)$ for nontrivial χ and $s>1$ (where $L(s,\chi)\neq 0$ by Theorem 2.4 and Proposition 3.2): since $L(s,\chi)\to 1$ as $s\to\infty$, put

$$\log_2 L(s,\chi) = -\int_s^\infty \frac{L'(t,\chi)}{L(t,\chi)}\,dt,$$

which converges because $L'(t,\chi)/L(t,\chi) = O(e^{-c't})$. (Here $\log_2$ is just a name, not the base-2 logarithm.) If $L(1,\chi)\neq 0$, this integral extends continuously to $s=1$, so $\log_2 L(s,\chi)$ stays bounded as $s\to 1^+$.

Proposition 3.6

If $s>1$, then $e^{\log_2 L(s,\chi)} = L(s,\chi)$. Moreover

$$\log_2 L(s,\chi) = \sum_p \log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big).$$

Proof.

Step First identity. The derivative of $e^{-\log_2 L(s,\chi)}L(s,\chi)$ is $e^{-\log_2 L}(-\frac{L'}{L}L + L') = 0$, so it is constant. As $s\to\infty$, $\log_2 L\to 0$ and $L\to 1$, so the constant is 1.

Step Second identity. Exponentiating the right side and using Proposition 3.1(i) and Theorem 2.4,

$$e^{\sum_p \log_1(\frac{1}{1-\chi(p)p^{-s}})} = \prod_p \frac{1}{1-\chi(p)p^{-s}} = L(s,\chi).$$

So both sides have the same exponential, and their difference is $2\pi i M(s)$ with $M(s)\in\mathbb{Z}$. Both sides are continuous in $s>1$ (the series converges uniformly on $[\delta,\infty)$, $\delta>1$, by (iii) of Proposition 3.1). Hence M is continuous and integer-valued, thus constant. As $s\to\infty$ both sides tend to 0, so $M\equiv 0$. ∎

By Proposition 3.1(ii),

$$\log_2 L(s,\chi) = \sum_p \log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big) = \sum_p \frac{\chi(p)}{p^s} + O\Big(\sum_p \frac{1}{p^{2s}}\Big) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So if $L(1,\chi)\neq 0$, then $\log_2 L(s,\chi)$ is bounded as $s\to 1^+$, hence so is $\sum_p \chi(p)p^{-s}$. Theorem 2.3, and with it Dirichlet's theorem, therefore reduces to:

8.3.3 Non-vanishing of the L-function

Theorem 3.7

If $\chi\neq\chi_0$, then $L(1,\chi)\neq 0$.

The proof splits into two cases. A Dirichlet character is **real** if it takes only real values $(0,\pm 1)$ and **complex** otherwise; equivalently, χ is real iff $\chi=\bar\chi$.

Case I: complex Dirichlet characters

Lemma 3.8

If $s>1$, then $\prod_\chi L(s,\chi)\geq 1$, the product being over all Dirichlet characters mod q. In particular the product is real-valued.

Proof. By Proposition 3.6 (and its analogue for χ_0, proved the same way from the Euler product) $L(s,\chi) = \exp\big(\sum_p \log_1 \frac{1}{1-\chi(p)p^{-s}}\big)$. Hence

$$\prod_{\chi} L(s,\chi) = \exp\Big(\sum_{\chi}\sum_{p}\sum_{k\ge 1}\frac{1}{k}\frac{\chi(p^k)}{p^{ks}}\Big) = \exp\Big(\sum_{p}\sum_{k\ge 1}\frac{1}{kp^{ks}}\sum_{\chi}\chi(p^k)\Big),$$

using $\chi(p)^k = \chi(p^k)$; the rearrangement is justified by absolute convergence. By Lemma 2.2 with $\ell = 1$, $\sum_\chi \chi(p^k) = \varphi(q)\delta_1(p^k) \ge 0$. So the exponent is real and non-negative, and the product is ≥ 1. ∎

Lemma 3.9

The following hold:
(i) If $L(1,\chi) = 0$, then $L(1,\bar\chi) = 0$.
(ii) If χ is nontrivial and $L(1,\chi) = 0$, then $|L(s,\chi)| \le C|s-1|$ for $1 \le s \le 2$.
(iii) For the trivial character, $|L(s,\chi_0)| \le \frac{C}{|s-1|}$ for $1 < s \le 2$.

Proof. (i) $L(1,\bar\chi) = \overline{L(1,\chi)}$, since the series has real terms n^{-s} multiplying $\chi(n)$.

(ii) By the mean value theorem, since $L(\cdot,\chi)$ is C^1 on $(0,\infty)$ (Proposition 3.4): $|L(s,\chi) - L(1,\chi)| \le \sup_{[1,2]}|L'| \cdot |s-1|$.

(iii) From Proposition 3.3, $|L(s,\chi_0)| \le \zeta(s) \le 1 + \frac{1}{s-1} \le \frac{2}{s-1}$ by (8.1). ∎

Proof of Theorem 3.7 for complex χ. Suppose $L(1,\chi) = 0$. By Lemma 3.9(i), $L(1,\bar\chi) = 0$ too, and $\bar\chi \ne \chi$. So in $\prod_{\chi'} L(s,\chi')$ at least two factors are $O(|s-1|)$ as $s \to 1^+$ (Lemma 3.9(ii)), the factor for χ_0 is $O(1/|s-1|)$ (Lemma 3.9(iii)), and all other factors are bounded (they are continuous on $[1,2]$). Hence the product is $O(|s-1|) \to 0$, contradicting Lemma 3.8. ∎

Case II: real Dirichlet characters

For real χ only one factor $L(s,\chi)$ could vanish, which the pole of $L(s,\chi_0)$ can cancel; a different idea is needed. Dirichlet's method of *summation along hyperbolas* is first illustrated on the divisor problem.

Sums vs. integrals.

Proposition 3.10

If N is a positive integer, then:
(i) $\displaystyle\sum_{1\le n\le N}\frac{1}{n} = \int_1^N \frac{dx}{x} + O(1) = \log N + O(1)$.
(ii) More precisely, there is a real number γ, **Euler's constant**, with $\displaystyle\sum_{1\le n\le N}\frac{1}{n} = \log N + \gamma + O(1/N)$.

Proof. It suffices to prove (ii). Let $\gamma_n = \frac{1}{n} - \int_n^{n+1}\frac{dx}{x}$. Since $1/x$ is decreasing, $0 \le \gamma_n \le \frac{1}{n} - \frac{1}{n+1} \le \frac{1}{n^2}$, so $\sum\gamma_n$ converges to some γ, and $\sum_{n>N}\gamma_n \le \sum_{n>N} n^{-2} \le \int_N^\infty x^{-2}dx = \frac{1}{N}$. Then

$$\sum_{n=1}^{N}\frac{1}{n} - \int_1^N\frac{dx}{x} = \sum_{n=1}^{N}\gamma_n + \int_N^{N+1}\frac{dx}{x} = \gamma - \sum_{n>N}\gamma_n + \int_N^{N+1}\frac{dx}{x},$$

and the last two terms are $O(1/N)$. ∎

Proposition 3.11

If N is a positive integer, then

$$\sum_{1\le n\le N}\frac{1}{n^{1/2}}=\int_1^N\frac{dx}{x^{1/2}}+c'+O(N^{-1/2})=2N^{1/2}+c+O(N^{-1/2}).$$

Proof. [Supplement] Let $\gamma_n=n^{-1/2}-\int_n^{n+1}x^{-1/2}dx$. Then $0\le\gamma_n\le n^{-1/2}-(n+1)^{-1/2}\le\frac{1}{2}n^{-3/2}$ by the mean value theorem for $x^{-1/2}$. So $\sum\gamma_n=c'$ converges, with $\sum_{n>N}\gamma_n\le\frac{1}{2}\int_{N-1}^{\infty}x^{-3/2}dx=O(N^{-1/2})$. As before,

$$\sum_{n\le N}n^{-1/2}-\int_1^N x^{-1/2}dx=c'-\sum_{n>N}\gamma_n+\int_N^{N+1}x^{-1/2}dx=c'+O(N^{-1/2}),$$

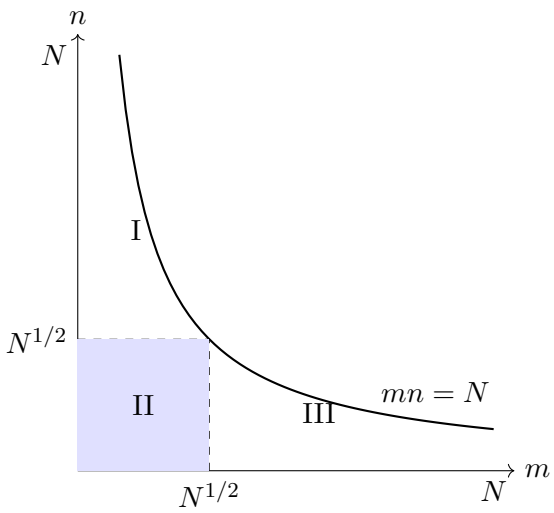
and $\int_1^N x^{-1/2}dx=2N^{1/2}-2$; take $c=c'-2$. ∎

Hyperbolic sums. For F defined on pairs of positive integers let $S_N=\sum\sum_{mn\le N}F(m,n)$. It can be computed in three ways:

(a) along hyperbolas: $S_N=\sum_{1\le k\le N}\left(\sum_{nm=k}F(m,n)\right)$;

(b) vertically: $S_N=\sum_{1\le m\le N}\left(\sum_{1\le n\le N/m}F(m,n)\right)$;

(c) horizontally: $S_N=\sum_{1\le n\le N}\left(\sum_{1\le m\le N/n}F(m,n)\right)$.



Split the region $\{mn\le N\}$ into $\mathrm{I}=\{1\le m<N^{1/2},\ N^{1/2}<n\le N/m\}$, $\mathrm{II}=\{1\le m\le N^{1/2},\ 1\le n\le N^{1/2}\}$, and $\mathrm{III}=\{N^{1/2}<m\le N/n,\ 1\le n<N^{1/2}\}$, with corresponding sums $S_{\mathrm{I}},S_{\mathrm{II}},S_{\mathrm{III}}$.

Let $d(k)$ be the number of positive divisors of k, so $d(k)=\sum_{nm=k}1$.

Theorem 3.12 (Dirichlet's divisor theorem)

For positive integers N,

$$\frac{1}{N}\sum_{k=1}^{N}d(k)=\log N+O(1).$$

More precisely, $\frac{1}{N}\sum_{k=1}^{N}d(k)=\log N+(2\gamma-1)+O(N^{-1/2})$.

Proof. Let $S_N=\sum_{k\le N}d(k)$, the hyperbolic sum of $F\equiv 1$.

Step First estimate. Summing vertically, $S_N = \sum_{m\le N}[N/m] = \sum_{m\le N}(N/m + O(1)) = N\sum_{m\le N}\frac{1}{m} + O(N)$, and Proposition 3.10(i) gives $S_N/N = \log N + O(1)$.

Step Refined estimate. By symmetry $S_{\mathrm{I}} = S_{\mathrm{III}}$, so $S_N = 2(S_{\mathrm{I}} + S_{\mathrm{II}}) - S_{\mathrm{II}}$. Summing vertically and using Proposition 3.10(ii),

$$S_{\mathrm{I}} + S_{\mathrm{II}} = \sum_{m\le N^{1/2}}[N/m] = N\sum_{m\le N^{1/2}}\frac{1}{m} + O(N^{1/2}) = N\log N^{1/2} + N\gamma + O(N^{1/2}).$$

Also $S_{\mathrm{II}} = [N^{1/2}]^2 = N + O(N^{1/2})$. Hence $S_N = N\log N + (2\gamma - 1)N + O(N^{1/2})$; divide by N. ∎

Non-vanishing for real characters. Let χ be a nontrivial real Dirichlet character, $F(m,n) = \frac{\chi(n)}{(nm)^{1/2}}$, and $S_N = \sum\sum_{mn\le N} F(m,n)$.

Proposition 3.13

The following hold:
(i) $S_N \ge c\log N$ for some constant $c > 0$.
(ii) $S_N = 2N^{1/2}L(1,\chi) + O(1)$.

This proves Theorem 3.7 for real χ: if $L(1,\chi) = 0$, then (ii) says $S_N = O(1)$, contradicting (i).

Summing along hyperbolas, $\sum_{nm=k}\frac{\chi(n)}{(nm)^{1/2}} = \frac{1}{k^{1/2}}\sum_{n|k}\chi(n)$. Part (i) follows from:

Lemma 3.14

For every $k \ge 1$, $\sum_{n|k}\chi(n) \ge 0$, and $\sum_{n|k}\chi(n) \ge 1$ if $k = \ell^2$ for some $\ell \in \mathbb{Z}$.

Proof.

Step Prime powers. If $k = p^a$, the divisors are $1, p, \dots, p^a$, and $\sum_{n|k}\chi(n) = 1 + \chi(p) + \chi(p)^2 + \cdots + \chi(p)^a$. Since $\chi(p) \in \{0, 1, -1\}$, this equals $a + 1$ if $\chi(p) = 1$; 1 if $\chi(p) = -1$ and a is even; 0 if $\chi(p) = -1$ and a is odd; and 1 if $\chi(p) = 0$ (i.e. $p \mid q$). It is always ≥ 0, and ≥ 1 when a is even.

Step General k. If $k = p_1^{a_1}\cdots p_N^{a_N}$, every divisor is $p_1^{b_1}\cdots p_N^{b_N}$ with $0 \le b_j \le a_j$, so by multiplicativity

$$\sum_{n|k}\chi(n) = \prod_{j=1}^{N}(1 + \chi(p_j) + \cdots + \chi(p_j^{a_j})).$$

Each factor is ≥ 0; if k is a square all a_j are even and each factor is ≥ 1. ∎

Proof of Proposition 3.13(i). By the lemma, $S_N = \sum_{k\le N} k^{-1/2}\sum_{n|k}\chi(n) \ge \sum_{\ell\le N^{1/2}}\frac{1}{(\ell^2)^{1/2}} = \sum_{\ell\le N^{1/2}}\frac{1}{\ell} \ge c\log N$, by Proposition 3.10(i). ∎

Lemma 3.15

For all integers $0 < a < b$:
(i) $\displaystyle\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}} = O(a^{-1/2})$;
(ii) $\displaystyle\sum_{n=a}^{b}\frac{\chi(n)}{n} = O(a^{-1})$.

Proof. Summation by parts as in Proposition 3.4, with $s_n = \sum_{k\le n}\chi(k)$, $|s_n|\le q$ (Lemma 3.5):

$$\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}} = \sum_{n=a}^{b-1} s_n\big[n^{-1/2}-(n+1)^{-1/2}\big] + \frac{s_b}{b^{1/2}} - \frac{s_{a-1}}{a^{1/2}},$$
$$\Big|\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}}\Big| \le q\sum_{n=a}^{b-1}\big[n^{-1/2}-(n+1)^{-1/2}\big] + \frac{2q}{a^{1/2}} \le \frac{3q}{a^{1/2}},$$

since the sum telescopes. For (ii) the same argument with n^{-1} in place of $n^{-1/2}$ gives the bound $3q/a$. ∎

Proof of Proposition 3.13(ii). Write $S_N = S_{\mathrm{I}} + (S_{\mathrm{II}} + S_{\mathrm{III}})$.

Step Region I, summed vertically.

$$S_{\mathrm{I}} = \sum_{m<N^{1/2}}\frac{1}{m^{1/2}}\Big(\sum_{N^{1/2}<n\le N/m}\frac{\chi(n)}{n^{1/2}}\Big).$$

By Lemma 3.15(i) the inner sum is $O(N^{-1/4})$, and by Proposition 3.11, $\sum_{m<N^{1/2}} m^{-1/2} = O(N^{1/4})$. So $S_{\mathrm{I}} = O(1)$.

Step Regions II and III, summed horizontally.

$$S_{\mathrm{II}}+S_{\mathrm{III}} = \sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}}\Big(\sum_{m\le N/n}\frac{1}{m^{1/2}}\Big) = \sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}}\Big(2\frac{N^{1/2}}{n^{1/2}} + c + O\Big(\frac{n^{1/2}}{N^{1/2}}\Big)\Big)$$

by Proposition 3.11. Split into three sums:

$$2N^{1/2}\sum_{n\le N^{1/2}}\frac{\chi(n)}{n} + c\sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}} + O\Big(\frac{1}{N^{1/2}}\sum_{n\le N^{1/2}}1\Big).$$

The third is $O(1)$. The second is $O(1)$ by Lemma 3.15(i) with $a=1$. For the first, Lemma 3.15(ii) gives $\sum_{n\le N^{1/2}}\frac{\chi(n)}{n} = L(1,\chi) - \sum_{n>N^{1/2}}\frac{\chi(n)}{n} = L(1,\chi) + O(N^{-1/2})$. Altogether $S_N = 2N^{1/2}L(1,\chi)+O(1)$. ∎

This completes the proof of Theorem 3.7, hence of Theorem 2.3, and by (8.2) $\sum_{p\equiv \ell\,(q)} p^{-s}\to\infty$ as $s\to 1^+$: there are infinitely many primes $p\equiv \ell \pmod q$, which is Dirichlet's Theorem 2.1.

Appendix. Integration

A.1 Definition of the Riemann integral

Motivation. The whole book uses only the Riemann integral. This appendix records its definition and basic properties, the approximation of integrable functions by continuous ones (used for Fourier series of integrable functions), and Lebesgue's criterion: a bounded function is integrable exactly when its discontinuities form a set of measure zero.

Let f be a bounded real-valued function on $[a,b]$. A **partition** P of $[a,b]$ is a finite sequence $a = x_0 < x_1 < \cdots < x_N = b$; let $I_j = [x_{j-1}, x_j]$ and $|I_j| = x_j - x_{j-1}$. The upper and lower sums are

$$\mathcal{U}(P,f) = \sum_{j=1}^{N} \Big[\sup_{x\in I_j} f(x)\Big]|I_j|, \qquad \mathcal{L}(P,f) = \sum_{j=1}^{N} \Big[\inf_{x\in I_j} f(x)\Big]|I_j|.$$

A partition P' is a **refinement** of P if $P' \supset P$. Adding points does not increase $\mathcal{U}$ nor decrease $\mathcal{L}$: if P' refines P,

$$\mathcal{L}(P,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P,f).$$

(Splitting I_j into two pieces replaces $\sup_{I_j} f \cdot |I_j|$ by a sum of two terms with smaller or equal sups.) Hence for any P_1, P_2, the common refinement $P' = P_1 \cup P_2$ gives $\mathcal{L}(P_2,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P_1,f)$: every lower sum is $\le$ every upper sum. With $\mathcal{U} = \inf_P \mathcal{U}(P,f)$ and $\mathcal{L} = \sup_P \mathcal{L}(P,f)$ we get $\mathcal{L} \le \mathcal{U}$.

Definition

f is **(Riemann) integrable** if for every $\epsilon > 0$ there is a partition P with $\mathcal{U}(P,f) - \mathcal{L}(P,f) < \epsilon$. Then $\mathcal{U} = \mathcal{L}$, and this common value is $\int_a^b f(x)\,dx$. For complex $f = u + iv$, f is integrable if u and v are, and $\int f = \int u + i\int v$.

Equivalently, with Riemann sums $\mathcal{R}(P,f) = \sum f(x_j^*)|I_j|$ $(x_j^* \in I_j)$ and mesh $\max |I_j|$, f is integrable iff $\mathcal{R}(P,f)$ converges as the mesh tends to 0. Continuous functions are integrable: by uniform continuity, for small mesh $\sup_{I_j} f - \inf_{I_j} f < \epsilon$ on every I_j, so $\mathcal{U} - \mathcal{L} \le \epsilon(b-a)$.

A.1.1 Basic properties

Proposition 1.1

If f and g are integrable on $[a,b]$, then:
(i) $f+g$ is integrable, and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
(ii) If $c \in \mathbb{C}$, then cf is integrable and $\int_a^b cf = c\int_a^b f$.
(iii) If f, g are real-valued and $f \le g$, then $\int_a^b f \le \int_a^b g$.
(iv) If $c \in [a,b]$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof. (i) Assume f, g real. For any P, $\mathcal{U}(P, f+g) \le \mathcal{U}(P,f) + \mathcal{U}(P,g)$ and $\mathcal{L}(P, f+g) \ge \mathcal{L}(P,f) + \mathcal{L}(P,g)$, since the sup of a sum is at most the sum of the sups. Given ϵ, take P_1, P_2 with $\mathcal{U} - \mathcal{L} < \epsilon$ for f and g respectively, and a common refinement P_0. Then $\mathcal{U}(P_0, f+g) - \mathcal{L}(P_0, f+g) < 2\epsilon$, so $f+g$ is integrable. Its integral I satisfies

$$I \le \mathcal{U}(P_0, f+g) \le \mathcal{U}(P_0,f) + \mathcal{U}(P_0,g) \le \int f + \int g + 2\epsilon,$$

and similarly $I \ge \int f + \int g - 2\epsilon$ from the lower sums.

(ii) For real $c \geq 0$, $\mathcal{U}(P, cf) = c\mathcal{U}(P, f)$ and $\mathcal{L}(P, cf) = c\,\mathcal{L}(P, f)$. For $c = -1$, $\mathcal{U}(P, -f) = -\mathcal{L}(P, f)$ and $\mathcal{L}(P, -f) = -\mathcal{U}(P, f)$. Complex c then follows by splitting into real and imaginary parts, using (i).

(iii) $g - f \geq 0$ gives $\mathcal{L}(P, g - f) \geq 0$, so $\int (g - f) \geq 0$; apply (i) and (ii).

(iv) Refining by the point c, we may take partitions containing c; then $\mathcal{U}(P, f) = \mathcal{U}(P', f) + \mathcal{U}(P'', f)$ with P', P'' the induced partitions of $[a, c]$ and $[c, b]$, and the same for $\mathcal{L}$. Hence $\mathcal{U} - \mathcal{L}$ on each piece is $< \epsilon$, and the integrals add. ■

Consequences: if f is integrable then so is $|f|$, and $\left|\int f\right| \leq \int |f|$ (Lemma 1.2 with $\varphi(x) = |x|$; for complex f use $\varphi(u, v)$ on the bounded range, and write $\int f = e^{i\theta} \left|\int f\right|$, so $\left|\int f\right| = \int \mathrm{Re}(e^{-i\theta} f) \leq \int |f|$). Also the product of two integrable functions is integrable, since $fg = \frac{1}{4}[(f + g)^2 - (f - g)^2]$ and squares are integrable by Lemma 1.2.

Lemma 1.2

If f is real-valued and integrable on $[a, b]$, and φ is a real-valued continuous function on $\mathbb{R}$, then $\varphi \circ f$ is integrable on $[a, b]$.

Proof. Let $\epsilon > 0$ and $|f| \leq M$. Since φ is uniformly continuous on $[-M, M]$, choose $\delta > 0$ (with $\delta < \epsilon$) so that $|y_1 - y_2| < \delta$ implies $|\varphi(y_1) - \varphi(y_2)| < \epsilon$. Choose P with $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \delta^2$. Let I_j be the subintervals of P, and split the indices into Λ, where $\sup_{I_j} f - \inf_{I_j} f < \delta$, and Λ', the rest. On I_j with $j \in \Lambda$, the oscillation of $\varphi \circ f$ is $\leq \epsilon$. For $j \in \Lambda'$,

$$\delta \sum_{j \in \Lambda'} |I_j| \leq \sum_{j \in \Lambda'} \Big(\sup_{I_j} f - \inf_{I_j} f \Big) |I_j| < \delta^2,$$

so $\sum_{\Lambda'} |I_j| < \delta$. With $B = \sup_{[-M,M]} |\varphi|$,

$$\mathcal{U}(P, \varphi \circ f) - \mathcal{L}(P, \varphi \circ f) \leq \epsilon(b - a) + 2B\delta < \epsilon(b - a + 2B).$$

■

Proposition 1.3

A bounded monotonic function f on $[a, b]$ is integrable.

Proof. We may take $a = 0$, $b = 1$ and f increasing. For the uniform partition P_N with $x_j = j/N$,

$$\mathcal{U}(P_N, f) - \mathcal{L}(P_N, f) = \sum_{j=1}^{N} \frac{1}{N} [f(x_j) - f(x_{j-1})] = \frac{f(1) - f(0)}{N} \to 0.$$

■

Proposition 1.4

Let f be a bounded function on $[a, b]$. If $c \in (a, b)$, and for all small $\delta > 0$ the function f is integrable on $[a, c - \delta]$ and $[c + \delta, b]$, then f is integrable on $[a, b]$.

Proof. Let $|f| \leq M$ and $\epsilon > 0$. Choose $\delta > 0$ with $4\delta M \leq \epsilon/3$, then partitions P_1 of $[a, c - \delta]$ and P_2 of $[c + \delta, b]$ with $\mathcal{U} - \mathcal{L} < \epsilon/3$ on each. Let $P = P_1 \cup P_2$, a partition of $[a, b]$ that has $[c - \delta, c + \delta]$ as one subinterval. Then

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \leq (\mathcal{U}(P_1, f) - \mathcal{L}(P_1, f)) + 2M \cdot 2\delta + (\mathcal{U}(P_2, f) - \mathcal{L}(P_2, f)) < \epsilon.$$

■

By induction, a bounded function on $[a,b]$ that is continuous except at finitely many points is integrable.

A function on the circle (equivalently, a 2π-periodic function on $\mathbb{R}$) is **integrable** if it is integrable on every finite interval, i.e. on $[-\pi,\pi]$.

Lemma 1.5

Suppose f is integrable on the circle and $|f| \le B$. Then there is a sequence $\{f_k\}$ of continuous functions on the circle with

$$\sup_{x\in[-\pi,\pi]} |f_k(x)| \le B \quad \text{for all } k, \qquad \int_{-\pi}^{\pi} |f(x) - f_k(x)|\,dx \to 0 \quad \text{as } k \to \infty.$$

Proof. Assume f real (otherwise treat the real and imaginary parts separately; the bound becomes $2B$, and one can then radially truncate to B without increasing the error, since $z \mapsto$ the nearest point of $\{|w| \le B\}$ is 1-Lipschitz).

Step Step function. Given ϵ, choose a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $\mathcal{U} - \mathcal{L} < \epsilon$, and let $f^*(x) = \sup_{x_{j-1}\le y\le x_j} f(y)$ for $x \in [x_{j-1}, x_j)$. Then $|f^*| \le B$ and

$$\int_{-\pi}^{\pi} |f^* - f| = \int_{-\pi}^{\pi} (f^* - f) \le \mathcal{U} - \mathcal{L} < \epsilon.$$

Step Smoothing the jumps. For small $\delta > 0$ let $\tilde f = f^*$ at distance $\ge \delta$ from every x_j. Near x_j $(1 \le j \le N-1)$ let $\tilde f$ be linear on $[x_j - \delta, x_j + \delta]$ with $\tilde f(x_j \pm \delta) = f^*(x_j \pm \delta)$. Near $-\pi$, let $\tilde f$ be linear on $[-\pi, -\pi + \delta]$ with $\tilde f(-\pi) = 0$, and similarly near π. Then $\tilde f(-\pi) = \tilde f(\pi) = 0$, so $\tilde f$ extends to a continuous 2π-periodic function with $|\tilde f| \le B$. It differs from f^* only on N intervals of length $\le 2\delta$, so $\int |f^* - \tilde f| \le 2B \cdot N \cdot 2\delta < \epsilon$ for δ small.

Hence $\int |f - \tilde f| < 2\epsilon$. Take $\epsilon = 1/k$ and $f_k = \tilde f$. ∎

A.1.2 Sets of measure zero and discontinuities of integrable functions

A set $E \subset \mathbb{R}$ has **measure** 0 if for every $\epsilon > 0$ there are countably many open intervals I_k with $E \subset \bigcup I_k$ and $\sum |I_k| < \epsilon$. Countable sets have measure 0 (cover the k-th point by an interval of length $\epsilon 2^{-k}$). If E is compact and has measure 0, finitely many of the I_k already cover E.

Lemma 1.6

The union of countably many sets of measure 0 has measure 0.

Proof. Let $E = \bigcup_i E_i$. Given ϵ, cover E_i by intervals $\{I_{k,i}\}_k$ with $\sum_k |I_{k,i}| < \epsilon/2^i$. Then $\{I_{k,i}\}_{k,i}$ is a countable cover of E with total length $< \sum_i \epsilon 2^{-i} = \epsilon$. ∎

Theorem 1.7 (Lebesgue)

A bounded function f on $[a,b]$ is integrable if and only if its set of discontinuities has measure 0.

Let $J = [a,b]$ and $I(c,r) = (c-r, c+r)$. The **oscillation** of f on $I(c,r)$ is $\operatorname{osc}(f,c,r) = \sup |f(x) - f(y)|$ over $x, y \in J \cap I(c,r)$, and the oscillation at c is $\operatorname{osc}(f,c) = \lim_{r\to 0} \operatorname{osc}(f,c,r)$ (a decreasing limit of non-negative numbers). Clearly f is continuous at c iff $\operatorname{osc}(f,c) = 0$. For $\epsilon > 0$ let $A_\epsilon = \{c \in J : \operatorname{osc}(f,c) \ge \epsilon\}$. Then the set of discontinuities is $D = \bigcup_{\epsilon>0} A_\epsilon = \bigcup_n A_{1/n}$.

Lemma 1.8

For $\epsilon > 0$, the set A_ϵ is closed, hence compact.

Proof. Suppose $c_n \in A_\epsilon$, $c_n \to c$, and $c \notin A_\epsilon$. Write $\operatorname{osc}(f, c) = \epsilon - \delta$ with $\delta > 0$. Choose r with $\operatorname{osc}(f, c, r) < \epsilon - \delta/2$ and n with $|c_n - c| < r/2$. Since $I(c_n, r/2) \subset I(c, r)$, $\operatorname{osc}(f, c_n, r/2) < \epsilon$, so $\operatorname{osc}(f, c_n) < \epsilon$, a contradiction. ∎

Proof of Theorem 1.7. Let $|f| \le M$.

Step Measure zero ⇒ integrable. Let $\epsilon > 0$. Since $A_\epsilon \subset D$ is compact with measure 0, it is covered by finitely many open intervals $I_1, \dots, I_N$ of total length $< \epsilon$. The complement in J of their union is compact, and each of its points z has an open interval F_z with $\sup_{x,y \in J \cap F_z} |f(x) - f(y)| \le \epsilon$ (since $z \notin A_\epsilon$, after shrinking). Choose a finite subcover $I_{N+1}, \dots, I_{N'}$ of these. The endpoints of $I_1, \dots, I_{N'}$ in J give a partition P of J. Each of its subintervals lies in some I_k: in the closure of one with $k \le N$ (contributing at most $2M$ times length) or in one with $k > N$ (oscillation $\le \epsilon$). Hence

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \le 2M \sum_{j=1}^{N} |I_j| + \epsilon(b - a) \le (2M + b - a)\epsilon.$$

Step Integrable ⇒ measure zero. By Lemma 1.6 it suffices to show each $A_{1/n}$ has measure 0. Given ϵ, take $P = \{x_0, \dots, x_N\}$ with $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \epsilon/n$. If $A_{1/n}$ meets the open interval $I_j = (x_{j-1}, x_j)$, then $\sup_{I_j} f - \inf_{I_j} f \ge 1/n$ there, so

$$\frac{1}{n} \sum_{j:\, I_j \cap A_{1/n} \neq \emptyset} |I_j| \le \mathcal{U}(P, f) - \mathcal{L}(P, f) < \frac{\epsilon}{n}.$$

Slightly enlarging these intervals, and adding tiny intervals around the finitely many points x_j, covers $A_{1/n}$ by open intervals of total length $\le 2\epsilon$. ∎

(Incidentally this gives another proof that fg is integrable when f and g are: the discontinuities of fg lie in the union of those of f and g.)

A.2 Multiple integrals

Motivation. Integrals over rectangles in $\mathbb{R}^d$ are defined exactly as in one variable. For continuous functions we need three facts: iterated integration (Fubini), the change of variables formula with the Jacobian factor $|\det Dg|$, and its special case, spherical coordinates.

A.2.1 The Riemann integral in $\mathbb{R}^d$

A closed rectangle is $R = [a_1, b_1] \times \cdots \times [a_d, b_d]$. If P_j is a partition of $[a_j, b_j]$, then $P = (P_1, \dots, P_d)$ is a partition of R, whose subrectangles $S = S_1 \times \cdots \times S_d$ have volume $|S| = |S_1| \cdots |S_d|$. Upper and lower sums

$$\mathcal{U}(P, f) = \sum_S \Big[\sup_S f\Big]|S|, \qquad \mathcal{L}(P, f) = \sum_S \Big[\inf_S f\Big]|S|,$$

refinements, integrability and $\int_R f(x)\,dx$ are defined exactly as in one dimension, with the same monotonicity under refinement. Continuous functions on R are integrable, by uniform continuity. If f is continuous on a closed ball B (or on a set whose boundary has measure zero in the d-dimensional sense), let $g = f$ on B and $g = 0$ outside. Then g is integrable on any rectangle $R \supset B$, its discontinuities lying on ∂B, and $\int_B f = \int_R g$ is independent of R.

A.2.2 Repeated integrals

Theorem 2.1 (Fubini, continuous case)

Let f be continuous on a closed rectangle $R = R_1 \times R_2 \subset \mathbb{R}^d$, $R_i \subset \mathbb{R}^{d_i}$, $d = d_1 + d_2$. Writing $x = (x_1, x_2)$, the function $F(x_1) = \int_{R_2} f(x_1, x_2)\, dx_2$ is continuous on R_1, and

$$\int_R f(x)\, dx = \int_{R_1} \Big(\int_{R_2} f(x_1, x_2)\, dx_2 \Big) dx_1.$$

Proof.

Step Continuity. $|F(x_1) - F(x_1')| \le \int_{R_2} |f(x_1, x_2) - f(x_1', x_2)|\, dx_2 \le |R_2| \sup_{x_2} |f(x_1, x_2) - f(x_1', x_2)|$, which is small when $|x_1 - x_1'|$ is small, by uniform continuity of f on R.

Step Identity. Let P_1, P_2 be partitions of R_1, R_2, with subrectangles S and T. Then

$$\sup_{S\times T} f \ge \sup_{x_1 \in S} \Big(\sup_{x_2 \in T} f(x_1, x_2) \Big), \qquad \inf_{S\times T} f \le \inf_{x_1 \in S} \Big(\inf_{x_2 \in T} f(x_1, x_2) \Big).$$

For each x_1, $\mathcal{L}(P_2, f(x_1, \cdot)) \le F(x_1) \le \mathcal{U}(P_2, f(x_1, \cdot))$, i.e.

$$\sum_T \Big[\inf_{x_2 \in T} f(x_1, x_2) \Big] |T| \le F(x_1) \le \sum_T \Big[\sup_{x_2 \in T} f(x_1, x_2) \Big] |T|.$$

Taking $\sup_{x_1 \in S}$ of the right side and using the first inequality,

$$\mathcal{U}(P_1, F) \le \sum_S \sum_T \Big[\sup_{S\times T} f \Big] |T||S| = \mathcal{U}(P, f),$$

and likewise $\mathcal{L}(P, f) \le \mathcal{L}(P_1, F)$, for $P = (P_1, P_2)$. Since f is integrable, $\mathcal{U}(P, f) - \mathcal{L}(P, f)$ can be made arbitrarily small, and so $\int_{R_1} F = \int_R f$. ∎

By induction, $\int_R f = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f\, dx_d \cdots dx_1$, in any order.

A.2.3 The change of variables formula

A map $g : A \to B$ between compact sets is a C^1 **diffeomorphism** if it is a bijection that extends to a C^1 map on an open neighbourhood of A, its inverse likewise extends to a C^1 map near B, and $\det Dg \neq 0$ on A.

Theorem 2.2 (change of variables)

Suppose A and B are compact subsets of $\mathbb{R}^d$ and $g : A \to B$ is a C^1 diffeomorphism. If f is continuous on B, then

$$\int_{g(A)} f(x)\, dx = \int_A f(g(y))\, |\det(Dg)(y)|\, dy.$$

Proof. [Supplement] We assume (as is implicit in the statement) that A and B have boundaries of measure zero, so both integrals are defined. Write $\mathrm{vol}(E) = \int_E 1$ for such sets E.

Step Linear maps. If L is invertible linear and R is a rectangle, then $\mathrm{vol}(L(R)) = |\det L|\, \mathrm{vol}(R)$. Both sides are multiplicative in L (the left because $\mathrm{vol}(L(E)) = c_L \mathrm{vol}(E)$ for all such E, by approximating E with rectangles). So it suffices to check this for elementary matrices. A permutation of coordinates or a scaling $x_i \mapsto c x_i$ maps rectangles to rectangles with the right volume. For a shear $x_1 \mapsto x_1 + t x_2$, Fubini (Theorem 2.1) shows that each slice in x_1 is just translated, so volume is preserved and $\det = 1$.

Step Local estimate. Fix $\epsilon \in (0,1)$. By uniform continuity of Dg and $(Dg)^{-1}$ on A, there is $\eta > 0$ such that for every cube Q in A of side $< \eta$ with centre a,

$$g(a) + Dg(a)\big((1-\epsilon)(Q-a)\big) \subset g(Q) \subset g(a) + Dg(a)\big((1+\epsilon)(Q-a)\big). \tag{A.1}$$

Indeed, $h = Dg(a)^{-1}(g - g(a))$ satisfies $|h(y) - (y-a)| \le \epsilon' |y-a|$ near a by the mean value inequality, with ϵ' uniform in a. Such an h maps $Q-a$ into $(1+\epsilon)(Q-a)$. Since h is a homeomorphism onto its image and moves the boundary of Q by less than the gap, the image covers $(1-\epsilon)(Q-a)$. By the linear case and (A.1),

$$(1-\epsilon)^d |\det Dg(a)|\,|Q| \le \operatorname{vol}(g(Q)) \le (1+\epsilon)^d |\det Dg(a)|\,|Q|.$$

Step Summation. Assume first $f \ge 0$; general f follows by linearity, writing f as a difference of non-negative continuous functions plus a constant. Partition a rectangle containing A into cubes Q_k of side $< \eta$. Let $\mathcal{K}$ be the cubes contained in A; the others meet ∂A and have total volume $\to 0$ as $\eta \to 0$, and so do their images, since g is Lipschitz. The images $g(Q_k)$ overlap only on boundaries, which have measure zero, so

$$\int_{g(A)} f = \sum_{k \in \mathcal{K}} \int_{g(Q_k)} f + o(1) = \sum_{k \in \mathcal{K}} f(g(a_k)) \operatorname{vol}(g(Q_k)) + o(1),$$

with a_k the centre of Q_k, using uniform continuity of f. By the local estimate this lies within a factor $(1 \pm \epsilon)^d$ of $\sum_k f(g(a_k))|\det Dg(a_k)||Q_k|$, which is a Riemann sum for $\int_A f(g(y))|\det Dg(y)|dy$. Letting $\eta \to 0$ and then $\epsilon \to 0$ gives the formula. ∎

The factor $|\det Dg|$ is the new infinitesimal volume element: for linear $g = L$, $|L(R)| = |\det L|\,|R|$.

A.2.4 Spherical coordinates

Spherical coordinates in $\mathbb{R}^d$ are $x = g(r, \theta_1, \dots, \theta_{d-1})$ with

$$\begin{aligned} x_1 &= r \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2} \cos\theta_{d-1}, \\ x_2 &= r \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2} \sin\theta_{d-1}, \\ &\vdots \\ x_{d-1} &= r \sin\theta_1 \sin\theta_2, \qquad x_d = r\cos\theta_1, \end{aligned}$$

where $0 \le \theta_i \le \pi$ for $1 \le i \le d-2$ and $0 \le \theta_{d-1} \le 2\pi$. The Jacobian determinant is $r^{d-1} \sin^{d-2}\theta_1 \sin^{d-3}\theta_2 \cdots \sin\theta_{d-2}$. Every $x \ne 0$ is uniquely $x = r\gamma$ with $r > 0$ and $\gamma \in S^{d-1}$. Define

$$\int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) = \int_0^\pi \cdots \int_0^\pi \int_0^{2\pi} f(g(1,\theta)) \sin^{d-2}\theta_1 \cdots \sin\theta_{d-2}\, d\theta_{d-1} \cdots d\theta_1.$$

Then the change of variables formula gives, for the ball $B(0,N)$,

$$\int_{B(0,N)} f(x)\, dx = \int_{S^{d-1}} \int_0^N f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma). \tag{A.2}$$

(The map g fails to be a diffeomorphism on a set of measure zero, which is handled by applying Theorem 2.2 on $\{\delta \le r,\ \delta \le \theta_i \le \pi - \delta\}$ and letting $\delta \to 0$.) The area of the unit sphere is $\omega_d = \int_{S^{d-1}} d\sigma$.

For the annulus $A(R_1, R_2) = \{R_1 \le |x| \le R_2\}$ and $\lambda \in \mathbb{R}$, (A.2) gives

$$\int_{A(R_1,R_2)} |x|^\lambda\, dx = \omega_d \int_{R_1}^{R_2} r^{\lambda+d-1} dr = \begin{cases} \dfrac{\omega_d}{\lambda+d}(R_2^{\lambda+d} - R_1^{\lambda+d}), & \lambda \ne -d, \\ \omega_d \log(R_2/R_1), & \lambda = -d. \end{cases}$$

Consequences: the integral of $|x|^\lambda$ over the ball $|x| \le 1$ is finite as $R_1 \to 0$ iff $\lambda > -d$, and over $|x| \ge 1$ it is finite as $R_2 \to \infty$ iff $\lambda < -d$.

A.3 Improper integrals. Integration over $\mathbb{R}^d$

Motivation. Fourier transforms require integrals over all of $\mathbb{R}^d$. For continuous functions of moderate decrease the integral is a limit over growing cubes (or balls); Fubini and spherical coordinates extend to this setting.

A.3.1 Integration of functions of moderate decrease

Let $Q_N = \{|x_j| \le N/2,\ 1 \le j \le d\}$. For continuous f on $\mathbb{R}^d$, if $\lim_{N\to\infty} \int_{Q_N} f$ exists we denote it by $\int_{\mathbb{R}^d} f(x)\,dx$. A continuous f is of **moderate decrease** if

$$|f(x)| \le \frac{A}{1+|x|^{d+1}} \quad \text{for some } A > 0.$$

(For $d = 1$ this is the definition of Chapter 5; the Poisson kernel $\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2}$ is an example.)

For such f the limit exists. Let $I_N = \int_{Q_N} f$. For $M > N$, $Q_M - Q_N$ lies in the annulus $A(aN, bM)$ with $a = 1/2$ and $b = \sqrt{d}/2$, since $\{|x| < N/2\} \subset Q_N \subset \{|x| \le \sqrt{d}N/2\}$. So, by the annulus formula with $\lambda = -d-1$,

$$|I_M - I_N| \le A \int_{aN \le |x| \le bM} |x|^{-d-1} dx = A\omega_d\Big(\frac{1}{aN} - \frac{1}{bM}\Big) \le \frac{C}{N}.$$

Thus $\{I_N\}$ is Cauchy. The same argument shows that $\lim_N \int_{B_N} f$ over balls exists and has the same value, since $|\int_{Q_N} f - \int_{B_{N/2}} f| \le C/N$.

The integral over $\mathbb{R}^d$ is linear, and is invariant under translations and rotations: $\int f(x+h)\,dx = \int f(x)\,dx$, and $\int f(R(x))\,dx = \int f(x)\,dx$ for rotations R. It also satisfies $\int f(\delta x)\,dx = \delta^{-d}\int f(x)\,dx$ for $\delta > 0$. Each follows from the corresponding statement on cubes or balls (Theorem 2.2, with $|\det| = 1$ or δ^d) by letting $N \to \infty$; translates of Q_N differ from Q_N by a set where $|f|$ contributes $O(1/N)$.

A.3.2 Repeated integrals

Theorem 3.1

Suppose f is continuous on $\mathbb{R}^2$ and of moderate decrease. Then $F(x_1) = \int_{\mathbb{R}} f(x_1, x_2)\,dx_2$ is of moderate decrease on $\mathbb{R}$, and

$$\int_{\mathbb{R}^2} f(x)\,dx = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} f(x_1, x_2)\,dx_2\Big)dx_1.$$

Proof.

Step Decrease of F. With $|f(x)| \le A/(1+|x|^3)$, split $|F(x_1)| \le \int_{|x_2|\le|x_1|} + \int_{|x_2|\ge|x_1|}$ of $A\,dx_2/(1+(x_1^2+x_2^2)^{3/2})$. On the first region the integrand is $\le A/(1+|x_1|^3)$, so that part is $\le 2A|x_1|/(1+|x_1|^3) \le A'/(1+|x_1|^2)$. On the second, the integrand is $\le A/(1+|x_2|^3)$, so that part is $\le A''\int_{|x_2|\ge|x_1|} dx_2/(1+|x_2|^3) \le A'''/(1+|x_1|^2)$. F is continuous: the integrals over $|x_2| \le N$ are continuous (Theorem 2.1) and converge uniformly in x_1, the tails being $\le \int_{|x_2|>N} A/(1+|x_2|^3)$.

Step Identity. Given $\epsilon > 0$, choose N so large that $|\int_{\mathbb{R}^2} f - \int_{Q_N} f| < \epsilon$. Also choose it so that $\int_{|x_1|>N/2} |F| < \epsilon$, and so that $\int_{|x_1|\le N/2}\int_{|x_2|>N/2} |f|\,dx_2\,dx_1 < \epsilon$; the last is possible since the inner integral is $\le \min\big(\frac{C}{1+|x_1|^2}, \frac{C}{N}\big)$. By Theorem 2.1 on Q_N,

$$\int_{Q_N} f = \int_{|x_1|\le N/2}\int_{|x_2|\le N/2} f\,dx_2\,dx_1,$$

which differs from $\int_{|x_1|\le N/2} F(x_1)dx_1$ by $< \epsilon$, which in turn differs from $\int_{\mathbb{R}} F$ by $< \epsilon$. Hence $|\int_{\mathbb{R}^2} f - \int_{\mathbb{R}} F| < 3\epsilon$. ■

The same holds in $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with the same proof.

A.3.3 Spherical coordinates

If f is of moderate decrease on $\mathbb{R}^d$, then for fixed $\gamma \in S^{d-1}$,

$$|f(r\gamma)r^{d-1}| \le \frac{Ar^{d-1}}{1+r^{d+1}} \le \frac{B}{1+r^2},$$

so $r \mapsto f(r\gamma)r^{d-1}$ is of moderate decrease on $[0,\infty)$. Letting $N \to \infty$ in (A.2) (the convergence of the inner integral is uniform in γ) gives

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

Combining this with rotation invariance $\int_{\mathbb{R}^d} f(R(x))\,dx = \int_{\mathbb{R}^d} f(x)\,dx$ yields rotation invariance of the surface measure:

$$\int_{S^{d-1}} f(R(\gamma))\,d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\,d\sigma(\gamma) \tag{A.3}$$

for continuous f on S^{d-1}. (Apply the previous formula to $F(x) = f(x/|x|)\varphi(|x|)$ with $\varphi \ge 0$ continuous, compactly supported in $(0,\infty)$, $\int_0^\infty \varphi(r)r^{d-1}dr = 1$.)